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Cambridge IGCSE Combined and Co-ordinated Sciences

Biology B1–B5 Combined Study Guide

Pages

Comprehensive Exam Preparation Guide

Biology Chapters B1–B5 — Cambridge IGCSE Biology Study Guide

Combined guide: B1 Cells and Organisms · B2 Movement into and out of Cells · B3 Biological Molecules · B4 Enzymes · B5 Plant Nutrition

Chapter Overview

This guide brings together the first five biology chapters of the Cambridge IGCSE Combined and Co-ordinated Sciences coursebook. Each chapter has its own notes, key formulas and concepts, textbook questions and practice exam questions, all with full answers. The Revision and Exam Preparation section at the end collects the exam focus points and chapter summaries in one place.

B1 Cells and Organisms

All learners study some content in this chapter.

In this chapter you will:

- Learn about the seven characteristics of living organisms
- Find out about the structure of the cells of bacteria, plants and animals
- Learn about the functions of each of the cells of bacteria, plants and animals
- Identify cell structures in diagrams
- Describe how the structures of some specialised cells are related to their functions
- State that new cells are produced by division of existing cells
- Describe the meaning of the terms cell, tissue, organ, organ system and organism
- Practise using the magnification equation
- Convert measurements between millimetres (mm) and micrometres (μm)

B2 Movement into and out of Cells

In this chapter you will:

- Learn about diffusion
- Investigate factors that affect the rate of diffusion
- Find out how osmosis is a special kind of diffusion, involving water
- Investigate osmosis, using plant tissues
- Learn about active transport
- Learn about water potential
- Explain how osmosis affects plant cells
- Explain how active transport happens

All learners study some content in this chapter. The big idea is that substances move into and out of cells in three ways: **diffusion** (any particles, down a concentration gradient), **osmosis** (water only, through a partially permeable membrane) and **active transport** (against a concentration gradient, using energy from respiration).

B3 Biological Molecules

In this chapter you will:

- Find out about the components of carbohydrates, proteins and lipids
- Use food tests to detect the presence of different biological molecules

All learners study all content in this chapter. The whole chapter is one topic: B3.01 Carbohydrates, fats and proteins.

B4 Enzymes

In this chapter you will:

- Learn about enzymes, and how they work as catalysts
- Investigate how temperature and pH affect the activity of enzymes
- Find out why enzymes are specific, and why temperature and pH affect them

All learners study some content in this chapter. In the syllabus, explaining enzyme action using the enzyme–substrate complex, explaining why enzymes are specific, and explaining (not just describing) the effects of temperature and pH are Supplement content. These parts are marked (Supplement) in this guide.

B5 Plant Nutrition

In this chapter you will:

- Learn about photosynthesis and the equation for it
- Find out how plants use the carbohydrates made in photosynthesis
- Learn how the structure of a leaf is adapted for photosynthesis
- Investigate the need for chlorophyll, light and carbon dioxide in photosynthesis
- Investigate how light and temperature affect the rate of photosynthesis
- Find out how a plant uses glucose produced in photosynthesis
- Learn about the functions of nitrate ions and magnesium ions in a plant
- Identify structures of the leaf in diagrams and images of a dicotyledonous plant
- Learn how the features of a leaf are adaptations for photosynthesis

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Important Terms

B1 Cells and Organisms

Term	Definition
Organism	A living thing, such as a human, a platypus, a bacterium or a mango tree
Cell	The smallest unit from which all organisms are made
Unicellular	Made of just a single cell; bacteria and yeast are unicellular
Multicellular	Made of many cells; humans and plants are multicellular
Movement	An action by an organism or part of an organism causing a change of position or place
Respiration	The chemical reactions in cells that break down glucose and other nutrient molecules to release energy for the cell to use
Sensitivity	The ability to detect changes in the internal or external environment and respond to them
Growth	A permanent increase in size and dry mass, by the growth of cells and the addition of new cells
Reproduction	The processes that make new organisms of the same species

Excretion	The removal from the body of the waste products of metabolism (unwanted or toxic substances) and of substances in excess of requirements
Nutrition	Taking in substances from the environment and using them for energy, growth and development
Metabolism	All the chemical reactions that take place in an organism's cells, including respiration
Metabolic reactions	Chemical reactions that take place in living organisms
Light microscope	A microscope that shines light through the specimen and uses glass lenses to magnify and focus the image; a very good one magnifies about ×1500
Electron microscope	A microscope that can magnify far more than a light microscope; needed to see ribosomes, and to see mitochondria clearly
Photomicrograph	A photograph taken using a light microscope
Electron micrograph	A picture taken using an electron microscope, such as Figure B1.10
Cell membrane	A very thin layer of protein and fat surrounding the cytoplasm of every cell; it controls what enters and leaves the cell
Partially permeable	Allows some molecules and ions to pass through, but not others
Cell wall	A tough layer outside the cell membrane; found in the cells of plants, fungi and bacteria
Cellulose	A polysaccharide that forms criss-crossing fibres; plant cell walls are made mainly of cellulose
Fully permeable	Allows all molecules and ions to pass through it
Cytoplasm	The jelly-like material that fills a cell; it is nearly all water, with many substances (especially proteins) dissolved in it
Vacuole	A fluid-filled space inside a cell, separated from the cytoplasm by a membrane
Cell sap	The fluid that fills the large vacuoles in plant cells; a solution of sugars and other substances
Vesicle	A very small vacuole
Nucleus	A structure containing DNA in the form of chromosomes (plural: nuclei)
Chromosome	A length of DNA, found in the nucleus of a cell; it contains genetic information in the form of many different genes
DNA	A molecule that contains genetic information, in the form of genes, that controls the proteins that are made in the cell
Chloroplasts	Small structures found inside some plant cells, inside which photosynthesis takes place
Chlorophyll	The green pigment inside chloroplasts that absorbs energy from sunlight for photosynthesis

Starch grains	Structures that consist of thousands of starch molecules, stored in some plant cells
Ribosome	A very small structure in a cell that uses information on DNA to make protein molecules
Mitochondrion	A small structure in a cell where aerobic respiration releases energy from glucose (plural: mitochondria)
Aerobic respiration	Chemical reactions in mitochondria that use oxygen to break down glucose and other nutrient molecules to release energy for the cell to use
Bacteria	Unicellular organisms whose cells do not contain a nucleus (singular: bacterium)
Prokaryotic cells	Cells with no nucleus; bacteria have prokaryotic cells
Circular DNA	The single large circle of DNA found in a bacterial cell instead of chromosomes in a nucleus; sometimes called a bacterial chromosome
Plasmids	Small, circular molecules of DNA found in many prokaryotic cells in addition to the main, much larger circle of DNA
Peptidoglycan	The substance that bacterial cell walls are made of, instead of cellulose
Specialised cell	A cell whose structure is modified to help it carry out a particular function effectively
Tissue	A group of similar cells that work together to perform a particular function
Epidermis	The outer layer of tissue on a plant; also the outer layer of an animal's skin
Organ	A structure made up of a group of tissues, working together to perform specific functions
Organ system	A group of organs with related functions, working together to perform body functions
Digestion	The breakdown of food
Magnification	How many times larger an image is than the actual object
Micrometre (μm)	A unit of length equal to one thousandth of a millimetre; 1 mm = 1000 μm and 1 μm = 1×10^{-6} m

B2 Movement into and out of Cells

Term	Definition
Particles	(In this context) the smallest pieces of which a substance is made; particles can be molecules, atoms or ions
Diffusion	The net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement
Net movement	Overall or average movement
Concentration gradient	An imaginary 'slope' from a high concentration to a low concentration
Random movement	Movement in any direction, purely by chance; particles travel in straight lines and change direction when they collide with other particles

Kinetic energy	The energy of movement; the energy for diffusion and osmosis comes from the kinetic energy of the moving particles, not from respiration
Rate of diffusion	How fast particles diffuse; it is faster at a higher temperature, with a larger surface area, a steeper concentration gradient and a shorter diffusion distance
Diffusion distance	The distance particles have to travel, e.g. the thickness of a membrane; the shorter the distance, the faster the diffusion
Surface area to volume ratio	The surface area of an object divided by its volume; cutting a cube into smaller cubes increases this ratio, so substances diffuse in faster
Independent variable	The variable that you change in an experiment
Dependent variable	The variable that you measure as you collect your results
Standardised variable	A variable that you keep the same throughout an experiment so that it cannot affect the results (also called a control variable)
Solvent	The liquid in which a substance dissolves; in the solutions in and around cells, the solvent is water
Solute	A substance that is dissolved in a solvent, e.g. sugar dissolved in water
Partially permeable membrane	A membrane that lets some molecules through (e.g. small water molecules) but not others (e.g. much larger sugar molecules); cell membranes are partially permeable
Osmosis	The diffusion of water molecules through a partially permeable membrane
Osmosis (in terms of water potential)	The net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane
Water potential	A way of describing how much water a solution contains compared with dissolved solute; a dilute solution has a high water potential, a concentrated solution has a low water potential, and pure water has the highest water potential of all
High water potential	An area where there are a lot of water molecules, that is, a dilute solution
Low water potential	An area where there are not many water molecules, that is, a concentrated solution
Water potential gradient	A difference in water potential between two areas; water moves down it, from a high to a low water potential
Turgid	A description of a plant cell that is tight and firm
Turgor pressure	The pressure of the water pushing outwards on a plant cell wall
Flaccid	A description of a plant cell that has lost its turgor pressure and is soft
Plasmolysed	A description of a cell in which the cell membrane tears away from the cell wall
Plasmolysis	What happens when a plant cell loses so much water by osmosis that its cell membrane tears away from the cell wall; it can kill the cell
Wilting	The drooping of a plant that happens when its cells become flaccid and the plant loses its firmness

Active transport	The movement of molecules or ions through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient) using energy from respiration
Carrier proteins (protein carriers)	Protein molecules in cell membranes that can use energy to change shape and move ions or molecules into or out of a cell
Root hair cell	A cell on the surface of a plant root that takes in water and mineral ions from the soil; it uses active transport to take in nitrate ions
Dialysis	A treatment for kidney failure in which unwanted substances diffuse out of the blood, through a very thin membrane, into dialysis fluid that then flows away
Indicator	A dye that changes colour depending on pH; cresol red is red-purple in alkaline conditions and turns yellow when acid is added
Anomalous result	A result that does not fit the pattern or trend shown by the other results

B3 Biological Molecules

Term	Definition
Biological molecules	Molecules that make up living cells, such as carbohydrates, fats and proteins; they are not often found in non-living things
Element	A substance made of only one kind of atom, e.g. carbon (C), hydrogen (H), oxygen (O) or nitrogen (N)
Compound	A substance made of two or more different elements chemically combined, e.g. glucose
Molecule	A particle made up of two or more atoms combined together, e.g. O_2 or $C_6H_{12}O_6$
Carbohydrates	Substances that include sugars, starch and cellulose; they contain carbon, hydrogen and oxygen, with about twice as many hydrogen atoms as carbon or oxygen atoms
Sugars	Carbohydrates that have relatively small molecules; they are soluble in water and they taste sweet
Glucose	A sugar that is used in respiration to release energy; formula $C_6H_{12}O_6$; the form in which carbohydrate is transported in the blood
Reducing sugar	A sugar, such as glucose, which turns Benedict's solution orange-red when they are heated together
Starch	A carbohydrate that is used as an energy store in plant cells; made of many glucose molecules in a long chain that coils into a spiral
Glycogen	A carbohydrate that is used as an energy store in animal cells; made of many glucose molecules in branched chains
Lipids	Substances containing carbon, hydrogen and oxygen; they are insoluble in water and are used as energy stores in organisms; another name for fats and oils
Fats	Lipids that are solid at room temperature
Oils	Lipids that are liquid at room temperature
Glycerol	One of the two kinds of smaller molecule that make up a fat; each fat molecule contains one glycerol molecule

Fatty acids	Smaller molecules that combine with glycerol to make fats and oils; each fat molecule contains three fatty acid molecules
Protein	A substance whose molecules are made of many amino acids linked together; each different protein has a different sequence of amino acids
Amino acids	Substances with molecules containing carbon, hydrogen, oxygen and nitrogen; there are 20 different amino acids found in organisms
Enzymes	Proteins that act as biological catalysts, speeding up chemical reactions in living organisms; all enzymes are proteins
Antibodies	Molecules secreted by white blood cells, which bind to pathogens and help to destroy them; antibodies are proteins
Pathogens	Microorganisms, such as bacteria, that cause disease
Haemoglobin	The red pigment that transports oxygen in mammalian blood; it is a protein
Keratin	The protein that forms hair (and fingernails)
Iodine solution	A solution of iodine in potassium iodide; it is orange-brown, and turns blue-black when mixed with starch
Benedict's solution	A blue liquid that turns orange-red when heated with a reducing sugar
Biuret reagent	A blue solution (also called biuret solution) that turns violet (purple) when mixed with proteins
Emulsion	A liquid containing two substances that do not fully mix; one of them forms tiny droplets dispersed throughout the other
Iodine test	The food test for starch: add iodine solution to the food; a blue-black colour means starch is present
Benedict's test	The food test for reducing sugars: heat the food with Benedict's solution in a water-bath; green, yellow or orange-red means reducing sugar is present
Biuret test	The food test for proteins: add biuret solution to the food and shake, with no heating; a violet (purple) colour means protein is present
Ethanol emulsion test	The food test for fats and oils: shake the food with ethanol, then pour the ethanol into water; a milky white emulsion means fat or oil is present
Water-bath	A container of water kept at a steady temperature, used to heat test-tubes gently and evenly, e.g. at about 80 °C for the Benedict's test

B4 Enzymes

Term	Definition
Catalyst	A substance that increases the rate of a chemical reaction and is not changed by the reaction
Biological catalyst	A catalyst made by living cells; enzymes are biological catalysts that speed up the reactions inside organisms
Substrate	The substance that an enzyme causes to react (for example, starch is the substrate of amylase)
Product	The new substance formed by a chemical reaction

Active site	The part of an enzyme molecule to which the substrate temporarily binds
Complementary	With a perfect mirror-image shape; the active site and its substrate have complementary shapes, so they fit together
Enzyme–substrate complex	The short-lived structure formed as the substrate binds temporarily to the active site of an enzyme
Specificity	Of enzymes, only able to act on a particular (specific) substrate
Lock and key model	The idea that a substrate fits an enzyme's active site like a key fits a lock; only a substrate with the complementary shape can fit
Carbohydrases	Enzymes that break down carbohydrates (for example, amylase, maltase, sucrase and lactase)
Protease	An enzyme that catalyses the breakdown of protein to amino acids
Lipases	Enzymes that break down lipids (fats and oils) into fatty acids and glycerol
Amylase	An enzyme that catalyses the breakdown of starch to maltose
Maltase	An enzyme that catalyses the breakdown of maltose to glucose
Sucrase	An enzyme that breaks down sucrose
Lactase	An enzyme that breaks down the milk sugar lactose into smaller sugar molecules that can be absorbed into the blood
Catalase	An enzyme that catalyses the breakdown of hydrogen peroxide to water and oxygen; it works inside all living cells
Hydrogen peroxide	A very dangerous substance made by many chemical reactions inside cells; catalase must break it down immediately
Lactose intolerance	Feeling ill (diarrhoea, abdominal pain, flatulence) after eating dairy foods, because the body does not make lactase and lactose is not broken down
Optimum	Best; for example, the optimum temperature of an enzyme is the temperature at which its activity is greatest
Optimum temperature	The temperature at which an enzyme works fastest; about 37 °C for enzymes from the human body
Optimum pH	The pH at which an enzyme works fastest; around pH 7 for most enzymes, but about pH 2 for a protease in the stomach
Denaturation	Damage to an enzyme's shape (including its active site) caused by high temperature or an unsuitable pH; the enzyme is denatured and can no longer bind its substrate or catalyse its reaction
Effective collision	A collision in which a substrate molecule enters an enzyme's active site, forms an enzyme–substrate complex and is converted to product
Range	The lowest to the highest value
Buffer solution	A liquid that has a known pH, and keeps that pH steady all the time, even if chemical reactions take place
Thymolphthalein	An indicator that is blue in alkaline conditions and loses its colour when the pH becomes less alkaline

B5 Plant Nutrition

Term	Definition
Photosynthesis	The process by which plants synthesise carbohydrates from raw materials using energy from light
Organic substances	Carbon-containing biological molecules such as carbohydrates, fats and proteins; plants can make them but animals must take them in as food
Raw materials of photosynthesis	Carbon dioxide (from the air) and water (from the soil), which react to make glucose and oxygen
Chloroplast	The part of a plant cell where photosynthesis happens; contains chlorophyll (spread out on membranes) and the enzymes for photosynthesis
Sucrose	A sugar whose molecules are made of glucose and another similar molecule (fructose) linked together; the form in which plants transport carbohydrate
Nitrate ions	Mineral ions absorbed from the soil through the root hairs; they supply the nitrogen needed to make amino acids, proteins and chlorophyll
Magnesium ions	Mineral ions absorbed from the soil; magnesium is needed to make chlorophyll
Nectar	A sweet liquid secreted by many insect-pollinated flowers, to attract their pollinators
Pollen grains	Small structures which contain the male gametes of a flower
Pollinator	An animal (insect, bat or bird) that carries pollen grains from one flower to another
Xylem vessels	Tubes that carry water and mineral ions from the roots up to the leaves
Phloem tubes	Tubes that carry sucrose (and amino acids) from one part of a plant to another
Vascular bundles	Collections of xylem tubes and phloem vessels running side by side, which form the veins in a leaf
Lamina	The broad, flat part (blade) of a leaf
Midrib	The main vein running down the centre of a leaf
Leaf stalk	The stalk that joins the lamina to the rest of the plant; it contains vascular bundles
Dicotyledonous (dicot) plant	A flowering plant whose leaves usually have a branching network of veins
Transverse section (TS)	A thin slice cut across a structure, such as a leaf, so that its layers of cells can be seen under a microscope
Cuticle	A thin, transparent, waterproof covering made of a waxy substance, on the upper surface of a leaf
Upper epidermis	The upper layer of tightly packed, transparent cells that reduces water vapour loss and lets light through
Palisade mesophyll	A layer of tall, narrow cells with very large numbers of chloroplasts near the top of the leaf; the main site of photosynthesis

Spongy mesophyll	A layer of photosynthesising cells with many air spaces between them, allowing gases to diffuse through the leaf
Air spaces	Gaps between spongy mesophyll cells through which carbon dioxide, oxygen and water vapour diffuse
Lower epidermis	The bottom layer of cells of a leaf; it contains the stomata and usually has no cuticle
Stoma (plural stomata)	An opening in the lower epidermis that lets carbon dioxide and oxygen diffuse in and out, and water vapour diffuse out
Guard cells	A pair of cells (containing chloroplasts) surrounding each stoma; they change shape to open and close it
Variegated leaf	A leaf that has some white areas (no chlorophyll) and some green areas (with chlorophyll)
Destarching	Leaving a plant in the dark for long enough for it to use up its starch stores
Control	A standard sample that you use as a comparison, to find the effect of changing a variable
Potassium hydroxide solution	A solution that absorbs carbon dioxide; used to remove carbon dioxide from the air around a leaf
Sodium hydrogencarbonate solution	A solution that provides carbon dioxide for aquatic plants; a higher concentration supplies more carbon dioxide
Rate of photosynthesis	How much photosynthesis happens in unit time, e.g. the number of oxygen bubbles released by an aquatic plant in one minute
Limiting factor	The factor in shortest supply (light intensity, carbon dioxide concentration or temperature) that stops the rate of photosynthesis going any higher
Heat shield	A transparent barrier between a lamp and a plant that lets light through but stops the lamp's heat changing the temperature
Glasshouse	A glass building in which temperature, light and carbon dioxide can be controlled so that crops photosynthesise quickly and give a high yield
Biofuels	Fuels made from plants grown for the purpose, which can be burnt to produce electricity or move vehicles

B1 Cells and Organisms

B1.01 Characteristics of living organisms

Before You Start

The chapter begins with two questions to discuss with a partner.

Q: The list below contains some features of living organisms: breathing, excretion, a blood system, a nervous system, sensitivity, growth, reproduction, movement, nutrition, respiration. Which of these features are found in all living organisms?

A: Excretion, sensitivity, growth, reproduction, movement, nutrition and respiration. These are the seven characteristics of living things. Breathing, a blood system and a nervous system are not found in all organisms: plants and bacteria do not breathe (breathing is not the same as respiration), and they have no blood system and no nervous system.

Q: If you have a microscope, how can you distinguish between a cell from a plant, and a cell from an animal?

A: Look for a cell wall. All plant cells have a cell wall outside the cell membrane, but animal cells never do. You could also look for chloroplasts and a large central vacuole: if a cell has chloroplasts, it is a plant cell. However, a cell without chloroplasts could still be a plant cell (for example, a root cell), so the cell wall is the most reliable clue.

What Is an Organism?

Biology is the study of **organisms**. An organism is a complete living thing, such as yourself, a platypus, a bacterium or a mango tree. There are very many different kinds of organism on Earth, but all of them share **seven characteristics** (Figure B1.01).

Some non-living things have some of these characteristics, but no non-living thing has all of them. For example, a car moves, takes in fuel and gives out waste gases, but it cannot grow or reproduce.

The Seven Characteristics of Living Organisms

A useful way to remember the seven characteristics is **MRS GREN**: Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition.

Characteristic	What it means	Example
Movement	All organisms are able to move to some extent. Most animals can move their whole body from place to place, and plants can slowly move parts of themselves	A bird flying; a plant's leaves slowly turning towards the light
Respiration	All organisms break down glucose and other substances inside their cells, to release energy that they can use. This energy is used to make other chemical reactions in their cells happen	Every living cell respires all the time, day and night
Sensitivity	All organisms pick up information about changes in their environment, and react to the changes. The changes may be in the internal environment (such as the temperature of the blood) or the external environment (such as the intensity of sunlight)	You pull your hand away from something hot; a plant shoot grows towards the light
Growth	All organisms begin small and get larger, by the growth of their cells and by adding new cells to their bodies.	A young kangaroo (joey) growing into an adult

	Growth is a permanent increase in size or dry mass	
Reproduction	Organisms are able to make new organisms of the same species as themselves	A kangaroo producing a joey; a mango tree producing seeds
Excretion	All organisms produce unwanted or toxic waste products as a result of their metabolic reactions, and these must be removed from the body	Breathing out carbon dioxide made by respiration; getting rid of urea in urine
Nutrition	Organisms take substances from their environment and use them to provide energy or materials to make new cells	A kangaroo eating plants; a plant making its own food by photosynthesis

Metabolism

The energy released by respiration is used to make the other chemical reactions in cells happen. All of these chemical reactions together, including respiration, are called **metabolism**. Each individual reaction is a **metabolic reaction**. Metabolism produces waste products, some of them toxic, which is why every organism needs to excrete.

Measuring Growth: Why Dry Mass?

Growth can be defined as a **permanent** increase in size or dry mass. In exam answers, use the Cambridge syllabus wording: growth is a permanent increase in size and dry mass. The word *permanent* matters: if you drink a glass of water you become heavier for a while, but that is not growth.

Measuring **dry mass** involves killing and drying the organism, then finding the mass of its body without any water. Because the organism has to be killed, this method is used more often for plants than for animals. Dry mass is a more reliable measure of growth than ordinary (fresh) mass, because the amount of water in an organism goes up and down all the time.

What Figure B1.01 Shows

Figure B1.01 is a scene of red kangaroos in the Australian outback: an adult kangaroo with a joey in its pouch, a second kangaroo bending down to feed on plants, a bird flying overhead and trees in the background. Each of the seven characteristics is printed around the picture with a line pointing to part of the scene. For example, the flying bird shows movement, the joey in the pouch shows reproduction, and the feeding kangaroo shows nutrition. The message is that every living thing in the scene, including the trees, carries out all seven characteristics.

Science in Context B1.01: How Many Cells in a Human Body?

In some organisms, it is possible to count the number of cells in their bodies. In organisms such as humans, we can only estimate the number, because it would be impossible to count them all. Cells also differ in their volume, mass and how closely or loosely they are packed together, so you cannot simply divide body mass by the mass of one cell.

Scientists recently calculated the numbers of different cells and cell types in different organs, and added them up. The latest estimate suggests that the body of an adult human contains around **37 trillion cells**, which is 37 000 000 000 000.

Q: Can you think of any advantages for larger organisms of consisting of multiple cells, rather than just one larger cell?

A: In a multicellular organism, different cells can become specialised to carry out particular functions (for example, nerve cells, muscle cells and red blood cells), so the organism can do many different jobs efficiently. Other advantages: if some cells are damaged or die, they can be replaced and the organism

survives; and small cells can exchange substances such as oxygen with their surroundings quickly, whereas in one giant cell substances would take too long to reach the middle.

Activity B1.01: Matching the Characteristics of Living Things with Their Descriptions

Aim: To learn the seven characteristics of living things and what each one means, by playing a memory card game.

Group size: A group of four or five.

You will need: 14 pieces of blank card, all exactly the same.

Method:

1. Write the seven characteristics of living things on seven of the pieces of card.
2. Write descriptions of each of the seven characteristics on the other seven cards.
3. Shuffle each set of cards. Place them face down in two rows of seven.
4. One person then selects a card from each row and turns them face up. If the name and description match, this person keeps the two cards. If they do not match, they place them face down again in the same positions.
5. Now the next person does the same.
6. Keep taking turns until all the cards have been taken by someone. The winner is the person with most cards at the end.

Expected results: Every name card is eventually paired with its correct description. Players who remember where cards are, and who know the descriptions well, collect the most pairs.

Conclusion: By the end you should be able to name all seven characteristics and match each one to its meaning. The cards must be identical so that nobody can recognise a card from its back.

Safety: There are no special hazards in this activity.

Questions B1.01 and B1.02 on this topic are answered in the Textbook Questions with Answers section.

B1.02 Cell structure

Organisms Are Made of Cells

All organisms are made of **cells**. Cells are very small, so large organisms contain millions of cells. Organisms made of many cells are **multicellular**. Some organisms are **unicellular**, which means that they are made of just a single cell. Bacteria and yeast are examples of single-celled organisms.

New Cells Come from Existing Cells

All cells are made from existing cells. New cells are formed when a fully grown cell **divides** into two. Cells never appear from non-living material. You can read about how cells divide in Chapter B12.

Microscopes

To see cells clearly, you need to use a microscope (Figure B1.02). The kind of microscope used in a school laboratory is called a **light microscope**, because it shines light through the piece of animal or plant you are looking at. It uses glass lenses to magnify and focus the image. A very good light microscope can magnify about **1500 times**.

Part of the light microscope (Figure B1.02)	What it does
Eyepiece	The lens at the top that you look through
Objective lenses	The lenses just above the specimen that magnify it; there are usually several, each giving a different magnification

Focusing knob	Turned to move the lenses and stage closer together or further apart until the image is sharp
Stage	Where the specimen is placed (usually on a glass slide)
Mirror	Reflects light up through the specimen

A photograph taken using a light microscope is called a **photomicrograph**. Some structures are too small to see clearly with a light microscope, so biologists use an **electron microscope**, which can magnify far more. A picture taken with an electron microscope is an **electron micrograph**.

Animal and Plant Cells Under a Light Microscope

Figure B1.03 (an animal cell): a cell with an irregular, rounded outline. It shows a thin cell membrane, cytoplasm, a large dark nucleus, several mitochondria, tiny dots labelled ribosomes, and a small vacuole or vesicle.

Figure B1.04 (a plant cell): a box-shaped cell. It shows a cell wall on the outside with the cell membrane pressed against it, a thin layer of cytoplasm around the edge containing the nucleus, green chloroplasts, mitochondria and ribosomes, and a very large vacuole filling most of the middle of the cell.

Ribosomes and mitochondria are drawn in these figures so that you know where they are. However, the book explains later that ribosomes are so small they can only be seen with an electron microscope, and mitochondria cannot be seen clearly without one.

Figure B1.05: a photomicrograph of cells from the trachea (windpipe) of a mammal. The cells have been stained (coloured) with a dye that makes the nucleus look darker than the cytoplasm. The cell membrane, cytoplasm and nucleus are labelled.

Figure B1.06: a photomicrograph of cells from a moss plant. The cell walls form a clear network, and each cell contains many green chloroplasts and a nucleus. You cannot see the cell membranes, because they are pressed tightly against the inside of the cell walls.

Cell Membrane

Every cell has a **cell membrane** around the outside. The cell membrane is a very thin layer of protein and fat. It is very important to the cell because it **controls what goes in and out** of it.

The cell membrane is **partially permeable**. This means that it will let some substances through but not others. The cell membrane separates the contents of the cell from its environment (surroundings).

It is difficult to see the cell membrane in a plant cell, because it is pressed tightly against the inside of the cell wall.

Cell Wall

All plant cells are surrounded by a **cell wall** made mainly of **cellulose**. Animal cells never have cell walls. Cellulose belongs to a group of substances called polysaccharides (Chapter B3).

Cellulose forms fibres which criss-cross over one another to form a very strong covering to the cell. Figure B1.07 is a three-dimensional photograph of a cell wall surface, showing these criss-crossing cellulose fibres. The cell wall:

- Helps to protect and support the cell
- Prevents the cell from bursting if it absorbs a lot of water and swells

Because of the spaces between the fibres, even very large molecules are able to go through the cellulose cell wall. The cell wall is **fully permeable**, so it does not control what enters the cell; that is the job of the cell membrane.

Cell walls are also found in the cells of fungi and bacteria, but these walls are not made of cellulose.

Cytoplasm

Cytoplasm is a clear jelly. It is nearly all water, and it contains many substances dissolved in it, especially proteins. Many different **metabolic reactions** (the chemical reactions of life) take place in the cytoplasm.

Vacuole

A **vacuole** is a fluid-filled space inside a cell, surrounded by its own membrane.

- **Plant cells** have very large, **permanent vacuoles**, which contain a solution of sugars and other substances called **cell sap**. When the vacuole is full, it presses outwards on the rest of the cell and helps it to keep its shape.
- **Animal cells** have much smaller vacuoles, called **vesicles**, which also contain solutions.

Nucleus

The **nucleus** (plural: nuclei) is where the genetic information is stored. The information is kept on the **chromosomes**, which are inherited from the organism's parents. The chromosomes are made of **DNA**. The information carried on the DNA determines the kinds of proteins that are made in the cell.

Figure B1.08 is a photomicrograph of some plant cells dividing. The nuclei have been coloured with a blue stain, and the things that look like spider legs are chromosomes. We can normally only see chromosomes when a cell is dividing.

Chloroplasts

Chloroplasts are never found in animal cells, but most of the cells in the green parts of plants have them. They contain the green colouring, or pigment, called **chlorophyll**. Chlorophyll absorbs energy from sunlight, and this energy is then used to make food for the plant by **photosynthesis** (Chapter B5).

Chloroplasts often contain **starch grains**, which have been made by photosynthesis. Animal cells never contain starch grains.

Figure B1.09 shows a chloroplast from a cell in a pea plant. The large blue structure inside the chloroplast is a starch grain. The black stripes inside the chloroplast are membranes where the chlorophyll is kept.

Ribosomes

Ribosomes are tiny structures found in almost all animal cells and plant cells. They are so small that we can only see them with an electron microscope.

Ribosomes are where the cell **makes proteins**. The instructions on the DNA molecules are used to link together long chains of amino acids in a particular sequence. You can read more about protein molecules in Chapter B4, and more about how a cell makes them in Chapter B6.

Mitochondria

All the structures above (cell membrane, cell wall, cytoplasm, vacuoles, nucleus and chloroplasts) can be seen with a good light microscope. But some smaller structures cannot be seen clearly unless we use an electron microscope. These include **mitochondria** (singular: mitochondrion).

Mitochondria are found in almost all plant and animal cells. They are the parts of the cell where **aerobic respiration** happens. This is how energy is released from glucose (Chapter B9).

Aerobic respiration is the main way in which cells get the energy that they need to stay alive. The more energy a cell needs, the more mitochondria it has. Muscle cells, for example, are packed full of mitochondria.

Figure B1.10 is an electron micrograph of a mitochondrion, magnified 20 000 times. It is a long, oval structure. The pink lines are membranes, which separate the inside of the mitochondrion from the rest of the cell; the inner membrane makes many folds across the inside.

Figure B1.11 is a drawing of a mitochondrion cut open, showing it in three dimensions. Two membranes separate the inside of the mitochondrion from the rest of the cell, and the inner membrane is folded.

Summary Table: Structures in Animal and Plant Cells

Structure	Function	Animal cell	Plant cell
Cell membrane	Partially permeable; controls what enters and leaves the cell	Yes	Yes
Cell wall	Fully permeable; protects and supports the cell; stops it bursting when it takes up water	Never	Yes (made of cellulose)
Cytoplasm	Jelly where many metabolic reactions take place	Yes	Yes
Nucleus	Contains DNA in chromosomes; the DNA determines which proteins the cell makes	Yes	Yes
Vacuole	Fluid-filled space; in plants, the large vacuole stores cell sap and helps the cell keep its shape	Only small vacuoles (vesicles)	Large permanent vacuole containing cell sap
Chloroplasts	Absorb energy from sunlight for photosynthesis	Never	In most cells of the green parts
Starch grains	Store starch made by photosynthesis	Never	Sometimes (often inside chloroplasts)
Ribosomes	Make proteins	Yes	Yes
Mitochondria	Aerobic respiration, releasing energy from glucose	Yes	Yes

Key idea: plant cells have everything an animal cell has, plus a cell wall, a large permanent vacuole and (often) chloroplasts.

Bacterial Cells

Bacteria (singular: bacterium) are unicellular organisms. Bacterial cells are rather different from the cells of animals and plants. Figure B1.12 is a diagram of a bacterium: a capsule-shaped cell with labels for the cell wall (made of peptidoglycan, not cellulose), cell membrane, cytoplasm, ribosomes, circular DNA and plasmids.

No nucleus. The most important difference between a bacterial cell and animal or plant cells is that bacteria do not have a nucleus. Bacterial cells are also known as **prokaryotic cells**. 'Pro' means 'before', and 'karyotic' means 'nucleus'. Prokaryotic cells appeared on Earth millions of years before cells with nuclei appeared.

Circular DNA. Instead of chromosomes inside a nucleus, bacteria have a circle of DNA, sometimes called a bacterial chromosome. The DNA has exactly the same function as in other cells: it provides instructions for making proteins.

Plasmids. Bacterial cells often have one or more smaller circles of DNA, called **plasmids**. Scientists can use plasmids in the genetic modification of cells and organisms.

Cell wall. Bacterial cells always have a cell wall. Unlike plant cells, this cell wall is not made of cellulose. But its function is the same as in plant cells: it helps to support the cell, and prevents it from bursting if the cell takes up a lot of water.

Cell membrane. A partially permeable cell membrane is pressed tightly against the inside of the bacterial cell wall. As in plant and animal cells, the cell membrane controls what enters and leaves the cell.

Cytoplasm and ribosomes. Bacterial cells have cytoplasm and ribosomes, with the same functions as in animal and plant cells.

Missing structures. Bacterial cells do not have mitochondria or chloroplasts.

Identifying Cell Structures in Diagrams and Micrographs

- **Cell wall:** a thick outer layer, often drawn as a double line, around plant and bacterial cells. An animal cell has only a thin single line (its membrane) around it.
- **Cell membrane:** a thin line around the cytoplasm. In plant cells it lies right against the inside of the wall, so it is hard to see.
- **Nucleus:** a large round or oval body, usually the darkest part of a stained cell.
- **Large vacuole:** a big, clear space in the middle of a plant cell that pushes the cytoplasm and nucleus out to the edges.
- **Chloroplasts:** green oval bodies in the cytoplasm of plant cells.
- **Mitochondria:** small, sausage-shaped bodies. In electron micrographs, look for a double membrane with a folded inner membrane that looks like stripes.
- **Ribosomes:** tiny dots in the cytoplasm, seen only in electron micrographs.
- **Vesicles:** small round spaces surrounded by a membrane.
- **Bacterial cell:** no nucleus; a loop of DNA lies in the cytoplasm, often with small circles (plasmids).

Activity B1.02: Comparing Animal Cells and Plant Cells

Aim: To make a display that compares the structures of animal cells and plant cells.

Group size: A group of three or four.

Method summary: Decide how you will present your comparison. You could use annotated drawings, construct a large comparison table, or make a presentation. Use the information in this chapter, and you might also look for more pictures on the internet.

Expected results: A good display shows the **similarities** (both have a cell membrane, cytoplasm, a nucleus, ribosomes and mitochondria) and the **differences** (only plant cells have a cellulose cell wall, a large permanent vacuole containing cell sap, chloroplasts and starch grains; animal cells have only small vacuoles called vesicles). Annotations should give the function of every structure.

Conclusion: Plant and animal cells share the same basic structures, but plant cells have three extra features linked to their way of life: a cell wall for support, a large vacuole that keeps the cell firm, and chloroplasts for photosynthesis.

Safety: There are no laboratory hazards. When searching online, use reliable websites and check that pictures are labelled correctly.

Reflection: How Will You Learn the Parts of Cells?

The book suggests several ways to learn the names of the parts of animal, plant and bacterial cells, and their functions:

- Looking at diagrams and reading about the structures
- Practising drawing your own diagrams and labelling them
- Getting a friend to test you by asking questions
- Making some revision cards for yourself, with the name of a structure on one side and its function on the other side

Q: What other ideas might you try?

A: For example: draw a cell from memory and then check it against the book; use a mnemonic or colour code (for instance, highlight the three plant-only structures in green); teach the structures to someone else; make a three-column table (animal, plant, bacterial) and fill it in from memory; or practise labelling unlabelled diagrams and micrographs from past papers.

Question B1.07 (comparing bacterial, animal and plant cells) is answered with a full comparison table in the Textbook Questions with Answers section.

B1.03 Specialised cells

What Does Specialised Mean?

Multicellular organisms, such as humans or plants, may contain vast numbers of cells, and not all of these cells are alike. In a human body almost all of our cells have the same features that are found in most animal cells: a cell membrane, cytoplasm and a nucleus. But most of our cells have a particular function to perform, and their structure is modified to help them carry out that function effectively. They are **specialised**.

The same is true in a flowering plant: all the cells have the basic characteristics of plant cells, but they have slightly different structures that relate to their specific functions.

Table B1.01: Examples of Specialised Cells

Table B1.01 lists seven specialised cells (five from animals and two from plants) and where you can find out more about each one.

Specialised cell	Where it is found	Function	Where you can find out more
Root hair cell	Near the tips of the roots of flowering plants	Absorption of water and mineral ions	Chapter B2
Palisade mesophyll cell	In the leaves of flowering plants	Photosynthesis	Chapter B5
Red blood cell	In the blood of mammals	Transporting oxygen	Chapter B7
Ciliated cell	Lining the trachea and bronchi of animals	Cilia move mucus upwards	Chapter B9
Neurone	In the nervous system of animals	Conducting electrical impulses	Chapter B10
Sperm cell	Produced in the testes of mammals	The male gamete in sexual reproduction	Chapter B11
Egg cell	Produced in the ovaries of mammals	The female gamete in reproduction	Chapter B11

How Structure Is Related to Function

You will study each of these cells in later chapters. For now, notice how each cell's structure suits its job:

- **Root hair cell:** has a long, thin extension (the root hair), which gives a large surface area for absorbing water and mineral ions from the soil.
- **Palisade mesophyll cell:** is packed with chloroplasts, so it can absorb a lot of light energy for photosynthesis.
- **Red blood cell:** is full of haemoglobin, which carries oxygen; it has no nucleus, leaving more room for haemoglobin.

- **Ciliated cell:** has tiny hair-like cilia on its surface that beat to sweep mucus (with trapped dust and bacteria) up the trachea and bronchi.
- **Neurone:** has a very long, thin fibre that carries electrical impulses over long distances around the body.
- **Sperm cell:** has a tail for swimming to the egg, and many mitochondria to release the energy needed for swimming.
- **Egg cell:** is large and contains a store of food for the embryo after fertilisation.

Tissues

Often, cells that specialise in the same activity are found together. A group of cells like this is called a **tissue**.

An example of a tissue is the layer of cells lining your stomach. These cells make enzymes to help digest your food. **Figure B1.13** shows a row of these stomach-lining cells sitting side by side on a basement membrane. Each cell has a cell membrane, cytoplasm and a nucleus, plus vacuoles containing digestive enzymes near its top. Enzymes are shown being secreted out of the cells into the stomach.

The stomach also contains other tissues. For example, there is a layer of muscle in the stomach wall, made of cells which can move. This **muscle tissue** makes the wall of the stomach move in and out, churning the food and mixing it up with enzymes.

Plants also have tissues. You may already have looked at some **epidermis** tissue from an onion bulb. Inside a leaf, a layer of cells makes up the **palisade tissue**. These cells are specialised to carry out photosynthesis.

Organs

A group of different tissues that carry out a function together is called an **organ**. The stomach is an organ, made of tissues such as the lining tissue and the muscle tissue. Other organs include the heart, the kidneys and the lungs.

In a plant, an onion bulb is an organ. A leaf is another example of a plant organ.

Organ Systems

The stomach is only one of the organs which help in the **digestion** (breakdown) of food. The mouth, the stomach and the intestines are all part of an **organ system** called the **digestive system**. Other examples:

- The heart is part of the **circulatory system**
- Each kidney is part of the **excretory system**

Levels of Organisation

The way in which organisms are built up can be summarised like this:

cells → tissues → organs → organ systems → organism

Level	Meaning	Animal example	Plant example
Cell	The smallest unit from which all organisms are made	A cell lining the stomach	A palisade mesophyll cell
Tissue	A group of similar cells working together to perform a particular function	The layer of cells lining the stomach; muscle tissue	Palisade tissue; epidermis

Organ	A group of different tissues working together to perform specific functions	Stomach, heart, kidney, lungs	Leaf, onion bulb
Organ system	A group of organs with related functions working together	Digestive system (mouth, stomach, intestines)	The shoot system (stem, leaves and flowers)
Organism	A complete living thing	A human	A mango tree

Exam tip: a tissue is made of *similar* cells, while an organ is made of *different* tissues. Mixing these up is a very common mistake.

B1.04 Sizes of specimens

Why We Need Magnification

Many of the structures that biologists study are very small. Cells, for example, are so small that we cannot see them without a microscope. The photographs and diagrams of cells in this chapter are all much larger than the actual cells.

We can tell someone how much bigger an image is than the actual object by giving its **magnification**. The magnification of an image is how many times larger it is than the real object.

The Magnification Equation

The key equation in this chapter is:

magnification = size of image ÷ size of actual object

It can be rearranged to find either of the other two quantities:

- **actual size = size of image ÷ magnification**
- **size of image = actual size × magnification**

A memory aid is a triangle with *image size* in the top half and *actual size* and *magnification* side by side in the bottom half. Cover up the quantity you want: if the two that are left are side by side, multiply them; if one is above the other, divide the top one by the bottom one.

Two Very Important Rules

1. **Make sure that all the numbers in your calculation have the same units.** It is often a good idea to convert everything to millimetres (mm) before you do anything else.
2. **Magnification is always written with a multiplication sign in front of it (×), and it has no units.** For example, write ×500, never 500 mm.

Millimetres and Micrometres

Some of the objects that we study in biology are so small that even millimetres are not a suitable unit for measuring them. Instead, we use **micrometres**. The symbol for a micrometre is **µm**.

- 1 mm = 1000 µm
- 1 µm = 1×10^{-6} m (one millionth of a metre)
- 1 m = 10^{6} µm (one million micrometres)

Conversion	What to do	Example
cm to mm	Multiply by 10	8 cm = 80 mm
mm to µm	Multiply by 1000	2.5 mm = 2500 µm
µm to mm	Divide by 1000	40 µm = 0.04 mm

m to µm	Multiply by 1 000 000	0.002 m = 2000 µm

Rule of thumb: changing to a *smaller* unit makes the number *bigger* (multiply); changing to a *bigger* unit makes the number *smaller* (divide).

Typical sizes: most plant and animal cells are about 10 to 100 µm across; bacteria and mitochondria are only a few µm long. If your answer for the size of a cell comes out as 50 000 µm (5 cm), check your working.

Worked Examples

Worked Example 1: Calculating Magnification (Same Units)

A drawing of an eraser is 120 mm long. The real eraser is 40 mm long. What is the magnification of the drawing?

1. Equation: magnification = size of image ÷ size of actual object
2. Both lengths are already in mm, so no conversion is needed
3. magnification = 120 ÷ 40 = 3
4. Answer: **×3**

Worked Example 2: Converting Units First

A photomicrograph shows a cheek cell that is 30 mm wide. The real cell is 60 µm wide. Calculate the magnification.

1. The units are different, so convert the image size to µm: 30 mm × 1000 = 30 000 µm
2. magnification = 30 000 µm ÷ 60 µm = 500
3. Answer: **×500**

Check by working in mm instead: 60 µm ÷ 1000 = 0.06 mm, and 30 ÷ 0.06 = 500. You get the same answer whichever unit you use, as long as both sizes are in the same unit.

Worked Example 3: Calculating the Actual Size (Question B1.09)

The mitochondrion in Figure B1.10 measures about 83 mm long on the page, and the magnification is ×20 000. What is its real length?

1. Convert the image size to µm: 83 mm × 1000 = 83 000 µm
2. actual size = size of image ÷ magnification = 83 000 µm ÷ 20 000
3. Answer: **4.15 µm**

Worked Example 4: Calculating the Image Size

A cell that is 25 µm wide is drawn at a magnification of ×2000. How wide is the drawing, in mm?

1. size of image = actual size × magnification = 25 µm × 2000 = 50 000 µm
2. Convert to mm: 50 000 µm ÷ 1000 = 50 mm
3. Answer: **50 mm**

Worked Example 5: Actual Size from a Diagram in cm (Practice Question 4)

A diagram of a flower is 8 cm across, and the magnification is ×4.

1. actual size = size of image ÷ magnification = 8 cm ÷ 4 = 2 cm
2. Convert to mm: 2 cm × 10 = 20 mm
3. Answer: **20 mm**

Worked Example 6: Magnification of a Photograph (Practice Question 5)

The jellyfish in the photograph is 89 mm across, and the real jellyfish is 50 mm across.

1. Both sizes are in mm, so magnification = 89 ÷ 50 = 1.78
2. Answer: **×1.8** (to 2 significant figures)

Common Mistakes in Magnification Questions

- **Mixing units:** dividing a length in mm by a length in µm gives an answer that is 1000 times too big or too small. Convert first.
- **Turning the equation upside down:** for an image that has been enlarged, the magnification must be bigger than 1. If you get a number smaller than 1, you have probably divided the actual size by the image size.
- **Multiplying instead of dividing:** to find the actual size you divide the image size by the magnification. Multiplying 8 cm by 4 to get 32 cm is the trap in Practice Question 4.
- **Adding units to a magnification:** ×500 is correct; ×500 mm is wrong.
- **Forgetting the × sign** in front of the magnification.
- **Measuring in cm and calling it mm:** check which marks on your ruler you are reading.

How to Measure an Image

1. Use a ruler marked in millimetres.
2. Measure the length the question asks for (for example, the maximum length or the diameter), in a straight line from one edge of the object to the other.
3. Line up the zero mark of the ruler, not the end of the ruler, with the edge of the object.
4. Record the measurement to the nearest millimetre, and write it down in your working; examiners give a mark for it.

Activity B1.03: Practising Using the Magnification Equation

Aim: To practise measuring objects and drawings, and using the magnification equation.

How it is organised: You work on your own for the first part of the activity, and then pair up with someone else for the second part.

You will need:

- Several small objects that are easy to measure and draw, for example an eraser, a paper clip and a button
- A ruler that can measure in mm
- Some plain paper, a sharp pencil and an eraser

Method, Part 1 (on your own): Step 1 is to measure each object carefully and write down the measurements. Do not let anyone else see these measurements at this point. The rest of the method continues on the next page of the book. In outline, you draw the objects on plain paper, and in the second part you and your partner use the magnification equation with each other's drawings, while your secret measurements are used to check the answers.

How the equation is used: Suppose a paper clip is 32 mm long and your drawing of it is 96 mm long. The magnification of your drawing is 96 ÷ 32 = ×3. If your partner is told that the drawing is ×3 and measures it as 96 mm, they can calculate the real length: 96 ÷ 3 = 32 mm, which matches your secret measurement.

Expected results: If everyone measures carefully in mm, the calculated sizes agree with the real measurements to within about a millimetre. Small differences come from measuring errors.

Conclusion: The magnification equation can be rearranged to find the magnification, the actual size or the image size, as long as all measurements are in the same units.

Safety: There are no special hazards; take care with sharp pencils.

Questions B1.08 and B1.09 (converting units and calculating the real size of a mitochondrion) are answered in the Textbook Questions with Answers section.

B1 Key Formulas and Concepts

1. **Magnification:** magnification = size of image ÷ size of actual object
2. **Actual size:** actual size = size of image ÷ magnification
3. **Image size:** size of image = actual size × magnification
4. **Same units first:** convert both sizes to the same unit (usually mm or μm) before dividing
5. **Millimetres and micrometres:** 1 mm = 1000 μm, so mm × 1000 = μm and μm ÷ 1000 = mm
6. **Centimetres:** 1 cm = 10 mm
7. **Metres:** 1 m = 1000 mm = 1 000 000 μm (10^6 μm), and 1 μm = 1 × 10^{-6} m
8. **Writing magnification:** always with a × sign and no units, for example ×1500
9. **Fitting objects end to end:** number that fit = total length ÷ length of one object, both in the same unit (for example 1000 μm ÷ 4.15 μm ≈ 241 mitochondria in 1 mm)
10. **Microscope power:** a very good light microscope magnifies about ×1500; electron microscopes magnify far more (Figure B1.10 is ×20 000)
11. **Levels of organisation:** cells → tissues → organs → organ systems → organism
12. **Seven characteristics (MRS GREN):** movement, respiration, sensitivity, growth, reproduction, excretion, nutrition
13. **Aerobic respiration (in mitochondria, details in Chapter B9):** glucose + oxygen → carbon dioxide + water, releasing energy
14. **Plant cell extras:** plant cell = animal cell structures + cell wall + large permanent vacuole + (often) chloroplasts
15. **Bacterial cell:** cell wall (not cellulose) + cell membrane + cytoplasm + ribosomes + circular DNA + (often) plasmids; no nucleus, no mitochondria, no chloroplasts

B1 Textbook Questions with Answers

Question B1.01

A student claimed that plants show fewer of the characteristics of living things than animals. Explain why this claim is wrong.

Answer: The claim is wrong because plants, like all living organisms, show all seven characteristics of living things. Plants simply carry out some of them more slowly or less obviously than animals, which may be why the student was misled.

- **Movement:** plants cannot move their whole body from place to place, but they slowly move parts of themselves; for example, many flowers open during the day and close at night.
- **Respiration:** every living plant cell respires all the time, day and night, releasing energy from glucose.
- **Sensitivity:** plants detect and respond to changes in their environment; for example, shoots grow towards light.
- **Growth:** plants grow throughout their lives, with a permanent increase in size and dry mass.
- **Reproduction:** plants make new plants of the same species, for example from seeds.
- **Excretion:** plants get rid of waste products of metabolism, such as oxygen from photosynthesis and carbon dioxide from respiration.
- **Nutrition:** plants make their own food by photosynthesis and take in water, carbon dioxide and mineral ions.

Question B1.02

Consider *two* organisms. For each, identify:

a) the characteristics of living things that the organism carries out all the time

b) the characteristics of living things that only happen at certain times.

Answer: Answers depend on the organisms you choose. This example uses a human and a mango tree.

a) Human: respiration (every cell respires all the time), excretion (carbon dioxide leaves in every breath out, and the kidneys make urine continuously) and sensitivity (the body is always detecting changes, such as in blood temperature). Mango tree: respiration (all its living cells, day and night), excretion (waste gases leave through the leaves) and sensitivity (it is always detecting light and gravity).

b) Human: growth (mainly until adulthood), reproduction (only adults, and only occasionally), nutrition (we take in food at mealtimes) and movement of the whole body (only when we choose to move). Mango tree: nutrition by photosynthesis (only in daylight), growth (mainly in the growing season), reproduction (flowers and fruit only at certain times of year) and movement (slow movements of leaves and flowers).

Question B1.03

Explain the difference between a unicellular organism and a multicellular organism, giving *one* example of each.

Answer: A unicellular organism is made of just one cell, which carries out all the characteristics of life by itself; an example is a bacterium (or yeast). A multicellular organism is made of many cells (large organisms contain millions of cells), and different cells can become specialised to carry out particular functions; an example is a human (or a plant such as a mango tree).

Question B1.04

Look at the micrographs in Figures B1.05 and B1.06. Suggest why a stain has been added to the animal cells, but we can easily see the plant cells without using a stain.

Answer: Animal cells do not have any coloured parts: the cytoplasm is a clear jelly and the cell membrane is very thin, so the cells and their structures are very difficult to see. The stain colours the cells and makes the nucleus look darker than the cytoplasm, so the structures show up. The moss plant cells contain green chloroplasts, which are already coloured, and their cell walls stand out clearly because they are much thicker than the animal cell membranes. So the plant cells can be seen easily without a stain.

Question B1.05

Name the part of a cell which has each of these functions:

a) makes proteins

b) contains the information for which proteins to make

c) photosynthesis

d) prevents the cell bursting when it takes up water

e) stores a solution of sugars and other solutes

f) controls what enters and leaves the cell.

Answer:

a) Ribosome

b) Nucleus (the DNA in its chromosomes carries the information)

c) Chloroplast

d) Cell wall

e) Vacuole (the large permanent vacuole of a plant cell, which contains cell sap)

f) Cell membrane

Question B1.06

Which of the structures that you have named in Question B1.05 are found in both animal and plant cells?

Answer: The ribosome, the nucleus and the cell membrane are found in both animal and plant cells. The chloroplast and the cell wall are found only in plant cells. A large permanent vacuole containing cell sap is also a plant-cell feature; animal cells only have small vacuoles called vesicles.

Question B1.07

Construct a table to compare the structure of a bacterial cell with animal and plant cells. Remember to include similarities as well as differences.

Answer:

Feature	Bacterial cell	Animal cell	Plant cell
Cell membrane	Yes (partially permeable, pressed against the cell wall)	Yes	Yes
Cell wall	Yes, but not made of cellulose	No	Yes, made of cellulose
Cytoplasm	Yes	Yes	Yes
Ribosomes	Yes	Yes	Yes
Nucleus	No	Yes	Yes
DNA	A single circle of DNA in the cytoplasm	In chromosomes inside the nucleus (not circular)	In chromosomes inside the nucleus (not circular)
Plasmids	Often present	No	No
Mitochondria	No	Yes	Yes
Chloroplasts	No	No	Often, in the green parts of the plant
Large permanent vacuole containing cell sap	No	No (only small vesicles)	Yes

Similarities: all three types of cell have a cell membrane, cytoplasm, ribosomes and DNA; bacterial and plant cells both have a cell wall that supports the cell and stops it bursting. **Main difference:** bacterial cells have no nucleus (they are prokaryotic), and their DNA is circular.

Question B1.08

How many micrometres are there in 1 mm?

Answer: There are 1000 µm in 1 mm. To convert millimetres to micrometres, multiply by 1000.

Question B1.09

The mitochondrion in Figure B1.10 is magnified 20 000 times. (Figure B1.10 is an electron micrograph of one long, oval mitochondrion that stretches almost the whole width of the picture.)

a) Using a ruler, carefully measure the maximum length of the mitochondrion. Record your measurement in mm (millimetres).

b) Convert your answer to µm (micrometres).

c) Use this formula to calculate the real size of the mitochondrion in μm: Real size in μm = size of image in μm ÷ magnification

d) How many of these mitochondria could you line up end to end between two of the mm marks on your ruler?

Answer:

a) About 83 mm, measured in a straight line from one end of the mitochondrion to the other along its longest axis. If your own careful measurement is a millimetre or two different, use your value in parts b to d.

b) 83 mm × 1000 = 83 000 μm

c) Real size = 83 000 μm ÷ 20 000 = 4.15 μm

d) The gap between two mm marks is 1 mm = 1000 μm. Number of mitochondria = 1000 μm ÷ 4.15 μm = 240.96, so about 241 mitochondria (240 whole ones would fit completely).

B1 Practice Exam Questions with Answers

Question B1.P1

Identify the characteristic which is *not* shown by all living organisms. (1 mark)

A) excretion

B) movement

C) photosynthesis

D) respiration

Answer: C — photosynthesis. Only plants (and some other organisms such as algae) photosynthesise; animals, fungi and most bacteria do not. Excretion, movement and respiration are three of the seven characteristics shown by all living organisms.

Question B1.P2

The tongue-eating louse is a parasite of large fish. As its name suggests, it has an unusual way of gaining nutrition. It detects chemicals on the gills of a fish. From the gills it crawls into the fish's mouth. First, the louse sucks blood from the tongue. This causes the tongue to shrink and eventually fall off. Next, the louse takes the place of the tongue. Here the louse remains, feeding on food that the fish takes into its mouth. (7 marks)

a) Define *nutrition*. [2]

b) Apart from nutrition, explain how *two* features of the louse are characteristics of all living organisms. [2]

c) The tongue-eating louse can reproduce.

i) Define *reproduction*. [2]

ii) Suggest *one* reason why scientists know very little about the process of reproduction in this species. [1]

Answer:

a) Taking in substances (materials) from the environment [1]; and using them for energy, growth and development [1].

b) Sensitivity: the louse detects chemicals on the gills of a fish, so it detects and responds to a change in its environment [1]. Movement: the louse crawls from the gills into the fish's mouth, so it changes its position [1].

c) i) The processes that make new organisms [1]; of the same species (the same kind of organism) [1].

c) ii) Any one sensible reason [1], for example: it lives hidden inside the mouths of fish in the sea, so it is very hard to observe; it spends very little time outside its host; it is rare in the wild; it is difficult to keep and breed in a laboratory.

Question B1.P3

The diagram below shows a cell from a plant leaf. Identify which labelled part contains DNA. (1 mark)

A) the outer double line that forms the boundary of the cell

B) the small oval structure at the left-hand side of the cell

C) the narrow layer between the outer boundary and the large central space (top right)

D) the large central space that fills most of the cell

Answer: B — the small oval structure is the nucleus, which contains DNA in the form of chromosomes. (A is the cell wall, C is the cytoplasm and D is the large permanent vacuole.)

Question B1.P4

A diagram of a flower is 8 cm across. The magnification of the diagram is ×4. Identify the actual width of the flower. (1 mark)

A) 3.2 mm

B) 20 mm

C) 0.5 cm

D) 32 cm

Answer: B — 20 mm. Actual size = size of image ÷ magnification = 8 cm ÷ 4 = 2 cm, and 2 cm × 10 = 20 mm. D (32 cm) comes from multiplying instead of dividing, and C (0.5 cm) comes from dividing 4 by 8.

Question B1.P5

The photograph shows a jellyfish. A jellyfish belongs to the animal kingdom. (9 marks)

(The photograph is a black-and-white side view of a jellyfish: a see-through, dome-shaped bell with a frilly edge and short fringes hanging underneath. Measured across its widest part, the jellyfish in the printed photograph is 89 mm wide.)

a) The actual jellyfish is 50 mm in diameter. **Calculate** the magnification of the photograph. [3]

b) State *three* parts that you would find in a cell in a jellyfish *and* outline the function of each part. [6]

Answer:

a) Magnification = size of image ÷ size of actual object [1]; = 89 mm ÷ 50 mm [1]; = 1.78, so the magnification is ×1.8 to 2 significant figures [1]. The answer must have a × sign and no units. If your own measurement of the photograph is slightly different, use it in the same way.

b) Any three parts, each with its function (1 mark for the part and 1 mark for its function):

- Nucleus [1]: contains DNA in the form of chromosomes, carrying the information about which proteins the cell makes [1]
- Cell membrane [1]: partially permeable; controls what enters and leaves the cell [1]
- Cytoplasm [1]: the jelly where many metabolic reactions take place [1]
- Mitochondrion [1]: the site of aerobic respiration, which releases energy from glucose [1]
- Ribosome [1]: where proteins are made [1]

Do not give a cell wall, chloroplasts or a large vacuole containing cell sap: a jellyfish is an animal, so its cells do not have these.

Question B1.P6

This is an electron micrograph of a small part of a cell from the pancreas. This cell makes large quantities of protein molecules, which are stored in vesicles before being exported from the cell. (8 marks)

(In the micrograph, label A points to a large, pale, finely grainy area at the top left. Label B points to a long, pale, sausage-shaped structure in the centre with fine stripes across its inside. Label C points to the dark, closely packed wavy lines that fill most of the background. Several round, dark grey structures are labelled 'vesicle containing protein molecules made by the cell'.)

a) Structure A contains molecules that determine the type of proteins made in the cell. Identify structure A. [1]

b) Use the information above to explain why the cell has large numbers of structures B and C. [6]

c) State *one* way you can tell that these are not bacterial cells. [1]

Answer:

a) The nucleus [1].

b) Any six from:

- B is a mitochondrion [1]
- Mitochondria are the site of aerobic respiration [1]
- Aerobic respiration releases energy from glucose [1] (do not write that mitochondria produce energy)
- Making (synthesising) proteins requires energy [1], and so does moving the vesicles and exporting the proteins from the cell
- C shows ribosomes; the dark, grainy lines are membranes covered in ribosomes [1]
- Ribosomes are the site of protein synthesis, where amino acids are linked together to make proteins [1]
- The cell makes large quantities of protein, so it needs many ribosomes to make them and many mitochondria to supply the energy [1]

c) Any one [1]: the cell has a nucleus (A); it has mitochondria (B); it has vesicles. Bacterial cells have no nucleus, no mitochondria and no vesicles.

B2 Movement into and out of Cells

B2.01 Diffusion

Before You Start: Particles in Solids, Liquids and Gases

The particle theory helps us to explain how matter behaves. Diffusion depends on particles being able to move, so it is worth checking that you remember how particles behave in each state of matter.

Q: Here are six statements about particles. Some of them describe solids, some describe liquids and some describe gases. Which are which?

A: Gases: statements 1 and 4. Solids: statements 2 and 6. Liquids: statements 3 and 5.

Statement	State of matter
1. The particles are far apart and rarely collide with each other.	Gas
2. The particles vibrate on the spot.	Solid
3. The particles are close to one another but can move around.	Liquid
4. The particles move freely in all directions.	Gas
5. The particles stay in contact and slide past one another as they move.	Liquid
6. The particles are in fixed positions, close to one another.	Solid

Because particles in gases and liquids can move around freely, diffusion happens in gases, solutions and mixtures of liquids. In a solid the particles only vibrate in fixed positions, so they cannot spread out.

Science in Context B2.01: Filtering Blood

We each have two kidneys. Their job is to remove unwanted substances from the blood. The kidneys produce a solution of the unwanted substances, which is removed from the body as **urine**. Sometimes kidneys stop working. Then the unwanted substances build up in the blood, and some of them can be quite dangerous, so another way must be found to remove them.

Figure B2.01 is a photograph of a boy in a hospital bed attached to a **dialysis machine**. This is how dialysis works:

1. Blood from a vein in his arm flows into the machine.
2. Inside the machine, a liquid called **dialysis fluid** flows past his blood. There is a very thin membrane between the blood and the dialysis fluid.
3. The unwanted substances in his blood diffuse through the membrane into the dialysis fluid (from a high concentration in the blood to a low concentration in the fluid).
4. The dialysis fluid, now carrying the unwanted substances, flows away.
5. His cleaned blood flows back into his body.

It takes several hours to remove all the unwanted substances from his blood, and he has to do this several times a week. Dialysis is a real-life use of **diffusion through a membrane**.

Q: (1) If the boy could have a kidney transplant, he would not have to spend time attached to a dialysis machine. Why do you think not everyone who needs a kidney transplant can have one?

A: There are not enough donated kidneys for everyone who needs one, so there are long waiting lists. A donated kidney must also be a close match to the patient's tissues, otherwise the patient's immune system will reject it, so a suitable match may not be available. Some patients may also be too ill to survive a major operation.

Q: (2) The dialysis fluid flows constantly through the machine. Can you suggest why this works better than just having stationary fluid in the machine?

A: Flowing fluid carries the unwanted substances away as soon as they diffuse into it, and fresh fluid containing none of these substances keeps arriving. This keeps the concentration of unwanted substances in the fluid low, so a steep concentration gradient is maintained between the blood and the fluid, and diffusion out of the blood continues quickly. In stationary fluid the unwanted substances would build up until their concentration was the same as in the blood. There would then be no concentration gradient, and net diffusion would stop.

What Is Diffusion?

Everything — including living cells — is made of atoms, molecules and ions. In this chapter we call all of these **particles**. Particles are always moving. When they can move freely, particles tend to spread themselves out evenly. This happens with gases, solutions and mixtures of liquids, and it is called **diffusion**.

Definition: Diffusion is the net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement.

The Rotten Egg Example (Figure B2.02)

Imagine a rotten egg full of very smelly **hydrogen sulfide** gas in one corner of a room.

- **Figure B2.02a:** at the start there is a high concentration of hydrogen sulfide molecules in one corner (near the egg) and none in the rest of the room.
- **Figure B2.02b:** given time, the molecules spread evenly through all the space available.
- Soon you cannot tell where the smell first came from, because the whole room smells of rotten egg.

Random Movement and Net Movement

- Diffusion happens because the molecules move **randomly**. They do not purposefully move from one place to another.
- Each molecule just moves by chance, changing direction when it bumps into another molecule. Some molecules might even go back into the rotten egg.
- **Net movement** means overall or average movement. Some molecules go from the egg into the room and a few go from the room back into the egg, but overall most of them go from the egg into the room.
- The energy for this movement is the **kinetic energy** of the particles themselves. Cells do not have to supply any energy from respiration for diffusion.

Three Conditions for Diffusion

The definition tells us that diffusion needs: (1) particles that can **move**, (2) a **concentration gradient**, and (3) **random movement** of the particles. Question B2.01 asks you to explain why each one is needed.

Concentration Gradients (Figure B2.03)

A place where a lot of molecules are close together has a **high concentration** of molecules. In the example, the rotten egg had the highest concentration of hydrogen sulfide molecules and the rest of the room had a low concentration.

Figure B2.03 uses a slope to picture this. Lots of particles are crowded at the top of the slope (high concentration of particles), there are only a few at the bottom (low concentration of particles), and an arrow points down the slope. We can imagine particles moving 'down the slope', from where there is a high concentration to where there is a low concentration. Another word for 'slope' is 'gradient', so we call this a **concentration gradient**. The overall result of diffusion is that particles move 'down' their concentration gradient.

Activity B2.01: Modelling Diffusion

Aim: To model diffusion, using people to represent the particles of a gas.

You will need: a fairly big, empty space to move around in, such as a playground or a large hall.

Method summary:

1. Work as a whole class. Each person represents a particle in a gas.
2. Start with most people in one corner of the space. Do not crowd too closely, because gas particles do not touch one another. Face any way you like.
3. Each person moves in a straight line in the direction they are facing.
4. Keep going until you hit something (another 'particle' or the wall of the 'container'). When you collide, change direction and keep going in a new straight line.
5. Continue for a few minutes, then stop and look around.

Safety: Real gas particles move very quickly, but do not move too fast, or there might be uncomfortable collisions.

Expected result: by the end, the 'particles' are spread out fairly evenly over the whole space.

Q: (1) What has happened to the 'particles'? Are they still concentrated in one corner?

A: No. The 'particles' have spread out across the whole space and are now fairly evenly spread, instead of being crowded into one corner.

Q: (2) Try to explain why this has happened.

A: Each person moved in a straight line and changed direction at random every time they collided with someone or something. Nobody was trying to leave the corner, but because most of the people started in the corner, by chance far more people moved out of the crowded corner than moved into it. This net movement continued until the people were evenly spread out. This models diffusion: particles moving randomly produce a net movement from a high concentration to a low concentration.

Q: (Reflection) You have now used two different models of diffusion: the picture of a slope in Figure B2.03 and moving people representing particles. Which one of these models best helped you to understand diffusion? Why?

A: Either answer is fine as long as you give a reason. The people model shows that particles move randomly, collide and change direction, and that the net movement happens by chance rather than on purpose. The slope model shows clearly which way the net movement goes (from high to low concentration) and helps you picture why a steeper gradient gives faster diffusion. Both models have limits: real particles move far faster than people, and particles do not 'roll down' a slope on purpose.

Diffusion Into and Out of Cells

- When substances diffuse into or out of a cell, they must pass through the **cell membrane** (see Question B2.02).
- **Oxygen** and **carbon dioxide** enter and leave cells by diffusion through the cell membrane, down a concentration gradient. For example, a respiring cell uses up oxygen, so there is less oxygen inside the cell than outside and oxygen diffuses in. The same cell makes carbon dioxide, so there is more carbon dioxide inside than outside and carbon dioxide diffuses out.

Factors Affecting the Rate of Diffusion

Diffusion can happen at different rates (speeds). How fast particles move from an area of high concentration to an area of low concentration depends on four factors:

Factor	What makes diffusion faster	Why
Temperature	A higher temperature	Particles have more kinetic energy, so their random movement is quicker

Surface area	A larger surface area	There is a greater area over which particles can move from one place to another
Concentration gradient	A steeper gradient (a large difference in concentration between two areas)	Particles move randomly towards the area of low concentration, and fewer move randomly in the opposite direction, so the net movement is greater
Diffusion distance	A shorter distance (e.g. a thin membrane)	Particles have less distance to travel

It is possible to investigate the effect of all four factors on the rate of diffusion in the laboratory, as the next two Experimental Skills boxes show.

Experimental Skills B2.01: Investigating How Surface Area Affects Diffusion

Aim: To use pieces of agar jelly to represent cells, and find out how the surface area of the 'cells' affects the diffusion of a substance (acid) into them.

You will need: a dish of agar jelly made with a small amount of a weak alkali (such as ammonium hydroxide) and coloured with a dye called cresol red; a sharp knife; plastic or blunt-ended forceps (tweezers); a ruler to measure in mm; two large test-tubes (more than 1 cm in diameter); a stopwatch; some dilute hydrochloric acid.

Safety: Take care with the acid and wear safety glasses in case of splashes. Clean up any spills immediately using plenty of cold water. Take care with the sharp blade: cut with the blade facing away from you, with the jelly on a firm surface.

Method:

1. Cut two cubes of agar jelly, with sides of 1 cm.
2. Cut one of the cubes into eight smaller cubes, each with sides of 0.5 cm.
3. Put the large cube into one test-tube and the eight small cubes into the other test-tube.
4. Add equal volumes of dilute hydrochloric acid to each tube, making sure all the cubes are covered with acid.
5. Start the stopwatch. Time how many seconds it takes for the cubes in each tube to become fully yellow.

How it works: cresol red is an indicator. It is red-purple in the alkaline jelly but turns yellow when the jelly becomes acidic. As the acid diffuses into a cube, the yellow colour spreads inwards from the surface. When a cube is completely yellow, the acid has reached its centre.

The numbers behind the experiment:

Measurement	One large cube	Eight small cubes
Side length	1 cm	0.5 cm
Volume of one cube	$1 \times 1 \times 1 = 1\ cm^3$	$0.5 \times 0.5 \times 0.5 = 0.125\ cm^3$
Total volume	$1\ cm^3$	$8 \times 0.125 = 1\ cm^3$
Surface area of one cube	$6 \times (1 \times 1) = 6\ cm^2$	$6 \times (0.5 \times 0.5) = 1.5\ cm^2$
Total surface area	$6\ cm^2$	$8 \times 1.5 = 12\ cm^2$
Surface area to volume ratio	6 : 1	12 : 1
Distance from surface to centre	0.5 cm	0.25 cm

Expected results: the eight small cubes turn fully yellow in a shorter time than the single large cube.

Q: (1) The jelly cubes contained a weak alkali and cresol red, which is an indicator. Explain why the cubes changed colour.

A: Hydrochloric acid diffused into the jelly, from a high concentration in the acid outside the cubes to a low concentration (none) inside them. The acid neutralised the alkali in the jelly, so the jelly became acidic. Cresol red changes colour with pH, so it turned from red-purple (alkaline) to yellow (acidic). The colour change shows where the acid has diffused to.

Q: (2) How did the total volume of the eight small cubes compare with the volume of the one larger cube? Was it greater, smaller, or the same?

A: The same. One large cube has a volume of 1 × 1 × 1 = 1 cm^3. Each small cube has a volume of 0.5 × 0.5 × 0.5 = 0.125 cm^3, so eight small cubes have a total volume of 8 × 0.125 = 1 cm^3.

Q: (3) How did the total surface area of the eight small cubes compare with the surface area of the one larger cube? Was it greater, smaller, or the same?

A: Greater (twice as big). The large cube has a surface area of 6 × (1 × 1) = 6 cm^2. Each small cube has a surface area of 6 × (0.5 × 0.5) = 1.5 cm^2, so eight small cubes have a total surface area of 8 × 1.5 = 12 cm^2.

Q: (4) Copy and complete these sentences, choosing the correct words: "The time taken for the eight small cubes to completely change colour was longer/shorter than the time for the one single cube. This is because the surface area/volume of the small cubes was greater/smaller than that of the single cube."

A: The time taken for the eight small cubes to completely change colour was **shorter** than the time for the one single cube. This is because the **surface area** of the small cubes was **greater** than that of the single cube.

Q: (5) Write a conclusion for your experiment. Include the words diffusion and surface area in your conclusion.

A: The larger the surface area, the faster the rate of diffusion. The eight small cubes had the same total volume as the large cube but twice the surface area (12 cm^2 compared with 6 cm^2), so acid could diffuse into them across a larger area, and it also had a shorter distance to travel to reach the centre of each cube. This is why the small cubes changed colour completely in a shorter time.

Experimental Skills B2.02: Investigating How Temperature, Concentration Gradient or Distance Affect Diffusion

In Experimental Skills B2.01, cubes of alkaline agar jelly and an indicator were used to measure the time taken for an acid to diffuse through the cubes. In this box you **plan** how to use the same technique to investigate a different factor. The model answers below plan an investigation into **temperature**; notes for the other two factors follow at the end.

Q: (1) As well as surface area, diffusion is also affected by temperature, concentration gradient and distance. Choose one of these variables to investigate.

A: Any of the three is fine. This plan investigates temperature.

Q: (2) The variable that you change in your experiment is called the independent variable. What is the independent variable in your experiment?

A: The temperature of the hydrochloric acid that the cubes are placed in, measured in °C.

Q: (3) Think about how you can change your independent variable. Decide how many different values you will have.

A: Stand the tubes of acid in water-baths (or beakers of water) at different temperatures and check each one with a thermometer. Use at least five values to show a clear pattern, for example 10 °C (iced water), 20 °C (room temperature), 30 °C, 40 °C and 50 °C. Leave each tube of acid in its water-bath for about 10 minutes before adding the cube, so that the acid reaches the right temperature.

Q: (4) The variable that you measure, as you collect your results, is the dependent variable. What is the dependent variable in your experiment?

A: The time taken for an agar cube to turn completely yellow, measured in seconds. The shorter the time, the faster the rate of diffusion.

Q: (5) Describe how you will measure your dependent variable.

A: Start the stopwatch at the moment the cube is dropped into the acid. Watch the cube against a white background and stop the stopwatch as soon as no red-purple colour is left anywhere in the cube. Record the time in seconds.

Q: (6) It is important that no variables, other than the one you are investigating, affect your experiment. Which variables will you standardise (keep the same) in your experiment? How will you do this?

A: Keep all of these the same:

- Size of the cubes: cut every cube with 1 cm sides, measured with a ruler.
- Type of jelly: cut all the cubes from the same dish of agar, so they all contain the same amount of alkali and indicator.
- Concentration of the acid: use acid from the same bottle.
- Volume of the acid: use a measuring cylinder to put the same volume (for example 10 cm^3) into each tube, enough to cover the cube.
- Size of the tubes: use the same type of tube each time.
- End-point: the same person decides when each cube is 'fully yellow', against the same white background.

Q: (7) Write a short, clear description of how you will do your experiment.

A: A clear method could be:

1. Set up water-baths at 10, 20, 30, 40 and 50 °C.
2. Put 10 cm^3 of dilute hydrochloric acid into each of five large test-tubes and stand one tube in each water-bath for 10 minutes.
3. Cut five 1 cm cubes from the same dish of alkaline agar jelly containing cresol red.
4. Use forceps to drop one cube into the tube at 10 °C and start the stopwatch.
5. Stop the stopwatch when the cube is completely yellow and record the time.
6. Repeat for each of the other temperatures.
7. Repeat the whole experiment twice more and calculate a mean time for each temperature.

Wear safety glasses, take care with the acid and the knife, and do not heat the acid above about 50 °C.

Q: (8) Draw a results chart that you could use to fill in your results. Include headings, but do not put in any results.

A: The first column holds the independent variable and the next columns hold the dependent variable, with space for repeats and a mean. Every heading includes its unit, and units appear only in the headings:

Temperature / °C	Time to turn fully yellow, trial 1 / s	Time to turn fully yellow, trial 2 / s	Time to turn fully yellow, trial 3 / s	Mean time / s
10				
20				
30				
40				
50				

Q: (9) Predict what you would expect to find in your experiment.

A: As the temperature increases, the time taken for a cube to turn fully yellow will decrease, which means diffusion is faster at higher temperatures. This is because the acid particles have more kinetic

energy at higher temperatures, so their random movement is quicker.

If you choose a different variable:

Variable chosen	Independent variable and how to change it	Prediction
Concentration gradient	The concentration of the hydrochloric acid; make up five different dilutions of the acid and keep the temperature and cube size the same	The more concentrated the acid, the steeper the concentration gradient, so the shorter the time for the cube to turn yellow
Distance	The distance the acid must travel; cut cubes of different sizes (for example sides of 0.5 cm, 1.0 cm and 1.5 cm)	The bigger the cube, the further the acid must diffuse to reach the centre, so the longer the time for the cube to turn fully yellow

B2.02 Osmosis

Water Inside and Outside Cells

Every cell in an organism's body has water inside it and outside it. Many different substances are dissolved in this water, and their concentrations may be different inside and outside the cell. This creates concentration gradients, down which water and solutes will diffuse, if they are able to pass through the membrane.

Remember: in a solution, the **solvent** is the liquid that does the dissolving (here, water) and the **solute** is the substance dissolved in it (for example, sugar).

How Osmosis Happens (Figure B2.04)

It is easiest to understand osmosis with just one solute. **Figure B2.04** shows a **dilute sugar solution** on the left separated from a **concentrated sugar solution** on the right by a membrane with very small holes, or pores, in it. The figure shows:

- Many small water molecules on both sides, with more of them on the left.
- Large sugar molecules: only a few on the left (dilute) and many on the right (concentrated).
- Arrows showing water molecules passing through the pores in both directions, but more often from left to right.
- The caption reminds you that it is important to think about the diffusion of each kind of molecule separately.

Step by step:

1. Water molecules are very small. Each one is made of two hydrogen atoms and one oxygen atom. Sugar molecules are many times larger.
2. The holes in the membrane are big enough to let water molecules through, but not sugar molecules. A membrane like this is **partially permeable**, because it lets some molecules through but not others.
3. There is a concentration gradient for sugar (higher on the right). Without the membrane, sugar molecules would diffuse into the dilute solution until they were evenly spread, but they cannot, because the holes are too small for them.
4. There is also a concentration gradient for water. On the left there is a high concentration of water molecules. On the right the concentration of water molecules is lower, because a lot of the space is taken up by sugar molecules.
5. Because there are more water molecules on the left, at any one moment more of them 'hit' a hole in the membrane and move through to the right than go the other way (right to left).
6. Over time there is an overall, or net, movement of water from left to right. This is **osmosis**.

Osmosis is really just a special kind of diffusion. It is the diffusion of water molecules in a situation where the water molecules, but not the solute molecules, can pass through a membrane.

Definition: Osmosis is the diffusion of water molecules through a partially permeable membrane.

Water Potential (Figure B2.05)

It can be confusing to talk about the 'concentration' of water molecules, because the word concentration normally means the concentration of the solute dissolved in the water. It is much better to use a different term, **water potential**:

- A **dilute solution** (where there is a lot of water) has a **high water potential**.
- A **concentrated solution** (where there is less water) has a **low water potential**.
- Pure water has the highest water potential of all.
- A difference in water potential between two areas is a **water potential gradient**. Water molecules diffuse down this gradient, from a high water potential to a low water potential.

Figure B2.05 pictures a water potential gradient as a slope. On the left, at the top of the slope, there are lots of water molecules (high water potential). A partially permeable membrane crosses the slope, and on the right, at the bottom, there are only a few water molecules (low water potential). Water molecules tend to move down the slope, from high to low water potential.

Definition (in terms of water potential): Osmosis is the net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane.

Feature	Dilute solution	Concentrated solution
Amount of solute	Little	A lot
Amount of water	A lot	Less
Water potential	High	Low
In osmosis, water moves...	Out of it, towards the concentrated solution	Into it, from the dilute solution

Water potential and osmosis are very important because they affect how and when organisms take up water. The uptake of water by plants is covered in Chapter B7, and the uptake of water by humans in Chapter B6.

Osmosis and Animal Cells (Figures B2.06 and B2.07)

Osmosis happens whenever two solutions (or a solution and pure water) are separated by a partially permeable membrane. **Cell membranes are partially permeable**: they let water through easily, but other molecules and ions often cannot get through. This means osmosis happens across cell membranes.

Animal Cell in Pure Water (Figure B2.06)

The cytoplasm inside the cell is a fairly concentrated solution. The proteins and many other substances dissolved in it are too large to get through the cell membrane, but water molecules can get through. This is just like Figure B2.04: the pure water outside is the dilute solution, the cytoplasm is the concentrated solution, and the cell membrane is the partially permeable membrane. So osmosis occurs:

1. Water molecules diffuse from the pure water (high water potential) into the cytoplasm (lower water potential), through the partially permeable cell membrane.
2. As more and more water enters, the cell swells.
3. The cell membrane has to stretch as the cell gets bigger.
4. Eventually the strain is too much and the cell **bursts**.

Figure B2.06 shows an animal cell surrounded by pure water, with an arrow showing water entering through the cell membrane.

Animal Cell in a Concentrated Solution (Figure B2.07)

If the solution outside is **more concentrated than the cytoplasm**, it has a lower water potential than the cytoplasm. Water molecules diffuse **out** of the cell through the cell membrane. As the water leaves, the cytoplasm shrinks, so the cell gets smaller. Figure B2.07 shows a shrunken, wrinkled cell with an arrow showing water leaving.

Note: if the solution outside has the same water potential as the cytoplasm, water enters and leaves at the same rate, so there is no net movement and the cell stays the same size.

Osmosis and Plant Cells in Pure Water (Figure B2.08)

Plant cells do not burst in pure water. Plant cells are surrounded by a **cell wall**, which is **fully permeable**: it lets all molecules go through it. Although it is not easy to see, a plant cell also has a **cell membrane** just inside the wall, like an animal cell, and this membrane is partially permeable.

1. A plant cell in pure water takes in water by osmosis through its partially permeable cell membrane, in the same way as an animal cell.
2. As the water goes in, the cytoplasm and vacuole swell.
3. The plant cell has a very strong cell wall, much stronger than the cell membrane. The cytoplasm presses out against the cell wall, but the wall resists and presses back on the contents.
4. This stops the plant cell from bursting.

Figure B2.08 shows a plant cell in pure water, with the cytoplasm and large vacuole pushed right up against the cell wall and an arrow showing water entering through the cell membrane.

A plant cell in this state is rather like a blown-up tyre — tight and firm. It is said to be **turgid**. The pressure of the water pushing outwards on the cell wall is called **turgor pressure**.

Why turgor matters:

- In a plant leaf, cells with plenty of water are all turgid. The pressure of the water pushing outwards on the cells' walls keeps the whole group of cells strong and firm, which helps to support the leaves.
- Turgor pressure helps a plant that has no wood in it to stay upright.
- Plant cells are usually turgid.

Plant Cells in a Concentrated Solution (Figures B2.09 and B2.10)

Now imagine a plant cell placed in a solution that is **more concentrated than its cytoplasm**:

1. Water leaves the cell by osmosis, out of the cytoplasm and vacuole, through the partially permeable cell membrane.
2. The cytoplasm and vacuole shrink.
3. The cell wall is strong and stays in position, so the cytoplasm pulls away from the cell wall, pulling the cell membrane away from the wall too.

Flaccid: when a plant cell loses a lot of water, it also loses its turgor pressure. The contents no longer push outwards on the cell wall, so the cell becomes soft and floppy. If the cells in a plant become flaccid, the plant loses its firmness and begins to **wilt**.

Plasmolysed: when the cell membrane tears away from the cell wall, the cell is said to be plasmolysed. Plasmolysis can kill a plant cell, because the cell membrane is damaged as it tears away from the cell wall.

Figure B2.09 is a diagram of a plant cell in a concentrated solution. The labels show the concentrated solution outside the cell, the less concentrated solution inside the cell, the cell wall (still the same size and shape) and the cell membrane, which has pulled away from the wall. The gap between the cell wall and the shrunken cell membrane is labelled **space X**.

Figure B2.10 is a photograph of onion cells that have been placed in a concentrated solution. The cytoplasm has lost so much water, and shrunk so much, that it has pulled away from the cell wall. The labels point to the cytoplasm, the cell wall, the cell membrane that has pulled away from the cell wall, and the space between the cell membrane and the cell wall.

Animal and Plant Cells Compared

Cell	In pure water	In a concentrated solution
Animal cell	Water enters by osmosis; the cell swells and bursts, because there is no cell wall to stop it	Water leaves by osmosis; the cytoplasm shrinks and the cell gets smaller
Plant cell	Water enters by osmosis; the cytoplasm and vacuole swell and press on the cell wall; the cell becomes turgid but does not burst	Water leaves by osmosis; the cytoplasm and vacuole shrink; the cell becomes flaccid and may become plasmolysed

Experimental Skills B2.03: Investigating the Effect of Osmosis on Potato Strips

Aim: To place pieces of plant tissue (potato) into different solutions and measure the effect on their length.

You will need: a large potato; a sharp knife or other way to peel the potato; a cork borer to cut cylinders from the potato (or use the knife to cut strips); a ruler to measure in mm; three containers such as small beakers; some distilled water, a dilute sugar solution and a concentrated sugar solution.

Safety: Take care when using the sharp knife or potato peeler.

Method:

1. Peel the potato. Use a cork borer to cut six cylinders. Cut the ends off each cylinder so that they are all the same length (40 mm is a good length) and record this length. Without a cork borer, cut rectangular pieces instead.
2. Pour the distilled water into one beaker and the two sugar solutions into the other two beakers. Label each beaker with the concentration of its solution.
3. Place two potato strips into each liquid, making sure they are completely covered.
4. Leave the strips in their liquids for about 20 minutes. While you wait, construct a results chart and fill in the original lengths.
5. After 20 minutes, take the strips out, measure the length of each one again and record the new lengths.
6. Calculate the change in length of each strip. Say whether it got longer (a positive change) or shorter (a negative change).

Figure B2.11 shows the two shapes you can cut: cylinders 40 mm long, or rectangular blocks 10 mm × 10 mm × 40 mm.

A suitable results table (headings with units; original lengths filled in before the experiment):

Liquid	Strip	Length at start / mm	Length after 20 minutes / mm	Change in length / mm
Distilled water	1	40		
Distilled water	2	40		
Dilute sugar solution	1	40		
Dilute sugar solution	2	40		
Concentrated sugar solution	1	40		

Concentrated sugar solution	2	40		

Change in length = length after 20 minutes − length at start. To compare strips that started at different lengths, you can also work out the percentage change = (change in length ÷ length at start) × 100%.

Peer assessment: swap results tables with a partner and score each feature 2 marks (really good), 1 mark (quite well done) or 0 marks (not done at all, or very poor):

- The results table is drawn with a ruler.
- There are clear headings for all the columns, with units.
- There are no units written anywhere except in the headings.
- It is really easy to understand the results table.
- The changes in length are shown in the results table.
- The changes in length have been correctly calculated, and say whether the strips increased or decreased in length.

Add up the marks you give your partner (the maximum is 12). If you have not given full marks, help your partner to see how to improve their table.

Expected results:

Liquid	Expected change in length	Why
Distilled water	Longer (positive change); the strip feels firm	Water enters the cells by osmosis and they become turgid
Dilute sugar solution	Small change, either slightly longer or slightly shorter	Depends on whether the solution has a higher or lower water potential than the potato cells
Concentrated sugar solution	Shorter (negative change); the strip feels soft and bendy	Water leaves the cells by osmosis and they become flaccid

Q: (1) What happened to the length of the potato strips that were placed in the distilled water?

A: They got longer (a positive change in length), and they also became firmer.

Q: (2) Use your knowledge of osmosis to explain why this happened.

A: Distilled water has a higher water potential than the solution inside the potato cells (their cytoplasm and vacuoles). So water moved into the cells by osmosis, through their partially permeable cell membranes, down the water potential gradient. The cytoplasm and vacuoles swelled and pushed outwards on the cell walls, so the cells became turgid. Each cell got slightly bigger, so the whole strip got longer and firmer. The cell walls stopped the cells from bursting.

Q: (3) What happened to the length of the potato strips that were placed in the concentrated solution?

A: They got shorter (a negative change in length), and they also became soft and floppy.

Q: (4) Use your knowledge of osmosis to explain why this happened.

A: The concentrated sugar solution has a lower water potential than the solution inside the potato cells. So water moved out of the cells by osmosis, through the partially permeable cell membranes, down the water potential gradient into the sugar solution. The cytoplasm and vacuoles shrank, the cells lost their turgor pressure and became flaccid (some may even have become plasmolysed), so the strip got shorter and softer.

B2.03 Active Transport

Why Cells Need Active Transport

Cells often need to take in substances that are only present in small quantities around them. If a substance has a **lower concentration outside the cell than inside**, we would expect it to diffuse **out** of the cell. If the cell needs this substance, it has to do something to make it go into the cell instead of diffusing out.

What Is Active Transport?

Cells can use energy to make substances move across their membranes **up** the concentration gradient (from a low concentration to a high concentration). This process is called **active transport**.

- The energy needed is provided by **respiration**.
- **Aerobic respiration**, in the **mitochondria** in the cell, releases energy for the cell to use. Some of this energy is used for active transport, moving substances against their concentration gradients.
- This means anything that slows down respiration (such as a shortage of oxygen) also slows down active transport.

Definition: Active transport is the movement of molecules or ions through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient) using energy from respiration.

Example: Root Hair Cells Taking In Nitrate Ions

- Root hair cells in plant roots take in **nitrate ions** from the soil (plants use nitrate to make proteins).
- The concentration of nitrate ions inside a root hair cell is usually **higher** than in the soil.
- So the diffusion gradient for nitrate ions is out of the root hair and into the soil.
- But root hair cells can still take nitrate ions in, using active transport.

The bottom of this page in the book has a photograph with no caption. It appears to show root hairs: long, thin extensions growing out from the surface of a root.

How Carrier Proteins Work (Figure B2.12)

There are special **carrier proteins** in the cell membrane of root hair cells. These proteins pick up nitrate ions from outside the cell and then change shape, pushing the ions through the cell membrane and into the cytoplasm. Figure B2.12 shows a carrier protein spanning the cell membrane, with 'outside the cell' above it and 'inside the cell' below it, in three stages:

1. The nitrate ion enters the carrier protein (from outside the cell).
2. The carrier protein changes shape. The energy needed for it to do this is provided by respiration in the cell.
3. The change of shape of the carrier protein pushes the nitrate ion into the cell.

The energy for the change of shape comes from the mitochondria, which carry out aerobic respiration.

Diffusion, Osmosis and Active Transport Compared

Feature	Diffusion	Osmosis	Active transport
What moves	Any particles (molecules or ions), e.g. oxygen, carbon dioxide	Water molecules only	Molecules or ions, e.g. nitrate ions
Direction	From a higher to a lower concentration (down a concentration gradient)	From a higher to a lower water potential (dilute to concentrated solution)	From a lower to a higher concentration (against a concentration gradient)
Membrane needed?	No (it happens in air and liquids), but in cells it crosses the cell membrane	Always needs a partially permeable membrane	Happens across a cell membrane

Energy source	Kinetic energy of the particles (no energy from respiration)	Kinetic energy of the water molecules (no energy from respiration)	Energy from respiration
Carrier proteins?	No	No	Yes, they change shape to move the particles
Example in this chapter	Hydrogen sulfide spreading through a room; dialysis; acid into agar jelly	Water entering potato cells; animal cells bursting in pure water	Root hair cells taking in nitrate ions

Project B2.01: Making a Good Cup of Tea

This project comes at the end of the chapter in the book, but it is about **diffusion** and uses the planning skills from Experimental Skills B2.02.

Tea leaves are dried leaves from a camellia plant (*Camellia sinensis* is the tea plant). When hot water is added to tea leaves, colour and flavour molecules diffuse out of the leaves and into the water. Working in a group of three or four, you investigate some factors that make a really good cup of tea.

Ideas for questions to investigate (you might think of your own):

- Does the temperature of the water affect the rate at which the molecules diffuse out of the tea bag?
- Does the temperature of the water affect the flavour of the tea?
- Do different kinds of tea bag affect the diffusion of substances out of the tea leaves and into the water?
- Does adding milk to the cup before you add the water affect the diffusion of molecules out of the tea leaves?

What to do: decide on a question, then plan the experiment. Change only one variable (the independent variable), think carefully about how you will measure the dependent variable, and standardise the other variables. Predict your results, and decide how you will record them and use them to answer your question.

Example plan (water temperature):

- Independent variable: water temperature, for example 20, 40, 60 and 80 °C.
- Dependent variable: how quickly the tea colours, for example the time until a mark under the cup can no longer be seen, or comparing the colour after 2 minutes against a colour chart.
- Standardised variables: the same brand and type of tea bag, the same volume of water, the same cup, and no stirring.
- Prediction: the hotter the water, the faster the colour molecules diffuse out of the leaves, because the particles have more kinetic energy.
- Safety: take care with hot water to avoid scalds. Never taste anything made in a science laboratory; flavour tests need clean kitchen equipment.

B2 Key Formulas and Concepts

1. Diffusion: higher concentration → lower concentration (down a concentration gradient) by random movement; the energy is the particles' kinetic energy, not respiration
2. Rate of diffusion is faster with a higher temperature, a larger surface area, a steeper concentration gradient and a shorter diffusion distance
3. Volume of a cube = side × side × side; surface area of a cube = 6 × side × side
4. One 1 cm cube: volume 1 cm^3, surface area 6 cm^2, ratio 6 : 1; eight 0.5 cm cubes: total volume 1 cm^3, total surface area 12 cm^2, ratio 12 : 1
5. Osmosis: higher water potential (dilute solution) → lower water potential (concentrated solution), through a partially permeable membrane

6. Dilute solution = a lot of water = high water potential; concentrated solution = less water = low water potential; pure water has the highest water potential
7. Animal cell: in pure water it swells and bursts; in a concentrated solution it shrinks
8. Plant cell: in pure water it becomes turgid (turgor pressure; the cell wall stops it bursting); in a concentrated solution it becomes flaccid and may become plasmolysed
9. Active transport: lower concentration → higher concentration (against a concentration gradient), through carrier proteins in the cell membrane, using energy from respiration
10. Change in length = final length − starting length (positive = longer, negative = shorter)
11. Percentage change = (change ÷ starting value) × 100%
12. Cell membrane = partially permeable; plant cell wall = fully permeable

B2 Textbook Questions with Answers

Question B2.01

Look at the definition of diffusion in the 'Key words' box at the start of this section. Explain why the following conditions are required for diffusion to occur:

a) movement of particles

b) a concentration gradient

c) random movement.

Answer:

a) Diffusion is the net movement of particles from one place to another, so the particles must be able to move. Their movement comes from their kinetic energy. If the particles could not move (for example, if they were held in fixed positions as in a solid), they could never spread out from a region of high concentration, so diffusion could not happen.

b) Diffusion is net movement from a region of higher concentration to a region of lower concentration, so there must be a difference in concentration. If the concentration were the same everywhere, the particles would still move about, but just as many would move one way as the other, so there would be no net movement and no diffusion.

c) The particles do not move purposefully towards the low concentration; they move randomly in all directions, changing direction when they collide. Because there are more particles in the region of high concentration, by chance more of them move out of that region than move into it. This random movement is what produces the net movement down the concentration gradient, until the particles are evenly spread out.

Question B2.02

When substances diffuse into and out of cells, which part of the cell must they move through?

Answer: They must move through the cell membrane. The cell membrane surrounds every cell, so any substance entering or leaving the cell has to cross it. (In a plant cell, substances also pass through the cell wall, but the cell wall is fully permeable, so it is the cell membrane that controls what enters and leaves.)

Question B2.03

These questions are about the diagram in Figure B2.04, which shows a dilute sugar solution (left) separated from a concentrated sugar solution (right) by a membrane with very small pores. What is the solvent in the sugar solution? What is the solute?

Answer: The solvent is water. The solute is sugar.

Question B2.04

(Figure B2.04: a dilute sugar solution and a concentrated sugar solution separated by a membrane with very small pores.) Explain why water molecules can move from one side of the membrane to the other, but sugar molecules cannot.

Answer: The membrane has very small holes (pores) in it. Water molecules are very small (each is made of just two hydrogen atoms and one oxygen atom), so they can fit through the pores. Sugar molecules are many times larger than water molecules, so they are too big to pass through the pores. The membrane is partially permeable: it lets water through but not sugar.

Question B2.05

(Figure B2.04: a dilute sugar solution on the left and a concentrated sugar solution on the right, separated by a partially permeable membrane.) In which direction is the net movement of water molecules?

Answer: From left to right, from the dilute sugar solution into the concentrated sugar solution. There are more water molecules on the left-hand side, so at any moment more of them hit a pore and pass through from left to right than pass from right to left.

Question B2.06

In Figure B2.04, which solution has the higher water potential? Explain your answer.

Answer: The dilute sugar solution (on the left) has the higher water potential. It contains more water molecules, because less of its space is taken up by sugar molecules, and a solution with a lot of water has a high water potential. The concentrated solution on the right contains less water, so it has a lower water potential.

Question B2.07

Copy and complete this sentence:

When an animal cell is placed in pure water, water …… the cell by …… through the partially permeable cell …….

Answer: When an animal cell is placed in pure water, water **enters** (moves into) the cell by **osmosis** through the partially permeable cell **membrane**.

Question B2.08

Animal cells burst if they are placed in pure water, but plant cells do not. Explain why.

Answer: Both kinds of cell take in water by osmosis in pure water, because the pure water has a higher water potential than the cytoplasm and the cell membrane is partially permeable. As water enters, the cell contents swell. An animal cell is surrounded only by its thin cell membrane, which stretches as the cell swells until the strain is too much and the cell bursts. A plant cell has a very strong cell wall outside its cell membrane. As the cytoplasm and vacuole swell, they press outwards on the cell wall, but the wall resists and presses back on the contents (turgor pressure). This stops the cell expanding any further, so the plant cell becomes turgid instead of bursting.

Question B2.09

Here are some descriptions of what happens to a plant cell when it is placed in a concentrated solution. They are in the wrong order. Write the descriptions in the correct order.

- The cytoplasm and vacuole shrink.
- The cell membrane is pulled away from the cell wall.
- Water moves out of the cell through the partially permeable cell membrane.

Answer: The correct order is:

1. Water moves out of the cell through the partially permeable cell membrane.
2. The cytoplasm and vacuole shrink.

3. The cell membrane is pulled away from the cell wall.

Water leaving the cell is what makes the cytoplasm and vacuole shrink, and because the strong cell wall stays in position, the shrinking contents pull the cell membrane away from the wall.

Question B2.10

Look at Figure B2.09, which shows a plant cell in a concentrated solution: the cytoplasm and vacuole have shrunk and the cell membrane has pulled away from the cell wall, leaving a gap labelled space X between the cell wall and the cell membrane. What fills space X? Explain your answer.

Answer: Space X is filled with the concentrated solution from outside the cell. The cell wall is fully permeable, so the outside solution (water and its dissolved solute) passes straight through it. When the cytoplasm and vacuole shrink and the cell membrane pulls away from the wall, the outside solution moves through the wall and fills the gap between the cell wall and the cell membrane. It is not cytoplasm, because the cytoplasm stays inside the cell membrane, and it is not air.

Question B2.11

A group of plant cells has been placed in a concentrated solution, and the cells are plasmolysed. Predict what will happen if the cells are now placed in pure water. Explain your answer, using the term water potential.

Answer: The cells will take up water by osmosis and become turgid again (as long as they have not been killed). Pure water has a higher water potential than the cell contents, because the cytoplasm and the solution in the vacuole are concentrated solutions with a low water potential. So water moves into the cells down the water potential gradient, through the partially permeable cell membranes. The cytoplasm and vacuole swell, the cell membrane is pushed back against the cell wall, and turgor pressure builds up, so the cells become turgid. However, if a cell membrane was badly damaged when it tore away from the cell wall, that cell may have died; its membrane would no longer be partially permeable, so it would not become turgid again.

B2 Practice Exam Questions with Answers

Question B2.P1

Identify which change would increase the rate at which a substance diffuses across a membrane. (1 mark)

A) a decrease in concentration gradient

B) a decrease in the surface area of the membrane

C) an increase in distance across the membrane

D) an increase in temperature

Answer: D — an increase in temperature. At a higher temperature, particles have more kinetic energy, so their random movement is faster and diffusion is faster [1]. A, B and C would all slow diffusion down: a smaller concentration gradient, a smaller surface area and a longer diffusion distance all decrease the rate of diffusion.

Question B2.P2

The table shows information about diffusion, osmosis and active transport. Identify which row is correct. (1 mark)

Each option is one row of the table, giving the entries for diffusion, osmosis and active transport in that order:

A) Diffusion: uses energy from respiration. Osmosis: always involves a partially permeable membrane. Active transport: uses protein carriers in membranes.

B) Diffusion: uses kinetic energy of molecules and ions. Osmosis: uses kinetic energy of water molecules. Active transport: involves movement up a concentration gradient.

C) Diffusion: uses protein carriers in membranes. Osmosis: involves movement up a concentration gradient. Active transport: always involves a partially permeable membrane.

D) Diffusion: always involves a partially permeable membrane. Osmosis: uses protein carriers in membranes. Active transport: uses energy from respiration.

Answer: B [1]. Diffusion is powered by the kinetic energy of the moving molecules and ions; osmosis is the diffusion of water, so it is powered by the kinetic energy of the water molecules; and active transport moves substances up (against) a concentration gradient. The other rows are wrong:

- A is wrong because diffusion does not use energy from respiration.
- C is wrong because diffusion does not use protein carriers, and osmosis moves water from a high to a low water potential, not up a gradient.
- D is wrong because diffusion does not need a membrane at all (for example, a gas spreading through a room), and osmosis does not use protein carriers.

Question B2.P3

A learner investigated how temperature affected the rate at which a dye moved through a piece of agar jelly. (11 marks)

She filled seven test-tubes with a solution of red dye. She placed each tube into a water-bath at a different temperature and left them for ten minutes.

She cut seven cubes of jelly. She placed each cube into one of the tubes containing the red dye. She measured the time taken for the cubes to change colour.

This graph shows her results. It plots Time taken / minutes (y-axis, 0 to 12) against Temperature / °C (x-axis, 0 to 60). There are seven plotted points and a straight line of best fit that slopes downwards from about 12 minutes at 0 °C to below 1 minute at 60 °C. Approximate values read from the graph:

Temperature / °C	0	10	20	30	40	50	60
Time taken / minutes	12.0	10.6	4.5	6.1	4.0	2.4	1.2

a) Give the name of the process by which the dye moved through the jelly, and describe this process. [3]

b) Suggest why the learner left the tubes of red dye in the water-baths for ten minutes. [1]

c) State two variables that the learner should standardise (keep the same) in her experiment. [2]

d) i) One of the learner's results is anomalous (inconsistent). Identify this result and explain why you have chosen this value. [2]

d) ii) Describe the trend or pattern shown by the results. [1]

d) iii) Use your knowledge of diffusion to explain the results. [2]

Answer:

a) Diffusion [1]. The net movement of dye particles (molecules) from a region of higher concentration (the dye solution) to a region of lower concentration (the jelly), i.e. down a concentration gradient [1]; as a result of the random movement of the particles, which comes from their kinetic energy [1].

b) So that the dye solution in each tube had time to reach the temperature of its water-bath before the jelly cube was added, so the dye was at the temperature being tested [1].

c) Any two of [1 each, maximum 2]: the size of the jelly cubes (same side length, so the same volume and surface area); the concentration of the dye solution; the volume of dye solution in each tube; the type of jelly (all cubes cut from the same batch); how the colour change is judged (the same end-point each time).

d) i) The result at 20 °C (about 4.5 minutes) [1]. It is far below the line of best fit (the line predicts about 8 minutes at 20 °C), and it does not fit the trend: it is shorter than the time at 30 °C, even though all the other results show the time getting shorter as the temperature rises [1].

d) ii) As the temperature increases, the time taken for the cubes to change colour decreases [1] (steadily, from about 12 minutes at 0 °C to about 1 minute at 60 °C).

d) iii) At higher temperatures the dye particles have more kinetic energy, so they move faster (their random movement is quicker) [1]. So the dye diffuses into the jelly faster (the rate of diffusion increases) and the cube changes colour in less time [1].

Question B2.P4

(8 marks)

a) Describe what is meant by osmosis. [2]

b) Compare the effect of immersing an animal cell and a plant cell in pure water. Explain the reasons for any differences that you describe. [6]

Answer:

a) The net movement (diffusion) of water molecules [1] from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane [1].

b) Any six of these points [1 each, maximum 6]:

- Similarity: both cells take in water by osmosis.
- Because the pure water has a higher water potential than the cytoplasm, water moves down a water potential gradient into the cell.
- In both cells, water enters through the partially permeable cell membrane.
- In both cells, the contents swell (the cytoplasm, and in the plant cell also the vacuole).
- Difference: the animal cell keeps swelling until it bursts.
- Reason: the animal cell has no cell wall; its thin cell membrane stretches until the strain is too much and it breaks.
- Difference: the plant cell does not burst; it becomes turgid (firm).
- Reason: the plant cell has a strong cell wall; the swollen contents press outwards on the wall (turgor pressure) and the wall presses back, stopping the cell expanding further.
- The cell wall is fully permeable, so it does not stop water entering; it only stops the cell from bursting.

Question B2.P5

This bar chart shows the concentration of potassium ions and sodium ions in a sample of pond water, and in the cells of a plant growing in the water. (9 marks)

The y-axis shows Concentration / arbitrary units (0 to 4) and the x-axis shows the ion (potassium and sodium). Light bars show pond water and dark bars show plant cells. Bar heights read from the chart:

Ion	Pond water / arbitrary units	Plant cells / arbitrary units
Potassium	0.5	4.0
Sodium	0.5	about 0.05 (a very short bar, almost zero)

a) Describe the differences between the concentrations of the ions in the pond water and in the plant cells. [3]

b) Use the data in the graph to suggest how the ions move between the pond water and the plant cells. Explain your answer. [2]

c) Describe how the process you have described in your answer to b takes place. [4]

Answer:

a) Any three of [1 each, maximum 3]:

- The concentration of potassium ions is much higher in the plant cells (4.0 arbitrary units) than in the pond water (0.5 arbitrary units), about 8 times higher.
- The concentration of sodium ions is lower in the plant cells (about 0.05 arbitrary units) than in the pond water (0.5 arbitrary units).
- In the pond water, the concentrations of potassium and sodium ions are the same (0.5 arbitrary units each).
- In the plant cells, there is far more potassium than sodium (4.0 compared with about 0.05).

b) Potassium ions are moved into the plant cells by active transport [1], because they have to move from a lower concentration in the pond water to a higher concentration in the cells, which is against their concentration gradient; diffusion alone would make them move out of the cells, not in [1]. (Sodium ions are kept at a low concentration inside the cells, so they are probably actively transported out, or prevented from entering, against their concentration gradient.)

c) Any four of [1 each, maximum 4]:

- There are carrier proteins in the cell membrane.
- A potassium ion from the pond water enters (binds to) a carrier protein on the outside of the membrane.
- The carrier protein changes shape.
- The energy for the change of shape comes from respiration (aerobic respiration in the mitochondria).
- The change of shape pushes the ion through the membrane and releases it into the cytoplasm.
- This moves the ion against its concentration gradient, from a low concentration to a high concentration.

B3 Biological Molecules

B3.01 Carbohydrates, Fats and Proteins

Before You Start: Elements, Compounds, Atoms and Molecules

The chapter begins by checking some chemistry ideas you need for this topic. Biological molecules are compounds made from just a few elements, so make sure you are clear about these words:

- **Atom** — the smallest particle of an element
- **Element** — a substance made of only one kind of atom, such as carbon (C), hydrogen (H), oxygen (O) or nitrogen (N)
- **Molecule** — a particle made up of two or more atoms combined together
- **Compound** — a substance made of two or more different elements chemically combined

Answers to the Before You Start questions:

Before You Start question	Answer
1a Oxygen gas is made up of pairs of oxygen atoms combined together. What is the formula for oxygen gas?	O_2
1b What do we call a particle made up of two or more atoms combined together?	A molecule
1c Is oxygen an element or a compound?	An element, because it contains only one kind of atom (oxygen atoms)
2a Glucose is made up of the elements that have the symbols C, H and O. What do C, H and O stand for?	Carbon, hydrogen and oxygen
2b One molecule of glucose contains 6 atoms of carbon, 12 atoms of hydrogen and 6 atoms of oxygen. Can you write the formula for glucose?	$C_6H_{12}O_6$
2c Is glucose an element, a compound or a mixture?	A compound, because three different elements are chemically joined together, in the same fixed proportions in every molecule

Science in Context: How Did Life on Earth Start?

One of the greatest mysteries of life on Earth is how it began. Living cells are made up of biological molecules, and these are not often found in non-living things. Scientists think that these molecules may have formed from simpler substances when the Earth was still very young, perhaps 4 billion years ago. The mixture of chemicals in the seas and atmosphere at that time, together with the energy provided by volcanic eruptions and lightning strikes, could possibly explain how the first molecules of life were produced.

Another possibility is that at least some of these molecules arrived on Earth from outer space. Meteorites are rocks that arrive on Earth from space, and they may have delivered molecules such as amino acids (the building blocks of proteins). If we can understand how life first arose from biological molecules, this could help us to answer another important question: is there life anywhere else in the universe?

Q: Discussion question 1: Most scientists think that all life on Earth today comes from other life. Suggest why scientists think it is unlikely that life could arise from non-living things today.

A: Conditions on Earth today are very different from those about 4 billion years ago. The atmosphere now contains a lot of oxygen, which reacts with and breaks down simple carbon compounds, and the mixture of chemicals in the seas and air has changed. The Earth is also full of living organisms, especially microorganisms such as bacteria, which would quickly use any new biological molecules as food or break them down before they could build up and join together. Finally, getting from simple molecules to even the simplest living cell probably took millions of years, so we would not expect to see it happen now.

What Is a Human Made Of?

The most abundant compound in our bodies is water. Almost all of the rest is made of the three main types of organic compound that make up the bodies of all living things: **carbohydrates, fats and proteins**.

Substance	Approximate percentage of total body mass
Water	62%
Protein	16%
Fat	16%
Carbohydrate	1%
Calcium and phosphorus (mostly in bones), plus other elements such as potassium, sodium, phosphate and chloride ions	5%

Notice how little of the body is carbohydrate. We use carbohydrate (glucose) quickly in respiration and store only a small amount of it, while fat forms a much bigger energy store.

Carbohydrates

- **Carbohydrates** include starch, cellulose and sugars
- Carbohydrate molecules are made from three elements: **carbon, hydrogen and oxygen** (C, H and O)
- One molecule of carbohydrate contains **about twice as many hydrogen atoms as carbon or oxygen atoms**. For example, glucose ($C_6H_{12}O_6$) has 12 hydrogen atoms but only 6 carbon atoms and 6 oxygen atoms

Sugars and Glucose

- **Sugars** are the simplest types of carbohydrate, with the smallest molecules. Sugars taste sweet, and they are soluble in water
- **Glucose** is one type of sugar. A glucose molecule is made of 6 carbon atoms, 12 hydrogen atoms and 6 oxygen atoms, so its formula is $C_6H_{12}O_6$
- **Shape**: the atoms in a glucose molecule are arranged to form a hexagon shape. Figure B3.01 shows glucose molecules drawn as small hexagons, because glucose molecules are quite small
- **Transport**: glucose is the form in which carbohydrates are transported around the human body. Glucose dissolves in blood plasma, which delivers it to every cell
- **Energy**: cells need glucose for energy, because glucose is used in respiration to release energy

Starch and Glycogen: Big Molecules Made from Glucose

Glucose molecules can link together in chains to form much larger molecules. Animals and plants do this in slightly different ways:

- **Glycogen** (in animals): glucose molecules link together to form glycogen, which is used as an energy store in animal cells. The liver helps to keep the concentration of glucose in the blood constant (you can read about this in Chapter B9)
- **Starch** (in plants): glucose molecules are linked together in a slightly different way to make starch, which is stored in plant cells. Starch can be broken down to form glucose again when the plant needs it
- **Cellulose**, which makes up plant cell walls, is another plant carbohydrate made from glucose molecules

Figure B3.02 shows a small part of a starch molecule. It is made from many glucose molecules (drawn as hexagons) linked together in a long chain, which coils into a spiral shape. Glycogen is also made of chains of glucose molecules, but its chains are branched rather than coiled.

The key idea to learn:

- many glucose molecules → starch (the energy store in plants)
- many glucose molecules → glycogen (the energy store in animals)

Carbohydrate	Size of molecule	Made from	Found in	Job
Glucose	Small (a sugar)	It is a single unit, the building block for the others	Plants and animals	Transported in blood plasma to cells; used in respiration to release energy
Starch	Very large	Many glucose molecules in a long chain that coils into a spiral	Plants, not animals	Energy store in plant cells
Glycogen	Very large	Many glucose molecules in branched chains	Animals, not plants	Energy store in animal cells, e.g. in the liver
Cellulose	Very large	Many glucose molecules	Plants, not animals	Makes up plant cell walls

Testing for Starch: The Iodine Test

- **Iodine solution** is a solution of iodine in potassium iodide. It is orange-brown
- **Positive result**: iodine solution turns blue-black when it is mixed with starch
- **Negative result**: it stays orange-brown if there is no starch
- **No heating** is needed: you just add a drop or two of iodine solution to a piece of food on a white tile

Figure B3.03 shows iodine solution being dropped onto a cut potato. The blue-black colour shows that starch is present in the potato.

Iodine test: orange-brown → blue-black means starch is present.

Testing for Reducing Sugars: Benedict's Test

- **Benedict's solution** is blue. It changes colour when it is heated with reducing sugars, such as glucose
- **Most sugars** can be detected using Benedict's solution
- **Heat is essential**: the mixture must be heated in a water-bath (about 80 °C), otherwise there is no colour change
- **Dissolve first**: the food is cut into small pieces and shaken with a little water, so that any sugar dissolves

The colour you get depends on the **concentration** of reducing sugar:

Colour after heating with Benedict's solution	Amount of reducing sugar

Blue (stays the same colour)	None, so the result is negative
Green	Very low concentration
Yellow	Slightly higher concentration
Orange-red (often called brick red)	Very high concentration

However, even low concentrations of glucose will eventually turn the blue colour to orange-red if they are heated for long enough. So, to compare foods fairly, heat them all for the same length of time, or time how long each one takes to change colour.

Figure B3.04 shows positive results of the Benedict's test. The tube on the left contains a small amount of reducing sugar, and the blue Benedict's solution has changed to green. The tube on the right contains a large amount of reducing sugar, and the mixture has turned orange-red.

Benedict's test: blue → green → yellow → orange-red as the amount of reducing sugar increases.

Good to know: not every sugar is a reducing sugar. Ordinary table sugar (sucrose) is not a reducing sugar, so it leaves Benedict's solution blue. This is why the book says that most sugars, rather than all sugars, can be detected with Benedict's solution.

Using a Water-Bath

The Benedict's test needs heat, and the safest way to heat the tubes is to stand them in a water-bath:

- **Electric water-bath**: can be set to a temperature above room temperature (about 80 °C for the Benedict's test)
- **Home-made water-bath**: heat water in a beaker over a burner, and stand the test-tube in the hot water (Figure B3.05)
- **0 °C water-bath**: fill a beaker with crushed ice
- **Keeping the temperature steady**: check the temperature with a thermometer, and either heat with the burner or add cold water to keep the temperature fairly constant

Figure B3.05 shows a boiling tube containing a mixture of Benedict's solution and food, standing in a beaker of water that is being heated over a burner. A water-bath heats the mixture gently and evenly; heating a tube directly in a flame could make the liquid boil suddenly and spit out.

Experimental Skills B3.01: Testing for Carbohydrates

Aim

To test several different foods to find out if they contain starch or reducing sugar.

You Will Need

- A selection of foods to test, kept separate from one another
- A knife to cut the foods into small pieces
- A white tile
- A small bottle of iodine solution, with a dropper
- A water-bath set at about 80 °C (an electric one, or one you make by heating water in a beaker over a burner, as in Figure B3.05)
- A thermometer to check the temperature of the water-bath
- Several large test-tubes (boiling tubes), and a beaker to stand them in
- Some Benedict's solution, with a dropper

Safety

- Take care when using a knife to cut food (this is the safety point printed in the book)
- Also sensible: wear eye protection, take care not to burn or scald yourself on the burner or the hot water, and point boiling tubes away from people

Method: Testing for Starch

1. Read through the instructions, and decide which foods you are going to test for starch.
2. Construct a results chart ready to fill in as you work, with a column for the food, a column for your observations and a column for your conclusion. Write down the colour that you see; do not write "no change".
3. Take a small piece of the first food and place it on the white tile. Add a drop or two of iodine solution. If the iodine solution stays brown, there is no starch in the food; if you see a dark blue or black colour, then starch is present.
4. Repeat with several other samples of different foods.

Method: Testing for Reducing Sugars

1. Read through the instructions and construct a results chart.
2. Take some of your first food and cut it into very small pieces. Put the pieces into a boiling tube and add a little water. Shake the tube to try to dissolve any sugar in the food.
3. Add enough Benedict's solution to cover the food. Shake the tube to mix the contents.
4. If you are using an electric water-bath, fill a beaker with water from the water-bath, stand the beaker in the water-bath and stand the test-tube in the beaker. If you have made your own water-bath, simply stand the test-tube in the beaker of hot water (Figure B3.05).
5. Watch as the mixture of food and Benedict's solution heats up. If it stays blue, there is no reducing sugar in the food; if it changes to red, there is reducing sugar in the food.
6. Repeat with several other samples of different foods.

Why these steps matter: cutting the food into very small pieces helps the substances inside it to dissolve in the water or reach the reagent, and the white tile makes the colour easy to see. Writing "no change" is not acceptable, because it does not tell anyone what colour you actually saw.

What the Results Look Like

Test	Negative result (substance absent)	Positive result (substance present)
Iodine test for starch	Iodine solution stays orange-brown	Dark blue or black (blue-black)
Benedict's test for reducing sugar	Stays blue after heating	Green, yellow or orange-red after heating, depending on how much reducing sugar there is

An example results chart (your own results depend on the foods you test):

Food	Test	Observation (colour seen)	Conclusion
Potato	Iodine	Blue-black	Starch present
Grapes	Iodine	Orange-brown	No starch
Grapes	Benedict's	Orange-red after heating	Large amount of reducing sugar
Cooking oil	Benedict's	Blue after heating	No reducing sugar

Conclusion

Foods that turn iodine solution blue-black contain starch. Foods that turn Benedict's solution green, yellow or orange-red when heated contain reducing sugar; the further the colour moves along the series towards orange-red, the more reducing sugar the food contains.

Answers to the Questions in This Box

Q: 1 Make a list of the foods that contained starch, and the foods that contained reducing sugar.

A: The list depends on which foods you tested. Typical results are: starch in potato, bread, rice, pasta and flour; reducing sugar in grapes, apples, onions, honey, fruit juice and milk. Foods such as cooking oil and table sugar (sucrose) give a negative result in both tests.

Q: 2 Do these foods come from plants, or from animals?

A: Most of them come from plants. Starch is the energy store in plant cells, so starchy foods such as potatoes, rice and wheat flour (used to make bread and pasta) all come from plants. Animals store carbohydrate as glycogen instead, so foods that come straight from animals, such as meat and eggs, do not contain starch. Most foods with reducing sugar also come from plants, such as fruits and onions (honey is made by bees from the sugary nectar of flowers). Milk is an exception: it comes from animals and contains a reducing sugar called lactose.

Activity B3.01: Comparing Reducing Sugar Content

When you tested foods for reducing sugar, you probably noticed that the red colour took a few minutes to appear. **The less time it takes for the red colour to appear, the more reducing sugar there is in the food.** In this activity (in a group of two or three), you discuss how to use this fact to compare the quantity of reducing sugar in three different foods, and write a list of instructions that someone else could follow.

A model set of instructions:

1. Choose three foods, for example grapes, apple and onion. Weigh out the same mass of each food (for example 2 g) and cut each one into equally small pieces.
2. Put each food into its own labelled boiling tube, add the same volume of water (for example 5 cm^3) to each, and shake each tube for the same length of time.
3. Add the same volume of Benedict's solution (for example 2 cm^3) to each tube and shake to mix.
4. Stand all three tubes in the same water-bath at about 80 °C at the same moment, and start a stopwatch.
5. Record the time taken for each mixture to turn red. If a tube has not turned red after 10 minutes, record "more than 10 minutes".
6. Repeat the whole experiment twice more and calculate a mean time for each food.
7. The food with the shortest mean time contains the most reducing sugar, and the food with the longest time contains the least.

Variables to mention in your instructions:

- **Independent variable** (what you change): the type of food
- **Dependent variable** (what you measure): the time taken for the red colour to appear
- **Controlled variables** (what you keep the same): mass of food, size of the pieces, volume of water, volume of Benedict's solution and the temperature of the water-bath

Peer assessment: swap your instructions with another group. Check whether their instructions are easy to follow and whether they would correctly show which food contains the most reducing sugar. Give feedback on what the group did well and one or two things that could be improved. Then think carefully about the feedback you receive, and what you would do differently if you planned a similar experiment in future.

Fats and Oils (Lipids)

Fats and oils are also known as **lipids**:

- A **fat** is a lipid that is **solid** at room temperature (for example, butter)
- An **oil** is a lipid that is **liquid** at room temperature (for example, olive oil)
- Lipids are **insoluble in water**

Fats and oils contain the same three elements as carbohydrates: **carbon, hydrogen and oxygen**. However, in fats each molecule contains **much less oxygen** than is found in a carbohydrate molecule.

Fat molecules are made of two kinds of smaller molecule: **glycerol** and **fatty acids**. Figure B3.06 shows a glycerol molecule drawn as a short upright bar and a fatty acid molecule drawn as a long zigzag line. In the fat molecule, **three fatty acid molecules are joined to one glycerol molecule**.

1 glycerol + 3 fatty acids → 1 fat (lipid) molecule

Why Organisms Need Fats

- **Cell membranes**: fats are important for making cell membranes
- **Energy stores**: fats are used as energy stores, and fat stores contain more energy per gram than carbohydrate stores
- **Insulation**: mammals often have a layer of cells containing fat droplets beneath the skin. This fat-containing tissue is both an energy store and a heat insulating layer

Figure B3.08 shows a bearded seal. Bearded seals have a really thick layer of fat underneath their skin, which helps them to survive in icy Arctic waters.

Testing for Fats and Oils: The Ethanol Emulsion Test

Fats and oils do not dissolve in water, but they do dissolve in **ethanol**. We can use this to detect them in food:

1. The food is cut into small pieces and shaken with ethanol. Any fats in the food dissolve in the ethanol.
2. The ethanol is poured into a clean tube containing water.
3. If there are fats in the ethanol, they form tiny droplets in the water, which give it a milky appearance (Figure B3.07).

The mixture of tiny droplets in water is called an **emulsion**, so this test is called the **ethanol emulsion test**. The milky colour appears because the fat cannot stay dissolved once the ethanol mixes with the water, so it comes out as millions of tiny droplets that scatter light.

Figure B3.07 shows a test-tube with a cloudy, milky white layer: this is what a positive result looks like. If there is no fat in the food, the liquid stays clear.

Ethanol emulsion test: clear → milky white emulsion means fat or oil is present.

Experimental Skills B3.02: Testing for Fats and Oils

Aim

To test a selection of different foods to find out if they contain fats or oils.

You Will Need

- A selection of foods to test, kept separate from one another
- A knife to cut the foods into small pieces
- Several very clean test-tubes
- A small bottle of ethanol, with a dropper
- Some distilled water

Safety

- This box does not print a safety note, but ethanol is highly flammable: keep it well away from flames, so no Bunsen burners should be lit nearby
- Take care when using a knife to cut food, and wear eye protection

Method

1. Read through the instructions and construct a results chart.
2. Take some of your first food and cut it into very small pieces. Put the pieces into a very clean test-tube and add a little ethanol. Shake the tube to try to dissolve any fats or oils in the food.

3. Take another clean test-tube and put some distilled water into it. Very carefully, pour the ethanol from the first tube into the distilled water. Do not let any of the food go in!

If you see a milky white colour appearing, this shows that there was fat in the food. The tubes must be very clean, and no food must go into the water, because traces of grease or small bits of food could make the water look cloudy and give a false positive result.

What the Results Look Like

Observation	Conclusion
A milky white (cloudy white) emulsion forms	Positive: fat or oil is present
The liquid stays clear and colourless	Negative: no fat or oil

Conclusion

Foods that produce a milky white emulsion contain fats or oils; foods that leave the water clear do not contain any fat that the test can detect.

Answers to the Questions in This Box

Q: 1 Which food(s) gave a positive result? Which gave a negative result?

A: This depends on the foods you tested. Typically, butter, margarine, cooking oil, cheese, peanuts and crisps give a positive result (a milky white emulsion), while foods such as apple, table sugar, egg white and boiled potato give a negative result (the liquid stays clear).

Q: 2 Use your observations to suggest how you could estimate which of the types of food contains the most fats and/or oils?

A: Compare how milky (cloudy) the emulsion is in each tube: the more fat or oil the food contains, the thicker and whiter the emulsion. To make the comparison fair, use the same mass of each food, cut into pieces of the same size, with the same volume of ethanol and the same volume of water, shaken for the same time. You could then rank the tubes in order of cloudiness, for example by holding each tube in front of a card with black writing on it and seeing which tube hides the writing most.

Proteins

Protein molecules contain four elements: **carbon, hydrogen, oxygen and nitrogen** (C, H, O and N). Some proteins, but not all, also contain a small quantity of the element **sulfur**.

- **Amino acids**: a protein molecule is made of a long chain of smaller molecules called amino acids (Figure B3.09)
- **20 kinds**: there are 20 different amino acids
- **Precise order**: each kind of protein has different amino acids linked together in a precise order (sequence)
- **One change matters**: if even one amino acid is changed, or moves into a different place in the protein molecule, then we get a different protein
- **Endless variety**: so there is an almost infinite number of different kinds of protein that can be made

Think of the 20 amino acids like the letters of an alphabet. By arranging letters in different orders you can make a huge number of different words; by arranging amino acids in different orders, cells make a huge number of different proteins.

Figure B3.09 shows a small part of a protein molecule as a chain of coloured blocks. Each block is one amino acid molecule, and the different colours represent different kinds of amino acid.

Many amino acids (20 kinds) joined in a specific sequence → one protein molecule

Functions of Proteins

Proteins have many different functions in organisms:

Protein	Function
Enzymes	All enzymes are proteins; they are biological catalysts that speed up reactions in cells (you will study enzymes in the next chapter)
Antibodies	Help to protect the body against pathogens; they are secreted by white blood cells, bind to pathogens and help to destroy them
Haemoglobin	The red pigment that transports oxygen in mammalian blood
Proteins in cell membranes	Proteins are important for forming cell membranes in all organisms
Keratin	The protein that makes up hair and fingernails in humans

Testing for Proteins: The Biuret Test

- **Biuret reagent** (also called biuret solution) is blue
- **Positive result**: it changes to violet (purple) when it is in contact with proteins; a pale lilac colour means only a little protein
- **Negative result**: it stays blue if there is no protein
- **No heating** is needed: just add the biuret solution to the food and shake

Figure B3.10 shows negative and positive results of the biuret test. The tube on the left has no protein, so it is still blue; the tube on the right contains protein, so it has turned violet.

Biuret test: blue → violet (purple) means protein is present.

Good to know: biuret reagent is a mixture of dilute copper sulfate solution and sodium hydroxide solution, and some exam questions describe adding these two solutions separately. Whichever way it is described, in an exam always say that the biuret test is the test for protein.

Experimental Skills B3.03: Testing for Proteins

Aim

To test a selection of different foods to find out if they contain proteins.

Using Control Tubes

You may find it helpful to start by testing something that you know contains protein, such as cooked egg white. Keep a tube showing a **negative result** (biuret reagent added to water, which will be blue) next to the tube showing a **positive result** (violet) in a test-tube rack. You can then compare each of your food tests with these two tubes to decide whether the result is negative or positive.

You Will Need

- A selection of foods to test, kept separate from one another
- A knife to cut the foods into small pieces
- Several clean test-tubes
- A small bottle of biuret solution with a dropper

Safety

- Take care when using a knife to cut food (this is the safety point printed in the book)
- Also sensible: biuret solution contains sodium hydroxide, which can harm skin and eyes, so wear eye protection and wash off any splashes with plenty of water

Method

1. Read through the instructions and construct a results chart.

2. Take some of your first food and cut it into very small pieces. Put the pieces into a clean test-tube and add some biuret solution. Shake the test-tube to mix the contents.
3. Look for a colour change from blue to violet.

What the Results Look Like

Observation	Conclusion
Stays blue, the same colour as the water control tube	Negative: no protein
Turns violet (purple), like the egg white tube	Positive: protein is present

Conclusion

Foods that turn biuret solution from blue to violet contain protein. Typical positive results come from egg white, meat, milk, cheese and beans.

Answers to the Questions in This Box

Q: 1 Do any of the foods you investigated contain no protein? How sure are you of this?

A: Any food that stayed blue, the same colour as the negative control tube, appeared to contain no protein; typical examples are cooking oil, table sugar and apple. However, we cannot be completely sure. The biuret test only shows protein if there is enough to give a visible colour change, so a food might contain a very small amount of protein that was not detected. A very pale violet can also be hard to tell apart from blue, especially if the food itself is coloured. We could be more confident by repeating the test, using more of the food, and comparing each tube carefully with the positive and negative control tubes against a white background.

Q: 2 Suggest one way you could improve the method to make the comparisons of the food types more valid.

A: Use the same mass of each food (for example, weigh 1 g of each), cut into pieces of the same size, and add the same volume of biuret solution to each tube, measured with a syringe or measuring cylinder rather than counted in drops. Then any difference in colour is caused only by the type of food. Other good improvements are to leave every tube for the same time before recording the colour, and to compare every tube with the same positive and negative control tubes.

Reflection: Remembering the Reagents

The book asks you to discuss with a partner good ways of remembering which reagent is used for which food test, and to decide which methods work best for you. Some ideas:

- Use a mnemonic: **"I Saw Big Sharks, Brave Penguins and Enormous Fish"** stands for Iodine for Starch, Benedict's for (reducing) Sugar, Biuret for Protein and Ethanol for Fat
- Remember **"Benedict's needs a bath"**: Benedict's is the only food test that is heated in a water-bath
- Remember **"proteins go purple"**: the biuret test turns purple (violet) when protein is present
- Picture the photos: the blue-black potato (starch), the orange-red tube (reducing sugar), the violet tube (protein) and the milky tube (fat)
- Make flashcards with the reagent on one side and the substance, method and colour change on the other, and test yourself regularly

Project B3.01: Modelling Biological Molecules

Working in a group of three or four, you make models of some biological molecules using paper and card. Different groups could make models of:

- **Starch and glycogen molecules**: both are made of glucose molecules linked in chains, but starch is coiled while glycogen is branched. Identical paper hexagons make good glucose molecules

- **Protein molecules**: made up of different sequences of amino acids. Use different colours or shapes for different amino acids, and show that changing the order makes a different protein
- **Fats and oils**: made of glycerol and fatty acids. Use one card strip for the glycerol, with three zigzag strips for the fatty acids attached to it

Comparing Carbohydrates, Fats and Proteins

Feature	Carbohydrates	Fats and oils (lipids)	Proteins
Elements	C, H, O (about twice as many H atoms as C or O atoms)	C, H, O (much less O than carbohydrates)	C, H, O, N (some also contain sulfur)
Smaller molecules they are made from	Starch and glycogen are made from glucose	Glycerol and fatty acids (1 glycerol + 3 fatty acids)	Amino acids (20 different kinds)
Examples	Glucose, starch, glycogen, cellulose	Butter (a fat), olive oil (an oil), fat under the skin	Enzymes, antibodies, haemoglobin, keratin
Uses in organisms	Glucose for respiration; starch (plants) and glycogen (animals) as energy stores	Cell membranes; energy stores; heat insulation	Enzymes, antibodies, oxygen transport, cell membranes, hair and nails
Food test	Iodine test (starch); Benedict's test (reducing sugars)	Ethanol emulsion test	Biuret test

Food Tests at a Glance

Food test	Tests for	What you do	Heat?	Negative result	Positive result
Iodine test	Starch	Add a drop or two of iodine solution to the food on a white tile	No	Stays orange-brown	Blue-black
Benedict's test	Reducing sugars, e.g. glucose	Shake the food with a little water, add Benedict's solution, then heat in a water-bath at about 80 °C	Yes	Stays blue	Green, yellow or orange-red, depending on the amount of reducing sugar
Biuret test	Proteins	Add biuret solution to the food and shake	No	Stays blue	Violet (purple)
Ethanol emulsion test	Fats and oils	Shake the food with ethanol, then pour the ethanol into water	No	Stays clear	Milky white emulsion

B3 Key Formulas and Concepts

1. Glucose: $C_6H_{12}O_6$ — 6 carbon, 12 hydrogen and 6 oxygen atoms, arranged in a hexagon shape
2. Carbohydrates contain C, H and O, with about twice as many hydrogen atoms as carbon or oxygen atoms
3. Fats and oils contain C, H and O, but much less oxygen than carbohydrates

4. Proteins contain C, H, O and N (some also contain sulfur)
5. Many glucose molecules → starch (energy store in plants) or glycogen (energy store in animals)
6. 1 glycerol + 3 fatty acids → 1 fat or oil (lipid) molecule
7. Many amino acids (20 kinds) in a specific sequence → a protein; a different sequence gives a different protein
8. Iodine test for starch (no heating): orange-brown → blue-black
9. Benedict's test for reducing sugars (heat in a water-bath at about 80 °C): blue → green → yellow → orange-red as the concentration of reducing sugar increases
10. Biuret test for proteins (no heating): blue → violet (purple)
11. Ethanol emulsion test for fats and oils: shake the food with ethanol, pour the ethanol into water → milky white emulsion
12. Human body composition by mass: about 62% water, 16% protein, 16% fat, 1% carbohydrate and 5% other substances
13. Fat stores contain more energy per gram than carbohydrate stores (roughly twice as much)

B3 Textbook Questions with Answers

Question B3.01

Name *one* carbohydrate that is found in both plants and animals.

Answer: Glucose. Plants make glucose in photosynthesis, and both plants and animals use glucose in respiration to release energy; in humans it is the form in which carbohydrate is transported in the blood plasma. (Starch is found only in plants and glycogen only in animals, so neither of these is a correct answer.)

Question B3.02

Animals that live in very cold places often have especially thick layers of fat underneath the skin (Figure B3.08). Suggest why.

(Figure B3.08: Bearded seals have a really thick layer of fat underneath their skin.)

Answer: The thick layer of fat under the skin acts as a heat insulating layer. Fat is a poor conductor of heat, so it reduces the amount of heat lost from the animal's warm body to its very cold surroundings, helping it to keep its body temperature high and constant. The fat is also an energy store: when food is scarce, the animal can use it in respiration to release energy, including heat to keep warm.

Question B3.03

Fat stores contain more energy per gram than carbohydrate stores. Suggest why birds that are about to go on long migrations build up stores of fat before they fly.

Answer: Flying a long distance needs a lot of energy, which is released by respiration in the bird's flight muscles, and the bird may not be able to feed for long periods during the journey, so it must carry its fuel with it. Fat stores contain more energy per gram than carbohydrate stores (roughly twice as much), so storing fuel as fat gives the bird the most energy for the least extra body mass. A lighter fuel store means the bird uses less energy carrying it, so it can fly further before it needs to stop and feed.

Question B3.04

Which element is found in all proteins, but not in carbohydrates and fats?

Answer: Nitrogen (N). All proteins are made of amino acids, which contain carbon, hydrogen, oxygen and nitrogen, but carbohydrates and fats contain only carbon, hydrogen and oxygen. (Sulfur is found in some proteins, but not in all of them, so it is not the answer.)

B3 Practice Exam Questions with Answers

Question B3.P1

Identify which elements are contained in starch. (1 mark)

A) carbon and hydrogen only

B) carbon, hydrogen and oxygen

C) carbon, hydrogen and nitrogen

D) carbon, oxygen and nitrogen

Answer: B — carbon, hydrogen and oxygen. [1] Starch is a carbohydrate made of many glucose molecules ($C_6H_{12}O_6$), and carbohydrates contain only C, H and O. A is wrong because it leaves out oxygen; C and D are wrong because starch contains no nitrogen (nitrogen is found in proteins).

Question B3.P2

The table shows the results of tests on four samples of food. Identify which food contains *only* fat. (1 mark)

Food	Iodine test	Benedict's test	Ethanol emulsion test	Biuret test
A	brown	blue	milky white	blue
B	blue-black	red	colourless	purple
C	brown	blue	colourless	blue
D	blue-black	red	milky white	purple

A) Food A

B) Food B

C) Food C

D) Food D

Answer: A — food A. [1] Food A gives a positive ethanol emulsion test (milky white), so it contains fat, and every other test is negative: iodine stays brown (no starch), Benedict's stays blue (no reducing sugar) and biuret stays blue (no protein). Food B contains starch, reducing sugar and protein but no fat; food C contains none of these substances; food D contains starch, reducing sugar, fat and protein.

Question B3.P3

Glucose, starch and glycogen are carbohydrates. (7 marks)

a) Identify which substance(s) are found in plants only, and not in animals? [1]

b) Give the elements that the molecules of all these substances contain. [1]

c) Describe the structure of a starch molecule. [2]

d) Glucose is a reducing sugar. Describe how you could test a liquid to find out if it contains glucose. [3]

Answer:

a) Starch (only). [1] Glucose is found in both plants and animals, and glycogen is found only in animals.

b) Carbon, hydrogen and oxygen (C, H and O). [1]

c) A starch molecule is made of many glucose molecules [1] linked together in a long chain, which coils into a spiral shape [1].

d) Add Benedict's solution to a sample of the liquid in a test-tube [1]. Heat the tube in a water-bath (at about 80 °C) for a few minutes [1]. If glucose is present, the colour changes from blue to green, yellow or orange-red (brick red); if the mixture stays blue, there is no glucose [1].

Question B3.P4

This question is about the biuret test and the ethanol emulsion test. (6 marks)

a) Biuret reagent is used to test for proteins. These statements about the biuret test are incorrect. Identify the errors so that you can write corrected versions of each statement.

i) To carry out the biuret test, add biuret solution to the food and heat gently. [1]

ii) If a food contains protein, biuret reagent turns from purple to blue. [1]

b) Ethanol is used to test for fats and oils.

i) Describe how you would carry out the ethanol test. [3]

ii) Describe what you would see if the test gave a positive result. [1]

Answer:

a) i) The error is heating: the biuret test does not need heating. Corrected statement: to carry out the biuret test, add biuret solution to the food and shake to mix, without heating. [1]

a) ii) The colours are the wrong way round. Corrected statement: if a food contains protein, biuret reagent turns from blue to purple (violet). [1]

b) i) Cut the food into very small pieces and put them into a clean test-tube with a little ethanol [1]. Shake the tube so that any fats or oils dissolve in the ethanol [1]. Pour the ethanol, without any of the food, into another clean test-tube containing distilled water [1].

b) ii) A milky white (cloudy white) emulsion forms in the water. [1]

B4 Enzymes

B4.01 Biological Catalysts

Before You Start: What Happens in a Chemical Reaction?

Enzymes control chemical reactions, so the chapter begins by checking what you already know about reactions. The book gives five statements and says that two of them are wrong.

Q: Here are some statements about chemical reactions. Two of them are incorrect. Can you identify these incorrect statements? When you think you have identified them, change them so that they are correct. In a chemical reaction: one or more substances are changed to one or more new substances; a large molecule may be split into smaller molecules; atoms of one element are rearranged to make atoms of other elements; small molecules may combine to form larger molecules; products are changed to reactants.

A: The two incorrect statements are:

- "Atoms of one element are rearranged to make atoms of other elements." This is wrong, because atoms are never changed into atoms of a different element in a chemical reaction. Corrected version: in a chemical reaction, atoms are rearranged to make new substances, but each atom stays as the same element.
- "Products are changed to reactants." This is the wrong way round. Corrected version: reactants are changed to products.

The other three statements are correct, and enzymes catalyse exactly these kinds of reactions: splitting large molecules into small ones (for example, starch into maltose) and joining small molecules into large ones (for example, amino acids into proteins).

Science in Context B4.01: The Dangers of Dairy Products

All mammals feed their young on milk (Figure B4.01 shows young guanacos feeding on milk from their mother). Milk is an excellent food. It contains carbohydrates, fats, proteins, minerals and vitamins.

Human babies are mammals too, so they feed on milk. To help them digest the carbohydrate in milk, they produce an enzyme called **lactase**. Lactase breaks a sugar called **lactose** into smaller molecules, which can then be absorbed into the blood.

Most people stop making lactase once they are old enough to eat food other than milk. Some people keep making lactase all their lives and can drink milk and eat dairy products with no problems. Everyone else is **intolerant to lactose**, and dairy foods make them feel ill:

1. They do not make lactase, so lactose is not broken down in the digestive system.
2. The lactose stays inside the alimentary canal (the tube that food passes through).
3. This makes water move into the alimentary canal by osmosis, and the extra water can cause diarrhoea.
4. Bacteria living in the alimentary canal feed on the lactose and produce gases as a waste product, which can cause abdominal pain and flatulence.

Solutions to lactose intolerance: the simplest solution is not to eat foods containing lactose. Another is to swallow a pill containing the enzyme lactase before eating dairy products, which may prevent the unpleasant symptoms.

Q: Discussion question 1 — If most people stop making lactase when they are adults, there must be some advantage in this. Discuss what this might be.

A: Possible ideas to discuss:

- Making an enzyme uses energy and raw materials (amino acids are needed to build the protein). An adult who no longer drinks milk does not need lactase, so stopping making it saves energy and amino acids for making other, more useful proteins.
- Long ago, before people kept animals for milk, adults had no milk to drink, so lactase was of no use after a child had stopped feeding on its mother's milk.
- Losing lactase may help a young mammal to stop feeding on its mother's milk (weaning), so the mother can save her energy and resources for her next baby.

What Is a Catalyst?

Many chemical reactions can be sped up by substances called **catalysts**. A catalyst alters the rate of a chemical reaction without being changed itself. Because it is not used up, a catalyst can be used again and again.

Enzymes as Biological Catalysts

Within any living organism, chemical reactions are taking place all the time. These are called **metabolic reactions**. Almost every metabolic reaction is controlled by catalysts called **enzymes**.

- Enzymes are **proteins** (so they are made of amino acids).
- Without enzymes, metabolic reactions would take place very slowly or not at all.
- Enzymes make sure that the rates of metabolic reactions are **fast enough to sustain life**.

Examples of Enzymes at Work

Digestion. Inside the intestines, large molecules are broken down to smaller ones in the process of digestion. These reactions are sped up by enzymes, and a different enzyme is needed for each kind of nutrient:

- starch → maltose (catalysed by amylase)
- protein → amino acids (catalysed by protease)

Plants. These enzymes are also found in plants. In germinating seeds, enzymes digest the food stores (mainly starch) to feed the growing seedling.

Catalase. Catalase works inside all the cells of living organisms. Many of the chemical reactions in cells produce hydrogen peroxide, which is a very dangerous substance and must be broken down immediately. Catalase breaks it down:

- hydrogen peroxide → water + oxygen (catalysed by catalase)

Building up molecules. Not all enzymes help to break things down. Many enzymes help to make large molecules from small ones. For example, enzymes help to link amino acids together to make proteins inside cells.

Naming Enzymes

Enzymes are named according to the reaction that they catalyse, and their names often end in **-ase**:

- Enzymes that break down **carbohydrates** are **carbohydrases**.
- Enzymes that break down **proteins** are **proteases**.
- Enzymes that break down **fats and oils (lipids)** are **lipases**.

Some enzymes have more specific names. The carbohydrase that breaks down **starch** is **amylase**, the one that breaks down **maltose** is **maltase**, and the one that breaks down **sucrose** is **sucrase**.

The substance that an enzyme changes is called its **substrate**. The substrate of amylase is starch, and the substrate of lipase is lipids.

Enzyme	Type of enzyme	Substrate	Product(s)
Amylase	Carbohydrase	Starch	Maltose

Maltase	Carbohydrase	Maltose	Glucose
Sucrase	Carbohydrase	Sucrose	Glucose and fructose
Lactase	Carbohydrase	Lactose	Smaller sugars (glucose and galactose)
Protease	Protease	Protein	Amino acids
Lipase	Lipase	Lipids (fats and oils)	Fatty acids and glycerol
Catalase	Not a digestive enzyme; works inside all cells	Hydrogen peroxide	Water and oxygen

Spelling warning: you have to be careful to spell the name of an enzyme and its substrate correctly and to write clearly. Maltose and maltase can look almost the same if you do not write carefully. A useful rule: names ending in **-ose** are sugars (maltose, sucrose, lactose), and names ending in **-ase** are enzymes (maltase, sucrase, lactase).

How Enzymes Work

Each type of enzyme has molecules with a very specific shape. The enzyme molecule has a 'dent' in it called the **active site**. Figure B4.02 shows how an enzyme catalyses a reaction:

1. The **substrate** molecule has a shape that is **complementary** to the active site of the enzyme (a perfect mirror-image shape).
2. The substrate fits into the active site, where the substrate and the enzyme bind together. For this to happen, the fit has to be perfect.
3. While the substrate is in the active site and bound to the enzyme, the enzyme changes the substrate into a new substance called the **product** (in Figure B4.02, the substrate is split into two product molecules).
4. The product breaks away from the enzyme.
5. The enzyme is now free, and unchanged, so it is ready to bind with another substrate molecule.

In Figure B4.02, one substrate molecule is broken into two product molecules. Enzymes can also catalyse reactions where two substrate molecules bind with the active site and are joined together to form a single product molecule.

Enzymes are very fast. In the few minutes it takes to read about how enzymes work, a single enzyme molecule could have catalysed millions of reactions. Catalase is the fastest enzyme known: one catalase molecule can break down an almost unbelievable 44 million hydrogen peroxide molecules in one second.

The Enzyme–Substrate Complex (Supplement)

The short-lived structure that forms as the substrate slots into the enzyme's active site has its own name: the **enzyme–substrate complex**. The whole process can be summarised as:

- enzyme + substrate → enzyme–substrate complex → enzyme + product(s)

The enzyme appears at both ends of this sequence, which shows that it is unchanged and can be used again.

Activity B4.01: Enzyme Specificity (Supplement)

Each enzyme can only catalyse reactions with one type of substrate. This is called enzyme **specificity**. The activity asks you to think about why enzymes are specific, discuss your ideas with a partner, and be ready to share them with the class.

Q: Why are enzymes specific?

A: The active site of an enzyme has a very specific shape. Only a substrate molecule with a complementary shape can fit into the active site and bind to it to form an enzyme–substrate complex. Molecules of other substances have different shapes, so they do not fit the active site and cannot be changed by that enzyme. This is like a key fitting a lock: only the key with the right shape will fit. For example, amylase breaks down starch, but it cannot break down proteins or fats, because their molecules do not fit its active site.

B4.02 Factors That Affect Enzymes

Optimum Temperature and Optimum pH

Enzymes act very quickly: each enzyme molecule can change many substrate molecules to products every second. But some factors affect this speed of action. Enzymes are very sensitive to **temperature** and **pH**.

Each kind of enzyme has a particular temperature and pH at which it works fastest. These are called the **optimum temperature** and the **optimum pH** for that enzyme. ("Optimum" means "best".)

Figure B4.03a: The Effect of Temperature on Enzyme Activity

Figure B4.03a is a graph of rate of reaction (y-axis, no numbers) against temperature / °C (x-axis, 0 to 60 °C). It shows that:

- At **0 °C** the rate of reaction is zero.
- From 0 °C to about 10 °C, the rate increases only slowly.
- From about 20 °C to about 35 °C, the rate increases steeply.
- The rate is greatest at about **37–38 °C**, just below 40 °C. This peak is labelled **optimum**.
- Above the optimum, the rate falls. It is still about three-quarters of the maximum at 50 °C, then it drops very steeply and reaches **zero at about 58–60 °C**.
- The fall above the optimum is much steeper than the rise below it.

Figure B4.03b: The Effect of pH on Enzyme Activity

Figure B4.03b is a graph of rate of reaction against pH (x-axis, pH 0 to 10). It has two curves:

Feature	A protease in the stomach	Most enzymes
Optimum pH (peak of curve)	About pH 2 (just below 2)	About pH 7
Lowest pH at which it works	About pH 0 (rises steeply from pH 0)	About pH 5.6
Highest pH at which it works	About pH 7 (activity is very low above pH 6)	About pH 10
Shape of curve	Steep rise to the peak, then a gradual fall	Very steep rise from pH 5.6 to 7, then a more gradual fall to pH 10

The graph shows that different enzymes have different optimum pH values. Most enzymes work best in neutral conditions (around pH 7), but the protease in the stomach works best in very acidic conditions.

Buffer Solutions

To investigate the effect of pH, you need to control the pH accurately. A **buffer solution** is a liquid that has a known pH and keeps that pH steady all the time, even if chemical reactions take place. You use a different buffer solution for each pH you want to test.

Experimental Skills B4.01: Investigating the Effect of pH on the Activity of Catalase

Aim

To find out how changing the pH affects the rate at which catalase breaks down hydrogen peroxide to water and oxygen.

How the Experiment Works

- The reaction is: hydrogen peroxide → water + oxygen (catalysed by catalase).
- Catalase is extracted from celery (or any other plant material, such as potato or leaves) by mashing it up in water, so that the catalase forms a solution in the water. It is best if this is done once for the whole class.
- A small square of filter paper is soaked in the catalase solution and pushed to the bottom of a beaker of hydrogen peroxide solution. The catalase on the paper produces oxygen, the oxygen bubbles collect on the paper, and the paper floats upwards (Figure B4.05).
- The faster the catalase works, the sooner the paper rises. So the **time taken for the paper to rise** is a measure of the rate of reaction: a shorter time means a faster rate.
- Buffer solutions are used to set the pH in each beaker.

Apparatus

- A blender or liquidiser, or a pestle and mortar
- Some fresh plant material, such as celery
- Distilled water
- Muslin or a sieve, and a filter funnel
- Some 3% hydrogen peroxide solution
- Five 50 cm^3 beakers
- Small measuring cylinders or syringes, to measure 5 cm^3 and 10 cm^3
- Five buffer solutions, each with a different pH
- 15 small squares of filter paper, all the same size (about 5 mm × 5 mm)
- Forceps (tweezers)
- A stopwatch or timer

Method

Before you start, practise using the forceps to pick up the little pieces of filter paper. It is not easy!

1. Cut up some celery, put it into a blender, add water and blend. (Without a blender, cut the celery into small pieces and grind it with water using a pestle and mortar.)
2. Pour the mixture through muslin in a filter funnel, or through a sieve, to separate most of the bits of celery from the liquid. (Filter paper would take too long.)
3. The liquid now contains catalase. Label a beaker 'catalase solution' and pour some of the liquid into it.
4. Copy the results table.
5. Write the pH values of your buffer solutions into the results table, and label each small beaker with the pH of one of the buffer solutions.
6. Measure 5 cm^3 of hydrogen peroxide solution into each labelled beaker.
7. Measure 10 cm^3 of the appropriate buffer solution into each beaker.
8. Use forceps to pick up a square of filter paper and dip it into the catalase solution (Figure B4.04).
9. Get the stopwatch ready. Still using the forceps, push the paper to the bottom of the hydrogen peroxide solution in one small beaker, let it go, and time how long it takes to rise to the top of the liquid (Figure B4.05). Record the time.
10. Repeat steps 8 and 9 twice more, using the same beaker, so you have three tries.
11. Repeat steps 8, 9 and 10 for each of the other small beakers.

12. Draw a line graph with pH on the x-axis and the mean time for the paper to rise on the y-axis. Join the points with ruled straight lines.

What the figures show: Figure B4.04 shows a filter paper square that has been dipped into the celery extract (catalase solution), held in forceps, ready to go into a beaker of hydrogen peroxide solution and buffer labelled pH 5.6. Figure B4.05 shows the same beaker with a stopwatch running, and oxygen bubbles collecting on the paper and making it float upwards.

Results Table

pH	1st try / s	2nd try / s	3rd try / s	Mean / s
One row for each buffer pH	Time for paper to rise	Time for paper to rise	Time for paper to rise	(1st + 2nd + 3rd) ÷ 3

Variables

Variable	In this experiment
Independent (what you change)	pH, set using five different buffer solutions
Dependent (what you measure)	Time taken for the filter paper to rise to the top of the liquid, in seconds
Controlled (kept the same)	Volume and concentration of hydrogen peroxide (5 cm^3 of 3% solution); volume of buffer (10 cm^3); size of the filter paper squares (about 5 mm × 5 mm); the same catalase solution for every square; temperature (all at room temperature); same size of beaker, so the depth of liquid is the same

Expected Results and Conclusion

- The paper rises fastest (shortest time) at the **optimum pH** of catalase. For catalase from celery or potato, this is usually close to pH 7.
- At pH values further from the optimum, in either direction, the paper takes longer to rise because the catalase works more slowly.
- At a very acidic or very alkaline pH, the paper may not rise at all, because the catalase has been denatured.
- So a graph of mean time against pH is roughly U-shaped, with its lowest point at the optimum pH. This is the opposite way up to a rate graph like Figure B4.03b, because **rate = 1 ÷ time**. For example, if the paper takes 20 s to rise, the rate is 1 ÷ 20 = 0.05 per second.
- **Conclusion:** pH affects the activity of catalase. It works fastest at its optimum pH and more slowly at pH values above or below this.

Safety

- Take care if you are using a blender or liquidiser: keep fingers away from the blades and make sure the lid is on before switching it on.
- Good practice: wear eye protection when using hydrogen peroxide (it is an irritant), and wash any splashes off your skin with water.

Answering the Experiment Questions

Q: (1) Write the word equation for the chemical reaction that took place in the small beakers.

A: hydrogen peroxide → water + oxygen (catalysed by catalase)

Q: (2) Explain, in your own words, why the pieces of filter paper rose to the top of the liquid in the beaker.

A: The filter paper was soaked in catalase. The catalase broke down the hydrogen peroxide in the beaker, producing oxygen gas. Bubbles of oxygen collected on the paper and made it buoyant (less dense than the liquid), so it floated up to the top. The faster the catalase worked, the faster the bubbles formed, and the sooner the paper rose.

Q: (3) Describe the relationship between the pH and the time taken for the filter paper to rise to the top of the liquid.

A: As the pH increases from the lowest value towards the optimum, the time taken for the paper to rise decreases. The shortest time is at the optimum pH (usually around pH 7). As the pH increases above the optimum, the time taken increases again. Quote the actual pH values and mean times from your own results table, with units (s). Remember that the shortest time shows the fastest rate of reaction.

Q: (4) Did you have any anomalous results? If so, explain why you decided that these results were anomalous. Explain how you dealt with these results when you drew your graph.

A: An anomalous result is one that does not fit the pattern: for example, one of the three tries at a pH is very different from the other two, or a mean does not fit the smooth trend of the other points. Possible causes are the paper sticking to the side of the beaker, the paper not being pushed right to the bottom, or more catalase soaking into one square than the others. To deal with it, leave the anomalous result out when calculating the mean (or repeat that try), and plot the mean of the remaining results. For example, if the three tries at one pH were 12 s, 13 s and 31 s, the 31 s result is anomalous, so the mean is (12 + 13) ÷ 2 = 12.5 s. If an anomalous point is plotted on the graph, circle it and do not join it into the line.

Q: (5) Use your results to estimate the optimum pH of the catalase that you used.

A: The optimum pH is the pH with the shortest mean time, which is the lowest point on your graph. For catalase from celery this is usually about pH 7. Because you only tested five pH values, you can only say that the true optimum is close to this value; it may lie between the pH values either side of your lowest point.

Q: (6) Suggest what you could do to obtain a more precise value for the optimum pH of this enzyme.

A: Repeat the experiment using more buffer solutions with pH values closer together around your estimated optimum (for example pH 6.0, 6.5, 7.0, 7.5 and 8.0, or even smaller steps). Do more repeats at each pH and calculate means.

Q: (7) Evaluate your experiment by identifying the main sources of error. Suggest how you could improve the experiment, to reduce these sources of error and give you more confidence that your results are correct.

A: Main sources of error and improvements:

- **Different amounts of catalase on each paper square:** the squares may not be exactly the same size or may be dipped for different times. Cut the squares accurately (or use a hole punch) and dip each one for the same time, for example 5 seconds.
- **Timing by eye:** reaction time when starting and stopping the stopwatch, and deciding exactly when the paper reaches the top. Use the same person and the same end point each time, or video the beakers and time from the recording.
- **The paper may stick to the side** of the beaker or not be released from the very bottom. Release it in the same way, in the centre of the beaker, each time.
- **Hydrogen peroxide is used up:** steps 8 to 10 reuse the same beaker three times, so the hydrogen peroxide becomes slightly less concentrated. Use fresh hydrogen peroxide and buffer for each try.
- **Temperature is not controlled:** stand the beakers in a water-bath at a constant temperature, for example 25 °C.
- **The catalase solution may lose activity over time:** use freshly prepared extract and keep it in the same conditions throughout.

- **Few data points:** test more pH values and do more repeats, then check that the repeats are close together.

Experimental Skills B4.02: Investigating the Effect of Temperature on the Activity of Lipase

Aim and Background

To investigate how temperature affects the activity of lipase. Lipase is an enzyme that catalyses the breakdown of lipids (fats and oils). Milk contains lipids.

- lipid → fatty acids + glycerol (catalysed by lipase)

Fatty acids are acids, so they decrease the pH. Sodium carbonate solution makes the mixture alkaline at the start. The indicator **thymolphthalein** is blue in alkaline conditions and loses its colour when the pH becomes less alkaline. So the **time taken for the blue colour to disappear** shows how quickly the lipase is producing fatty acids: the shorter the time, the faster the lipase is working.

Apparatus

- Fresh milk
- Lipase solution, with a concentration of about 5%
- Sodium carbonate solution, with a concentration of 0.05 mol dm^{-3}
- A small bottle of thymolphthalein indicator, with a dropper
- Five test-tubes and a rack to stand them in
- A test-tube holder
- A syringe or measuring cylinder to measure 10 cm^3
- A small syringe to measure 1 cm^3
- A stopwatch or timer
- One or more glass rods
- Water-baths at five different temperatures, with at least two below the optimum temperature for the enzyme and two above (Figure B4.06 shows an electric water-bath that you can set to a particular temperature, and a simple water-bath made from a beaker of water over a burner)
- A thermometer
- A waterproof marker to label the test-tubes

Method

1. Read through the instructions and draw a results chart that you can fill in as you work.
2. Label your test-tubes with your initials.
3. Measure 5 cm^3 of milk into each test-tube.
4. Measure 7 cm^3 of sodium carbonate solution into each test-tube.
5. Add five drops of thymolphthalein indicator to each test-tube. This makes the liquid go blue. Stir the contents with the glass rod. (Figure B4.07 shows one of these tubes, containing 5 cm^3 of milk, 7 cm^3 of sodium carbonate solution and 5 drops of thymolphthalein indicator, ready to go into the water-bath.)
6. Stand each test-tube in a water-bath at a different temperature. Leave the tubes for a few minutes so that the contents reach the same temperature as the water-bath. Measure this temperature and record it.
7. Using the small syringe, add 1 cm^3 of lipase to the first test-tube. Start timing immediately. Stir with a clean glass rod to mix the lipase into the milk.
8. Record the time taken for the blue colour to disappear, to the nearest second.
9. Repeat for the other four temperatures.

10. Draw a line graph with temperature on the x-axis and time taken on the y-axis. Join the points with ruled lines.

Results Chart

Temperature / °C	Time taken for blue colour to disappear / s
One row for each water-bath temperature	Time to the nearest second

Variables

Variable	In this experiment
Independent (what you change)	Temperature (five water-baths)
Dependent (what you measure)	Time taken for the blue colour to disappear, in seconds
Controlled (kept the same)	Volume and type of milk (5 cm^3); volume and concentration of sodium carbonate solution (7 cm^3 of 0.05 mol dm^{-3}); number of drops of indicator (5); volume and concentration of lipase solution (1 cm^3 of about 5%); time allowed for the tubes to reach the water-bath temperature; how the mixture is stirred

Expected Results and Conclusion

- At low temperatures the blue colour takes a long time to disappear.
- As the temperature increases towards the optimum (often somewhere around 35–45 °C for lipase), the time gets shorter.
- Above the optimum, the time gets longer again. At a high temperature (for example 60 °C or more), the colour may never disappear, because the lipase has been denatured.
- So the graph of time against temperature is roughly U-shaped. Its lowest point is at the optimum temperature.
- **Conclusion:** lipase works fastest at its optimum temperature; it works more slowly at lower temperatures, and it is denatured at high temperatures.

Safety

- If you have made your own water-bath, take care not to knock over the beaker of water. Stand up while you are working, so that if something does get knocked over you can quickly move out of the way.
- If the water-bath is very hot, do not touch the hot water, or a hot test-tube, with your hands. Use a test-tube holder.
- Good practice: wear eye protection, because sodium carbonate solution is an irritant.

Reflection

Q: The instructions for these experiments are quite long and complicated. How did you deal with that? What can you do in the future, to help you to understand complicated instructions and follow them successfully?

A: Useful strategies include:

- Read all the instructions through before starting, and highlight the quantities (5 cm^3 milk, 7 cm^3 sodium carbonate, 5 drops of indicator, 1 cm^3 lipase).
- Draw the results chart first (step 1 tells you to), so you know exactly what you are measuring.
- Turn the method into a short flow chart or numbered checklist, and tick off each step as you finish it.

- Set out and label the apparatus in the order you will use it.
- Work with a partner: one reads out each step while the other carries it out.
- Ask your teacher if any step is unclear before you begin, not halfway through.

Answering the Experiment Questions

Q: (1) Explain why the liquid containing the milk was blue at the start of the experiment.

A: The mixture contained sodium carbonate solution, which is alkaline. Thymolphthalein indicator is blue in alkaline conditions, so the liquid was blue.

Q: (2) Explain why the liquid in some of the tubes became colourless.

A: The lipase broke down the lipids (fats) in the milk into fatty acids and glycerol. The fatty acids lowered the pH, so the mixture became less alkaline, and thymolphthalein loses its colour when the pH becomes less alkaline. The colour only disappeared in tubes where the lipase was working. At a temperature where the lipase was denatured (very hot), few or no fatty acids were made, so the liquid stayed blue.

Q: (3) Describe how temperature affected the time taken for the liquid to become colourless.

A: As the temperature increased from the lowest temperature up to the optimum, the time taken for the liquid to become colourless decreased. Above the optimum, the time taken increased again, and at the highest temperature the liquid may not have become colourless at all. Quote your own temperatures and times, with units (°C and s).

Q: (4) According to your results, what is the optimum temperature for lipase?

A: The optimum temperature is the temperature at which the time taken was shortest (the fastest reaction). In most class experiments this is somewhere around 40 °C. The true optimum may lie between two of the temperatures you tested.

Q: (5) Your timer probably measured to 0.1 or even 0.01 seconds. Suggest why it is sensible to record the times to the nearest second.

A: The colour fades gradually, so it is hard to judge the exact moment it disappears, and you also have a reaction time (a fraction of a second) when starting and stopping the timer. The times are therefore not accurate to 0.1 s or 0.01 s. Recording those extra digits would suggest a precision that the measurements do not really have, so the nearest second is a sensible level of precision.

Q: (6) Evaluate your experiment, by identifying the main sources of error. How could you improve the experiment, so that you have more confidence in your results?

A: Main sources of error and improvements:

- **Judging the end point:** it is difficult to decide exactly when the blue colour has gone. Compare each tube with a colourless reference tube against a white background, or use a pH meter or colorimeter.
- **Lipase added at room temperature:** this cools or warms the mixture. Stand the lipase solution in the same water-bath before adding it.
- **Water-bath temperature changes:** a beaker over a burner is hard to keep steady. Use a thermostatically controlled electric water-bath, and check the temperature with a thermometer throughout.
- **Delay between adding lipase and starting the timer, and uneven stirring:** start the timer as the lipase goes in, and stir every tube in the same way.
- **Only one result at each temperature:** do three repeats at each temperature, calculate means, and test more temperatures, especially close to the optimum.

Self Assessment

The book asks you to rate how well you constructed your results chart, using a smiling face (did it really well), a straight face (made a good attempt and partly succeeded) or a sad face (did not try or did not

succeed), for each of these statements:

- I drew the results chart using a ruler for all the rows and columns.
- I included the units (s for the time measurements, and °C for the temperature measurements) in the headings of the table.
- I did not write the units again with the individual results in the table.
- I recorded each temperature reading to the nearest whole degree or 0.5 degree.
- I recorded each time reading to the nearest second.
- A friend confirms that my results chart is very easy to understand.

Denaturation

Temperature. Enzymes have an optimum temperature at which they work fastest:

- Enzymes from the human body generally have an optimum temperature of about **37 °C**.
- Enzymes from plants may have optimum temperatures much lower than this.
- Some bacteria, especially those that live in hot springs, have very high optimum temperatures: up to **80 °C** in some cases.

For many enzymes, a temperature above about **60 °C** completely stops them working. This is because the high temperature damages the enzyme. The enzyme is said to be **denatured**, and it cannot catalyse its reaction any more.

pH. Enzymes also have an optimum pH. For most enzymes this is around **pH 7**, but some enzymes work best at a much higher or lower pH. For example, in Chapter B6 you will meet an enzyme that works best in the acidic conditions of the human stomach, at a pH of about **2**. When an enzyme is placed in a liquid with a pH that is not its optimum pH, it is damaged. The further the pH is from the optimum, the more the enzyme is damaged, and well away from the optimum it is denatured and cannot catalyse its reaction.

Important: low temperatures do NOT denature enzymes. At 0 °C an enzyme has almost no activity because its molecules are moving very slowly, but if it is warmed up again it will start working. Denaturation is caused by high temperatures or by a pH far from the optimum.

Explaining the Effect of Temperature on Enzyme Activity (Supplement)

Use Figure B4.03a to follow these steps:

1. **At 0 °C, there is no activity.** Molecules have very little kinetic energy and move only slowly. Substrate molecules rarely collide with the enzyme, so they rarely enter its active site, and very few substrate molecules are converted to product.
2. **As temperature increases, activity increases.** The kinetic energy of the enzyme and substrate molecules increases. They move faster and collide with each other more frequently and with more energy, so effective collisions are more frequent. Each second, more substrate molecules collide with an active site, form enzyme–substrate complexes and are converted to product.
3. **At the optimum (about 37–38 °C in Figure B4.03a), activity is greatest.** There are the most effective collisions per second while the active site still has its correct shape.
4. **Above the optimum, activity decreases.** The kinetic energy of the enzyme begins to shake it apart. Its molecules begin to lose their shape, so the active site is no longer a true complementary shape to the substrate. When a substrate molecule collides with an active site it may not fit, so it cannot form an enzyme–substrate complex and is not converted to product.
5. **At about 60 °C, activity stops.** The active site is so out of shape that the enzyme has completely stopped working. It is completely denatured.

Explaining the Effect of pH on Enzyme Activity (Supplement)

Something similar happens when pH changes, but the explanation is about **shape**, not kinetic energy:

- Each enzyme molecule has a very specific shape, and its active site is the best fit for the substrate at the enzyme's optimum pH.
- As the pH moves away from the optimum, the enzyme's shape changes, so the active site fits the substrate less well and fewer enzyme–substrate complexes form.
- A pH well above or below the optimum causes the enzyme molecule to lose its shape completely (denature), so it can no longer bind with the substrate.

Comparing the Effects of Temperature and pH

Condition	Effect of temperature	Effect of pH
Below the optimum	Molecules have less kinetic energy, so there are fewer effective collisions; the enzyme is slow but NOT denatured	More acidic than the optimum: the enzyme's shape changes, so the active site fits the substrate less well; far below the optimum the enzyme is denatured
At the optimum	Fastest rate: most effective collisions while the active site still has its correct shape	Fastest rate: the active site is the best fit for the substrate
Above the optimum	The enzyme starts to lose its shape; the active site is no longer complementary; it is completely denatured at about 60 °C for many enzymes	More alkaline than the optimum: the enzyme's shape changes, so the active site fits the substrate less well; far above the optimum the enzyme is denatured

Project B4.01: An Enzyme Demonstration

Working in a group of three, you plan a short demonstration to explain what enzymes are and how they work. It can be 'live' with a commentary, or a video with a voiceover. Imagine that your audience knows nothing about enzymes: your task is to get them interested and help them begin to understand what enzymes are and what they do.

Choosing an enzyme: the best enzyme to use is **catalase**, because it is easy to see it working. Catalase is present in all animal and plant tissues, and when it breaks down hydrogen peroxide it releases oxygen.

- **Figure B4.08:** raw liver contains a lot of catalase, so it breaks down hydrogen peroxide and releases oxygen very quickly. The reaction can be quite dramatic, with froth pouring out of the flask.
- **Figure B4.09:** if hydrogen peroxide is added to raw potato, the catalase in the potato breaks it down and releases oxygen, which forms a white froth. This is a quieter reaction, which may be better for a demonstration.

Ideas for the demonstration: show the froth forming on raw potato; explain the active site with a simple model (for example, two shapes that fit together like a lock and key); compare raw potato with boiled potato, where the catalase has been denatured and no froth forms; show that the gas relights a glowing splint, proving it is oxygen.

Q: It is extremely important to consider safety as you plan. What are the risks that you need to be aware of? How will you reduce these risks?

A: Risks and how to reduce them:

- **Hydrogen peroxide** is an irritant and can harm the eyes and skin. Wear eye protection and gloves, use only dilute (3%) solution, and wipe up spills straight away.
- **Froth overflowing**, especially with liver. Use small amounts in a large container standing on a tray, and never put a lid on the container (the oxygen produced would build up pressure).
- **Raw liver** may carry harmful bacteria. Wear gloves, wash your hands afterwards, and dispose of the liver safely.
- **Knives** for cutting potato or liver. Cut on a board, cutting away from your body.

- **Boiling water** (if you boil potato for comparison). Handle hot liquids carefully and let them cool.
- **Glowing splints** are a fire risk. Tie back long hair and keep flammable materials away.

B4 Key Formulas and Concepts

1. **Catalyst:** a substance that increases the rate of a chemical reaction and is not changed by the reaction; enzymes are biological catalysts made of protein.
2. **How an enzyme works:** enzyme + substrate → enzyme–substrate complex → enzyme + product(s)
3. **Amylase:** starch → maltose
4. **Maltase:** maltose → glucose
5. **Protease:** protein → amino acids
6. **Lipase:** lipid (fats and oils) → fatty acids + glycerol
7. **Catalase:** hydrogen peroxide → water + oxygen
8. **Lactase:** lactose → smaller sugars (glucose + galactose), which are absorbed into the blood
9. **Rate and time:** rate = 1 ÷ time taken. A shorter time means a faster rate. Example: a time of 4 min gives a rate of 1 ÷ 4 = 0.25 per minute, and a time of 8 min gives 1 ÷ 8 = 0.125 per minute, so the first reaction is twice as fast.
10. **Optimum temperature:** about 37 °C for human enzymes; many enzymes are denatured above about 60 °C; some bacteria in hot springs have optimum temperatures up to 80 °C.
11. **Optimum pH:** about pH 7 for most enzymes; about pH 2 for the protease in the stomach.
12. **Range:** the lowest to the highest value; for example, most enzymes in Figure B4.03b work from about pH 5.6 to about pH 10.
13. **Naming rule:** enzyme names usually end in -ase (carbohydrase, protease, lipase, amylase); sugar names end in -ose (maltose, sucrose, lactose).
14. **Iodine test for starch:** blue-black means starch is present; brown (the colour of iodine solution itself) means no starch is present.
15. **Thymolphthalein:** blue in alkaline conditions; it loses its colour when the pH becomes less alkaline (for example, when fatty acids are made).

B4 Textbook Questions with Answers

Question B4.01

Explain why enzymes are called biological catalysts.

Answer: Enzymes are catalysts because they increase the rate of chemical reactions without being changed by the reactions, so each enzyme molecule can be used again and again. They are called biological catalysts because they are proteins made by living cells, and they catalyse the metabolic reactions that take place inside living organisms. Without enzymes, these reactions would happen very slowly or not at all, far too slowly to sustain life.

Question B4.02

You read about the enzyme lactase at the start of this chapter.

a) What type of enzyme is lactase?

b) Name the substrate of lactase.

Answer:

a) Lactase is a carbohydrase, because it breaks down a carbohydrate (the sugar lactose) into smaller sugar molecules.

b) Lactose, the sugar found in milk.

Question B4.03

Look at Figure B4.03a. Describe what is shown on the graph. (When you are describing a graph, it is a good idea to quote figures, with units.)

(Figure B4.03a is a graph of rate of reaction, with no scale, against temperature / °C from 0 to 60 °C. A single curve rises to a peak, which is labelled 'optimum', and then falls.)

Answer: The graph shows how the rate of an enzyme-controlled reaction changes with temperature:

- At 0 °C the rate of reaction is zero.
- As the temperature increases from 0 °C, the rate increases, slowly at first (up to about 10 °C) and then steeply between about 20 °C and 35 °C.
- The rate reaches a maximum at about 37–38 °C (just below 40 °C). This is the optimum temperature.
- Above the optimum, the rate decreases, gradually at first and then very steeply, until it reaches zero at about 58–60 °C.
- The rate falls much more steeply above the optimum than it rises below it.

Question B4.04

Look at Figure B4.03b.

a) What is the optimum pH of the protease in the stomach?

b) What is the optimum pH of most enzymes?

c) What is the range of pH values at which enzymes are able to work?

(Figure B4.03b is a graph of rate of reaction against pH, from pH 0 to pH 10. It has two curves: one for a protease in the stomach, and one labelled 'most enzymes'.)

Answer:

a) About pH 2. The peak of the protease curve is just below pH 2.

b) About pH 7.

c) For most enzymes, the range is from about pH 5.6 to about pH 10; below about pH 5.6 and above about pH 10 their activity is zero. The protease in the stomach has a different range, from about pH 0 to about pH 7. So, looking at both curves, enzymes can work across almost the whole range shown (about pH 0 to pH 10), but each enzyme works only over its own narrower range.

Question B4.05

Look at the graph that you drew from your results in Experimental skills B4.02 Investigating the effect of temperature on the activity of lipase. In your own words, explain the shape of the graph (use your biological knowledge and understanding to give reasons for the shape of the graph).

Answer: The graph of time taken for the blue colour to disappear (y-axis) against temperature (x-axis) should be roughly U-shaped. The time is long at low temperatures, falls to a minimum at the optimum temperature (often around 40 °C), and then rises again at higher temperatures. At the highest temperature the colour may not have disappeared at all. A short time means a fast rate, so this graph is the upside-down version of a rate graph like Figure B4.03a.

- **Low temperatures (long times):** lipase and lipid molecules have little kinetic energy and move slowly, so they rarely collide. Few lipid molecules enter the active site each second, so fatty acids are made slowly and it takes a long time for the pH to fall enough to make the thymolphthalein colourless.
- **Rising towards the optimum (times get shorter):** the molecules gain kinetic energy, move faster, and collide more often and with more energy. There are more effective collisions, so more enzyme–substrate complexes form and more fatty acids are made per second. The pH falls faster, so the colour disappears sooner.

- **At the optimum (shortest time):** the rate is greatest, because there are the most effective collisions while the active site still has its correct shape.
- **Above the optimum (times get longer):** the high kinetic energy starts to shake the lipase molecules apart. They lose their shape, so the active site is no longer complementary to the lipid molecules. Fewer substrate molecules fit and form enzyme–substrate complexes, so fatty acids are made more slowly.
- **Very high temperatures (no colour change):** the lipase is completely denatured, so no fatty acids are made, the pH does not fall and the liquid stays blue.

Question B4.06

Use your own words to explain the shapes of the curves for most enzymes, in Figure B4.03b.

(In Figure B4.03b, the curve for most enzymes is at zero below about pH 5.6, rises steeply to a peak at about pH 7, and then falls back to zero at about pH 10.)

Answer:

- **At the optimum pH (about pH 7):** the enzyme molecule has exactly the right shape, so its active site is the best fit for (complementary to) the substrate. Substrate molecules fit easily, many enzyme–substrate complexes form, and the rate of reaction is highest.
- **As the pH moves away from 7 (more acidic or more alkaline):** the change in pH begins to change the shape of the enzyme molecule, including its active site. The active site becomes less complementary to the substrate, so fewer substrate molecules can fit and bind, fewer enzyme–substrate complexes form, and the rate falls.
- **Well below or well above the optimum (below about pH 5.6 or above about pH 10):** the enzyme has lost its shape completely; it is denatured. The substrate can no longer bind with the active site, so the rate is zero.
- The curve rises more steeply on the acidic side (pH 5.6 to 7) than it falls on the alkaline side (pH 7 to 10). This shows that these enzymes are affected more by a small decrease in pH below 7 than by a similar increase above 7.

B4 Practice Exam Questions with Answers

Question B4.P1

Identify the smaller molecules from which an enzyme molecule is made. (1 mark)

A) amino acids

B) fatty acids

C) glucose

D) glycerol

Answer: A — amino acids. Enzymes are proteins, and protein molecules are made from long chains of amino acids [1]. Fatty acids and glycerol are the smaller molecules that make up lipids, and glucose is the smaller molecule that makes up starch.

Question B4.P2

(5 marks)

a) Complete the sentences about enzymes. Choose words from the list. [3]

Word list: chemical, proteins, metabolic, active, complementary

Enzymes are _____ that function as biological catalysts.

Enzymes are involved in all _____ reactions.

The part of an enzyme where a substrate binds is called its _____ site.

b) Describe why enzymes are important to all living organisms. [2]

Answer:

a) Enzymes are **proteins** that function as biological catalysts [1]. Enzymes are involved in all **metabolic** reactions [1]. The part of an enzyme where a substrate binds is called its **active** site [1]. ("Chemical" is not accepted for the second gap, because enzymes are not involved in all chemical reactions, only in metabolic reactions inside living things.)

b) Any two of these points:

- Enzymes speed up (catalyse) the metabolic reactions in living organisms [1].
- Without enzymes, these reactions would happen too slowly or not at all; enzymes make the reactions fast enough to sustain life [1].
- They allow essential processes to happen quickly enough, for example digesting food so that it can be absorbed, building proteins from amino acids, or breaking down toxic hydrogen peroxide (catalase) [1].

Question B4.P3

The figure below shows some stages in an enzyme-controlled reaction. (4 marks)

(The figure shows three stages, from top to bottom. Stage 1: an enzyme molecule (labelled 'enzyme') with a dent in one side, labelled Q, and a separate round molecule labelled P. Stage 2: the round molecule fitted into the dent of the enzyme, with the two together labelled R. Stage 3: the enzyme, with its dent empty again, next to the round molecule now split into two pieces, labelled S.)

Identify the parts labelled P, Q, R and S. [4]

P: _____

Q: _____

R: _____

S: _____

Answer:

P = substrate [1]

Q = active site [1]

R = enzyme–substrate complex [1]

S = products (the product molecules) [1]

Question B4.P4

Amylase is an enzyme that breaks down starch to maltose. (9 marks)

a) A student put some starch solution into a test-tube. He took a small sample of starch from the tube and added it to a drop of iodine solution on a white tile.

(The figure shows a test-tube of starch solution, a glass rod carrying a sample of starch solution, and a white tile with drops of iodine solution in its dimples.)

Next, he added some amylase solution to the starch solution. After five minutes, he took another sample from the tube, and tested it with iodine solution.

The table shows his results.

	Result of iodine test
At start	blue-black
After adding amylase solution	brown

Explain the student's results. [3]

b) The student investigated the effect of pH on the activity of amylase.

He placed the same volume of starch solution in five tubes. He added different buffer solutions to each tube. He then added amylase solution to each tube.

He took samples of the mixture from each tube and measured the time for all the starch to disappear.

His results are shown in the table.

pH	Time for starch to disappear / min
4	did not disappear
6	8
7	4
9	14
11	did not disappear

i) Use the student's results to describe the effect of pH on the activity of amylase. [3]

ii) The student kept the volume of starch solution the same in each tube. State three other variables that the student should have kept the same in his experiment. [3]

Answer:

a) Any three of these points:

- At the start, the sample turned the iodine solution blue-black, which shows that starch was present [1].
- Amylase broke down (digested) the starch [1] into maltose [1].
- After five minutes the iodine solution stayed brown, which shows that there was no starch left; maltose does not give a blue-black colour with iodine [1].

b) i) One mark for each of these points:

- Amylase is most active at pH 7, which is its optimum pH; the starch disappeared in the shortest time (4 min) [1].
- Activity decreases as the pH moves away from 7 in either direction: the starch took 8 min to disappear at pH 6 and 14 min at pH 9 [1].
- At pH 4 and pH 11 the starch did not disappear, so the amylase was not working (it was denatured) at these pH values [1].
- Working with rates: rate = 1 ÷ time, so the rate was 1 ÷ 8 = 0.125 per min at pH 6, 1 ÷ 4 = 0.25 per min at pH 7, and 1 ÷ 14 = 0.071 per min at pH 9. Amylase worked twice as fast at pH 7 as at pH 6 (8 ÷ 4 = 2), and 3.5 times as fast as at pH 9 (14 ÷ 4 = 3.5).

b) ii) Any three of these variables [1 each]:

- temperature (for example, keep all the tubes in the same water-bath)
- volume of amylase solution
- concentration of amylase solution
- concentration of starch solution
- volume of buffer solution
- how often samples were taken and tested (and the size of each sample)

Question B4.P5

The graph below shows how the activity of two enzymes, A and B, is affected by temperature. (12 marks)

(The graph plots rate of activity, with no scale, on the y-axis against temperature / °C on the x-axis, from 0 to 100 °C. Enzyme A: activity rises from zero at 10 °C to a peak at 40 °C, then falls steeply to zero at

about 58 °C. Enzyme B: activity rises from zero at 40 °C to a peak at 70 °C, then falls to zero at about 92 °C. The two peaks are at about the same height, and the two curves cross at about 55 °C.)

a) Compare the effect of temperature on enzyme A and enzyme B. [5]

b) Explain the effect of temperature on enzyme B from:

i) 40 to 60 °C [3]

ii) 70 to 90 °C. [4]

Answer:

a) Any five of these points (a comparison needs similarities and differences, and each statement should mention both enzymes):

- Similar pattern: for both enzymes, activity increases as temperature increases up to an optimum, and then decreases [1].
- Enzyme A has an optimum temperature of 40 °C, whereas enzyme B has an optimum of 70 °C, which is 30 °C higher [1].
- Enzyme A starts to work at 10 °C, but enzyme B shows no activity until 40 °C [1].
- Enzyme A stops working (is completely denatured) at about 58 °C, whereas enzyme B keeps working until about 92 °C [1].
- Both enzymes reach about the same maximum rate of activity [1].
- Both work over a similar range of about 50 °C: A from 10 °C to about 58 °C, and B from 40 °C to about 92 °C [1].
- Below about 55 °C enzyme A is more active than enzyme B, and above about 55 °C enzyme B is more active; they have the same activity at about 55 °C [1].
- For both enzymes, activity falls more steeply above the optimum than it rises below it [1].

b) i) Three of these points:

- The activity of enzyme B increases, from zero at 40 °C to about 70% of its maximum at 60 °C [1].
- As temperature increases, the enzyme and substrate molecules gain kinetic energy and move faster [1].
- They collide more frequently, so there are more effective collisions between substrate molecules and the active site [1].
- So more enzyme–substrate complexes form and more product is made per second [1]. (This range is below B's optimum, so B is not being denatured.)

b) ii) Four of these points:

- The activity of enzyme B decreases, falling from its maximum at 70 °C to a low level by 90 °C (and to zero at about 92 °C) [1].
- Above the optimum, the high kinetic energy makes the enzyme molecule vibrate so much that it begins to shake apart [1].
- The enzyme, including its active site, changes shape [1].
- The active site is no longer complementary to the substrate, so the substrate cannot fit or bind [1].
- Fewer enzyme–substrate complexes form, so less product is made per second [1].
- The enzyme becomes denatured; by about 92 °C it is completely denatured and has no activity [1].

B5 Plant Nutrition

B5.01 Photosynthesis

Before You Start

The chapter opens with three questions to discuss with a partner, about where the energy in food comes from.

Q: (1) What types of nutrient does your body use to obtain energy?

A: Mainly **carbohydrates** (such as glucose and starch) and **fats** (lipids). Proteins can also be used to release energy, but the body mainly uses them for growth and repair.

Q: (2) Energy cannot be created or destroyed. Where did the energy in the nutrients come from originally?

A: From the **Sun**. The energy was originally light energy in sunlight.

Q: (3) How did the energy get into the nutrients?

A: Plants absorbed the light energy using **chlorophyll** and transferred it into **chemical energy** in carbohydrates such as glucose, by photosynthesis. Plants use this glucose to make their other nutrients. Animals take in the energy when they eat plants, or eat other animals that have eaten plants.

Nutrition — Why Plants Are Different from Animals

All living organisms need to take many different substances — **nutrients** — into their bodies. Some are used to make new parts or to repair old parts. Others are used to release energy. Taking in useful substances is called **nutrition**.

In Chapter B3 you met the important groups of biological molecules: **proteins, fats and carbohydrates**. These are all **organic substances**.

Animals	Plants
Cannot make organic substances themselves	Can make organic nutrients from inorganic substances
Must take in organic substances as nutrients when they feed	Use carbon dioxide (from the air), water and mineral ions (from the soil)
Depend on plants, directly or indirectly, for food and energy	Make carbohydrates by photosynthesis, which means "making with light"

Like all reactions in organisms, photosynthesis depends on **enzymes** to help it take place.

The Photosynthesis Equation

In photosynthesis, **water and carbon dioxide react together to produce glucose and oxygen**. They will only do this in the presence of **sunlight** and **chlorophyll**.

- **Word equation:** carbon dioxide + water → glucose + oxygen (with "sunlight" written above the arrow and "chlorophyll" below it)
- **Balanced chemical equation:** $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

A balanced chemical equation shows the chemical formula of each reactant and product, and how many molecules of each are involved.

What the formulae tell us:

- **CO_2** (carbon dioxide): one molecule contains 1 atom of carbon (C) and 2 atoms of oxygen (O)
- **H_2O** (water): one molecule contains 2 atoms of hydrogen (H) and 1 atom of oxygen

- $C_6H_{12}O_6$ (glucose): one molecule contains 6 carbon atoms, 12 hydrogen atoms and 6 oxygen atoms
- O_2 (oxygen): one molecule contains 2 oxygen atoms

Because a glucose molecule contains 6 carbon atoms, **six carbon dioxide molecules are needed to make one glucose molecule**. There is the same number of each kind of atom on each side of the equation:

Atom	Reactants: $6CO_2 + 6H_2O$	Products: $C_6H_{12}O_6 + 6O_2$
Carbon (C)	6 × 1 = 6	6
Hydrogen (H)	6 × 2 = 12	12
Oxygen (O)	(6 × 2) + (6 × 1) = 18	6 + (6 × 2) = 18

	Reactants (raw materials)	Products
Substances	Carbon dioxide and water	Glucose and oxygen
Where they come from or go	Carbon dioxide diffuses in from the air; water comes from the soil	Glucose is used or stored by the plant; oxygen is released into the air (or into the water, for an aquatic plant)

Remember: Light and chlorophyll are **not** reactants. Light provides the energy and chlorophyll captures it, so they are written on the arrow, never after a "+" sign.

Science in Context B5.01: Using Solar Energy to Make Fuels

- As the human population grows, we use more and more fuel for homes, industries and vehicles. Renewable energy sources such as wind and solar power are being used more, but a lot of our energy still comes from **fossil fuels**.
- Burning fossil fuels produces **carbon dioxide**, so the quantity of carbon dioxide in the atmosphere is increasing. This contributes to **global warming**, so we urgently need other ways of providing energy.
- Plants use energy from sunlight to make carbohydrates, which provide energy for their cells. This way of providing energy does not produce carbon dioxide — it **uses it up**.
- In many parts of the world, plants are grown to provide **biofuels**, which are burnt to produce electricity or to move vehicles. But this takes up land that may be needed for food crops, or that would be better left as natural forests or other habitats for wildlife.
- Scientists are researching **artificial photosynthesis** to make fuels. Chlorophyll captures energy from sunlight and helps to transfer it into carbohydrates, so researchers are looking for other substances that can do the same job. One laboratory is using **tiny particles of gold** as a substitute for chlorophyll. It works, but only very slowly. Researchers think it will be at least 2030 before artificial photosynthesis can produce significant quantities of fuel.

Q: How might the large-scale use of artificial photosynthesis help to slow down climate change?

A: Artificial photosynthesis would use energy from sunlight to make fuels from carbon dioxide, so carbon dioxide would be removed from the atmosphere as the fuel is made. When the fuel is burned, it only releases the carbon dioxide that was taken in to make it, so there is no overall increase in atmospheric carbon dioxide. Fossil fuels, by contrast, release carbon that was locked away underground for millions of years. Replacing fossil fuels with these fuels would reduce the build-up of carbon dioxide (a greenhouse gas), slowing global warming and climate change. Unlike biofuel crops, it would not need large areas of land, so forests, which absorb carbon dioxide, could be left standing.

Activity B5.01: Modelling Photosynthesis

Work in a group of four or five. In this model, **small balls represent atoms** and **pieces of egg carton represent molecules**.

You will need: egg cartons cut into molecule-shaped pieces (six each of three different shapes, Figure B5.01a–c); one complete egg carton that can hold 12 eggs, labelled to hold a glucose molecule (Figure B5.01d); 36 small balls that fit into the egg cartons (for example, table tennis balls); red, black and blue pens.

Method:

1. Use the red pen to label 18 of the balls "O" (oxygen), the black pen to label 6 of them "C" (carbon), and the blue pen to label the other 12 "H" (hydrogen).
2. Take one ball representing oxygen and two balls representing hydrogen, and put them into an angled box to make a water molecule (Figure B5.02).
3. Make five more water molecules.
4. Use the remaining balls to make six carbon dioxide molecules, choosing the correct egg carton shapes.
5. Use your water molecules and carbon dioxide molecules to make a glucose molecule, by putting the balls into the correct holes in the base and lid of the 12-hole egg carton. Use oxygen from the water only — do not use oxygen from the carbon dioxide.
6. Use your leftover balls to make oxygen molecules.
7. If you have time, take apart your glucose and oxygen molecules and turn them back into carbon dioxide and water.

How the numbers work: six water molecules use 12 H and 6 O; six carbon dioxide molecules use 6 C and 12 O. The glucose molecule takes 6 C, 12 H and 6 O. The 12 O left over make six O_2 molecules. All 36 balls are used both before and after the "reaction" — the atoms are only rearranged.

Q: (1) Explain how this model represents the balanced equation for photosynthesis.

A: Each ball is an atom and each carton shape is a molecule. You start with six carbon dioxide molecules (6 C and 12 O) and six water molecules (12 H and 6 O), which match the reactants $6CO_2 + 6H_2O$. You rearrange exactly the same 36 balls into one glucose molecule (6 C, 12 H, 6 O) and six oxygen molecules (12 O), which match the products $C_6H_{12}O_6 + 6O_2$. No balls are added or lost, which shows that there is the same number of each type of atom on both sides — the equation is balanced. The numbers of each type of carton show the numbers in front of each formula.

Q: (2) Suggest how you could modify the model, or the way that you use the model, to show how energy in sunlight helps the reaction to take place.

A: For example, only allow the molecules to be taken apart and rebuilt while a torch or lamp ("sunlight") is shining on the group, and only by a group member holding a green card representing chlorophyll. That person could collect "energy tokens" from the lamp and hand one over for each bond that is broken and remade, then place the tokens inside the glucose carton. This shows that chlorophyll captures light energy and that the energy ends up stored in glucose.

Q: (3) You may hear people say that "photosynthesis changes carbon dioxide into oxygen". Use your model to explain why this is not correct.

A: In the model, no ball ever changes into a different kind of ball — a carbon atom stays a carbon atom. The carbon dioxide molecules are broken apart and their atoms are rearranged: the carbon atoms from carbon dioxide end up in the glucose molecule, not in oxygen. Water is also a reactant, and its hydrogen atoms end up in glucose too. So photosynthesis turns carbon dioxide **and water** into **glucose and oxygen**; carbon dioxide is not simply "changed into" oxygen. (Beyond the syllabus: experiments with labelled atoms show that, in real plants, the oxygen gas released actually comes from the water molecules, so the model is a simplification.)

Reflection: Some people find that handling objects and moving them around, as in this model, helps them to understand. Think about whether the model helped you to understand the photosynthesis equation and how — for example, counting the balls shows clearly why the numbers must balance. If it

did not help, suggest another method that might suit you better, such as drawing the atoms as coloured circles, building molecules from modelling clay, or watching an animation.

Chlorophyll — Capturing Energy from Sunlight

In the chloroplasts of a plant's leaves, water and carbon dioxide react together to make carbohydrates and oxygen. In other circumstances, you could shine light on a mixture of carbon dioxide and water for ever, and no carbohydrates or oxygen would ever appear. What is special about a plant's leaves is that they contain a green pigment called **chlorophyll**, which is present in their **chloroplasts**.

How chlorophyll works, step by step:

1. Chlorophyll **captures energy from sunlight** (Figure B5.03).
2. It immediately **passes the energy on (transfers it)** to water molecules and carbon dioxide molecules.
3. The energy makes these substances **react**, producing a kind of carbohydrate called **glucose** (and oxygen).
4. The glucose made in the leaves contains some of the energy that was originally in the sunlight.

So chlorophyll **transfers energy from light into energy in carbohydrates** (light energy becomes chemical energy). Figure B5.03 shows a forest described as "a giant carbohydrate factory", using sunlight as fuel and carbon dioxide and water as its raw materials.

How a Plant Uses Carbohydrates

Photosynthesis has two products: **glucose and oxygen**.

- The **oxygen** is released into the atmosphere (or into the water, if the plant is aquatic).
- The **glucose** (a carbohydrate) is used for many different purposes, described below.

Releasing Useful Energy

Some of the glucose is used to provide **energy** for activities that the plant's cells need to carry out. For example:

- Energy is needed to move **mineral ions into the root hairs by active transport**.
- Energy is needed to **build protein molecules from amino acids**, for growth.

The energy is released from glucose by **respiration** (Chapter B9).

Storing, to Use Later — Starch

Plants usually make much more glucose than they need to use for energy immediately. They store it by turning it into **starch**.

- Starch molecules are made by linking many glucose molecules together into a **long, spiral chain**.
- Millions of these molecules clump together to make **starch grains**. Figure B5.04 is a photomicrograph of potato tuber cells; iodine solution has been added, so the starch grains are stained blue.

Why starch is a good storage substance:

- Starch molecules are **not soluble in water**, so they do not get involved in the chemical reactions taking place inside the cells.
- They **do not affect the concentration of the solutions** inside the cell, so they do not cause water to enter or leave by **osmosis**.
- They can be **quickly and easily broken down to glucose** again when the plant needs it.

Making Sucrose, for Transport

- Plants can only make glucose in the parts that contain chlorophyll, which usually means the **leaves**. All the other parts of the plant must have carbohydrate delivered to them.

- Plants do not actually transport glucose. They first change it into a sugar with larger molecules, called **sucrose**.
- Sucrose is carried from one part of the plant to another inside **phloem tubes** (Chapter B7).
- When the sucrose reaches its destination, it can be changed back to glucose and used in **respiration** to release energy.

Making Cellulose, to Build Cell Walls

- As plants grow, they make new cells. Every cell needs a **cell wall**, so a growing plant must make **cellulose** molecules to form these walls.
- Cellulose is made by linking glucose molecules in long chains. The molecules are linked in a different way from starch, so the chains **stay straight** rather than coiling up into spirals.

Making Nectar, to Attract Pollinators

- Many plants reproduce sexually, producing male and female gametes in flowers.
- Unlike animal gametes, the male gametes of flowers **cannot move themselves** from place to place. Instead, they rely on **insects, bats or birds** to carry them, inside **pollen grains**, from one flower to another.
- No animal will do this unless it gets a reward, so flowers produce **nectar** that animals can feed on. Nectar contains different kinds of sugar, all made from the glucose that the plant has made by photosynthesis.
- Figure B5.05 shows insects visiting flowers to collect carbohydrate-rich nectar and protein-rich pollen. Without knowing it, they help the flower by pollinating it.

Making Amino Acids, to Make Proteins

- Plants use some of the glucose made in photosynthesis to make **amino acids**, which are then used to make **proteins**, for growth.
- Proteins contain carbon, hydrogen and oxygen, **and also nitrogen**. So to make amino acids from glucose, plants need a source of nitrogen.
- They get nitrogen from the soil in the form of **nitrate ions**. These are usually taken in by **active transport** through the root hairs, then transported to all parts of the plant, where they are combined with glucose to make amino acids.
- If a plant cannot get enough nitrate ions, it cannot synthesise proteins effectively, so it will **not grow quickly or strongly**.

Making Other Substances, such as Chlorophyll

- Glucose can also be used to make **chlorophyll**.
- Chlorophyll is **not a protein**, but it does contain **nitrogen**. It also contains **magnesium**. Therefore plants need to take in **magnesium ions**, as well as nitrate ions, to make chlorophyll.
- Without these ions, the plant's leaves look **yellow** rather than green. Figure B5.06 shows leaves on an orange tree growing in soil that does not contain enough magnesium ions; the tree cannot make enough chlorophyll, so the leaves are not completely green.
- If a plant cannot make lots of chlorophyll, it cannot photosynthesise well, and so it will not grow well.

Summary Table: Uses of Glucose

What the glucose is used for	Purpose	Key detail
Respiration	Releases useful energy	Energy for active transport of mineral ions and for building proteins

Making starch	Energy storage	Insoluble; does not affect osmosis; easily broken back down to glucose
Making sucrose	Transport	Carried in phloem tubes to parts that cannot photosynthesise
Making cellulose	Building cell walls	Straight chains of glucose; every new cell needs a wall
Making nectar	Attracting pollinators	A sugary reward for insects, bats or birds that carry pollen
Making amino acids, then proteins	Growth	Needs nitrogen from nitrate ions
Making chlorophyll	Absorbing light energy	Needs nitrogen (from nitrate ions) and magnesium (from magnesium ions)

Mineral Ions: Nitrate and Magnesium

	Nitrate ions	Magnesium ions
Element supplied	Nitrogen	Magnesium
Why needed	To make amino acids, and then proteins (nitrogen is also needed for chlorophyll)	To make chlorophyll
Deficiency symptoms	Weak growth, yellow leaves	Yellowing of leaves, often especially between the veins

Both deficiencies cause yellow leaves because both nitrogen and magnesium are needed to make chlorophyll. A plant short of either ion photosynthesises less and grows poorly.

B5.02 Leaves

A Leaf Is a Carbohydrate Factory

Photosynthesis happens inside **chloroplasts**. This is where the **enzymes** and **chlorophyll** that catalyse and supply energy for the reaction are found. In a typical plant, most chloroplasts are in the cells in the leaves, so a leaf is a factory for making carbohydrates. It is not surprising that most plants have leaves that are perfectly adapted to help photosynthesis take place as quickly and efficiently as possible.

The Structure of a Leaf

Figure B5.07a shows the parts of a leaf:

- **Lamina** — the broad, flat part of the leaf.
- **Leaf stalk** — joins the lamina to the rest of the plant. Inside the leaf stalk are collections of parallel tubes called **vascular bundles**.
- **Midrib** — the main vein running down the middle of the lamina.
- **Veins** — the vascular bundles branching through the lamina. The tubes in the vascular bundles carry substances to and from the leaf (Chapter B7).

Figure B5.07a also shows where a transverse section of the leaf and a cross-section of a vein are cut. Figure B5.07b shows the **network of veins** in a dicot leaf, which carries water to all parts of it.

Inside the leaf cells, photosynthesis happens in chloroplasts. The chlorophyll is **spread out on membranes** inside the chloroplasts, so that a lot of sunlight can reach it.

Why Most Leaves Are Broad and Thin

The raw materials for photosynthesis are carbon dioxide and water, and sunlight is needed to provide energy. Most — but not all — leaves have a **large surface area** and are **very thin**.

Feature	How it helps photosynthesis
Large surface area	Allows large amounts of sunlight to fall onto the leaf; also increases the rate at which carbon dioxide can diffuse into the leaf from the air
Thin	Sunlight can pass right through the leaf, so many cells inside it can photosynthesise; carbon dioxide can reach all the cells quite quickly by diffusion
Network of veins	Xylem vessels in the veins bring water from the soil to all parts of the leaf

Only **0.04%** of the air is carbon dioxide, so the structure of a leaf must allow carbon dioxide to move into the leaf really easily.

Water is brought to the leaf from the soil in tubes called **xylem vessels** (pronounced "zi-lem"). These run in the vascular bundles, which form the veins in the leaf. You can often see many tiny veins if you look closely at a leaf.

How the Raw Materials Reach a Palisade Cell

Figure B5.08 summarises how the raw materials for photosynthesis travel to a chloroplast in a palisade cell:

What is needed	How it reaches the chloroplasts
Sunlight (energy)	Passes through the transparent cuticle and upper epidermis
Carbon dioxide	Diffuses through the stomata from the atmosphere, then diffuses through the air spaces to the palisade cells
Water	Travels from the roots in xylem vessels, then travels to the chloroplasts by osmosis

Tissues in a Leaf

Although a leaf is thin, it is made up of several layers of cells. You can see these if you look at a **transverse section (TS)** of a leaf under a microscope (Figure B5.09). Each tissue in a leaf has its own function. From top to bottom:

Tissue or structure	What it is like	Function and adaptation
Cuticle	A waxy substance forming a thin, transparent, waterproof covering	Reduces water loss; transparent, so light passes straight through
Upper epidermis	A layer of transparent cells packed tightly together, with no chloroplasts	Reduces the quantity of water vapour escaping from the leaf; lets light through to the palisade cells
Palisade mesophyll	Tall, narrow cells containing very large numbers of chloroplasts, close to the top of the leaf	Main function is photosynthesis; being near the top, they get plenty of sunlight through the transparent epidermis above

Spongy mesophyll	Rounded, loosely packed cells with chloroplasts and many air spaces between them	Also photosynthesise; the air spaces let carbon dioxide and oxygen diffuse between the air and the cells inside the leaf, and let water vapour move from the cell surfaces to the outside of the leaf
Vascular bundle (vein)	Xylem vessels and phloem tubes side by side	Xylem brings water and mineral ions into the leaf; phloem carries sucrose away to the rest of the plant
Lower epidermis	A layer of cells that usually has no cuticle	The underside of the leaf does not often have sunlight falling on it, so it does not get as hot and does not lose as much water vapour
Stomata	Openings in the lower epidermis	Allow diffusion of carbon dioxide and oxygen in and out of the leaf; water vapour also diffuses out through them
Guard cells	A pair of cells around each stoma; they contain chloroplasts	Change their shape to open and close the stomata

The mesophyll layer is made up of the palisade mesophyll and spongy mesophyll together. Figure B5.09 also shows the inside of a palisade cell: a cell wall, cytoplasm, a nucleus, a large vacuole and many chloroplasts.

The Palisade Mesophyll Cell

The palisade mesophyll cell is the leaf cell that is best adapted for photosynthesis:

- It contains **very large numbers of chloroplasts**, so lots of chlorophyll is available to absorb light energy.
- It is **tall and narrow**, and the cells are packed side by side in a layer, so many cells fit into the top part of the leaf where light is brightest.
- It is **close to the upper surface**, just below the transparent cuticle and upper epidermis, so it gets plenty of sunlight.
- It is close to **air spaces** (supplying carbon dioxide by diffusion) and to the **veins** (supplying water, which moves to the chloroplasts by osmosis).

Identifying Leaf Structures in Photomicrographs

You need to identify leaf structures in diagrams and in photomicrographs of a dicotyledonous leaf, such as Figure B5.10 (part of a tea plant leaf, ×400). Tips for recognising each layer:

- **Upper epidermis** — the single row of cells forming the top surface; the cuticle is the thin line on top of it.
- **Palisade mesophyll** — the row of tall, narrow cells packed tightly side by side just below the upper epidermis; they look darker because they are full of chloroplasts.
- **Spongy mesophyll** — rounded, irregular cells with many spaces between them, in the lower part of the leaf.
- **Vascular bundle** — a cluster of cells (xylem and phloem), often near the middle of the leaf.
- **Lower epidermis** — the row of cells on the bottom surface. Look for a gap (a **stoma**) between two small **guard cells**, usually with a large air space just above it.

Summary: How the Leaf Is Adapted for Photosynthesis

Feature of the leaf	Why it is an adaptation for photosynthesis

Large surface area	Captures more sunlight; more carbon dioxide can diffuse in
Thin	Light reaches all the cells; short diffusion distance for carbon dioxide
Transparent cuticle and upper epidermis	Let light through to the palisade layer
Palisade cells near the top, packed with chloroplasts	Absorb as much light energy as possible
Spongy mesophyll with air spaces	Carbon dioxide can diffuse to all the photosynthesising cells
Stomata with guard cells	Let carbon dioxide in and oxygen out; can be closed to reduce water loss
Network of veins (xylem and phloem)	Xylem supplies water; phloem carries away the sucrose made
Chlorophyll spread out on membranes in chloroplasts	A lot of sunlight can reach it

B5.03 Factors Affecting Photosynthesis

What Affects the Rate of Photosynthesis?

The **rate** of photosynthesis is how fast it happens — for example, how much oxygen is given off in unit time. The rate is affected by several **environmental factors**:

- **The supply of the raw materials** — carbon dioxide and water
- **The quantity of sunlight**, which provides energy for the reactions
- **The temperature**, because this affects the activity of enzymes

As well as these environmental factors, the **quantity of chlorophyll** in the leaf also helps to determine how fast photosynthesis can take place.

Factor	Effect of increasing it	Reason
Light intensity	Rate increases, until another factor limits it	More energy is available for the reactions
Carbon dioxide concentration	Rate increases, until another factor limits it	More of a raw material is available
Temperature	Rate increases up to an optimum, then falls quickly	Enzymes work faster as temperature rises, but are denatured at high temperatures
Quantity of chlorophyll	More chlorophyll allows a faster rate	More light energy can be absorbed

Limiting Factors

Suppose a plant has plenty of light and a warm temperature, but very little carbon dioxide. Making the light brighter will not make it photosynthesise any faster, because the shortage of carbon dioxide is holding the rate back. The factor that is in shortest supply, and is holding back the rate, is called the **limiting factor**.

- On a graph of rate against a factor, the line **rises** while that factor is limiting, then **levels off (plateaus)** when some other factor becomes limiting.

- This explains why glasshouse growers add extra heat, light or carbon dioxide (Project B5.01), and why the students in Practice Question 6 put their syringes in **bright sunshine**: so that light would not limit the rate and they could see the effect of carbon dioxide.

Experimental Skills B5.01: Testing a Leaf for Starch

Aim: To find out whether a leaf contains starch. Leaves turn some of the glucose that they make by photosynthesis into starch, so if we find starch in a leaf, that tells us it has been photosynthesising. You also practise working safely, making careful observations and drawing conclusions.

You will need: a 250 cm^3 beaker; a burner, tripod and gauze; a boiling tube; some iodine solution, with a dropper; forceps; a glass rod; some alcohol (methylated spirits) — collect this only when you are ready to use it; a fresh green leaf that has been photosynthesising; a white tile.

Safety:

- Alcohol is flammable (burns easily).
- Do not collect the alcohol until you have turned off your burner.
- If your alcohol (or someone else's) does start to burn, put a damp cloth over it immediately.

Method (Figure B5.11):

1. Boil some water in a beaker. Take a green leaf from a healthy plant and drop it into the boiling water. This breaks down the cell membranes in the leaf. Leave the leaf in the boiling water for about 30 seconds.
2. Turn off the flame.
3. Put some alcohol into a boiling tube. Stand the tube of alcohol in the hot water.
4. Use the forceps to remove the softened leaf from the hot water and drop it into the tube of alcohol. Chlorophyll is soluble in alcohol, so it comes out of the leaf. Leave the leaf in the alcohol until all the chlorophyll has come out.
5. The leaf will now be brittle (it will break easily). Carefully remove it from the alcohol, and dip it into hot water again to soften it.
6. Spread out the leaf on a white tile and cover it with iodine solution. A blue-black colour shows that the leaf contains starch.

Step	Why it is done
Boil the leaf in water	Breaks down the cell membranes so that iodine solution can reach the starch inside the cells; it also kills the cells
Turn off the flame before using alcohol	Alcohol is flammable and must never be near a naked flame
Stand the tube of alcohol in hot water	Warms the alcohol safely so it dissolves the chlorophyll faster
Leave the leaf in alcohol	Removes the green chlorophyll, so the leaf goes pale and colour changes are easy to see
Dip the leaf in hot water again	Softens the brittle leaf so it can be spread out without breaking
Use a white tile	Makes the colour of the leaf easy to see
Add iodine solution	Turns blue-black where starch is present; stays orange-brown where there is none

Variables: Nothing is varied — this is a standard test that you will use in Experimental skills B5.02, B5.03 and B5.04.

Expected results: A leaf from a plant that has been photosynthesising turns **blue-black**. Any part without starch stays the **orange-brown** colour of the iodine solution.

Conclusion: A blue-black leaf contains starch, so it has been photosynthesising.

Q: (1) Describe the results of your experiment. What conclusion can you make?

A: When iodine solution was added, the pale leaf turned blue-black. (Iodine solution itself is orange-brown, and it would have stayed orange-brown if there was no starch.) Conclusion: the leaf contained starch, so it had been photosynthesising.

Q: (2) Cell membranes are partially permeable. They will not allow starch or iodine molecules to pass through them. Use this information to explain why it was necessary to boil the leaf.

A: The starch is inside the cells, surrounded by cell membranes. Iodine molecules cannot pass through intact cell membranes, so the iodine solution could not reach the starch and no colour change would be seen. Boiling the leaf breaks down the cell membranes, so the iodine solution can get into the cells and react with the starch.

Q: (3) Suggest why it is a good idea to remove the chlorophyll from the leaf, before adding iodine solution to it.

A: The dark green colour of chlorophyll would hide (mask) the colour of the iodine solution, so it would be difficult to tell whether the leaf had turned blue-black. Removing the chlorophyll leaves the leaf pale, so the colour change is easy to see.

Experimental Skills B5.02: Investigating the Necessity for Chlorophyll for Photosynthesis

Aim: To find out whether chlorophyll is needed for photosynthesis, using the starch test on a **variegated leaf** — a leaf that has some white areas and some green areas (Figure B5.12). You also practise recording observations with a labelled drawing.

You will need: the same apparatus as for Experimental skills B5.01 Testing a leaf for starch; a fresh variegated leaf, taken from a plant that has been photosynthesising.

Safety: Alcohol is flammable (burns easily). Do not collect the alcohol until you have turned off your burner. If your alcohol (or someone else's) does start to burn, put a damp cloth over it immediately.

Method:

1. Make a drawing of your variegated leaf. Do not colour it. Label the parts that are green and the parts that are white.
2. Test the leaf for starch, following the procedure described in Experimental skills B5.01.
3. Record your results by drawing and labelling the leaf after you have added iodine solution to it.

Variable or feature	In this investigation
Independent variable	Whether chlorophyll is present (green areas) or absent (white areas)
Dependent variable	Whether starch is present (colour with iodine solution)
Controlled variables	Both areas are on the same leaf, so they had the same light, carbon dioxide, water, temperature and time
Control	The green parts of the leaf

Expected results: The areas that were green turn **blue-black**; the areas that were white stay **orange-brown**. The pattern of blue-black on the second drawing matches the pattern of green on the first drawing.

Conclusion: Only the parts of the leaf that contain chlorophyll made starch, so chlorophyll is necessary for photosynthesis.

Q: (1) Which parts of the leaf contained starch?

A: Only the parts that were green (containing chlorophyll) — they turned blue-black. The white parts stayed orange-brown, so they contained no starch.

Q: (2) Write a brief conclusion for your experiment.

A: Starch was only made in the green parts of the leaf, which contain chlorophyll. The white parts, which have no chlorophyll, did not make starch. This shows that chlorophyll is necessary for photosynthesis.

Experimental Skills B5.03: Investigating the Necessity for Light for Photosynthesis

Aim: To use the starch test to compare the starch content of parts of a leaf that have received light with parts that have not. You also practise recording results with a labelled drawing and writing a short conclusion.

You will need: a healthy plant growing in a pot, which has been in a dark cupboard for two days; some black paper, scissors and paper clips; (after one day) the same apparatus as for Experimental skills B5.01 Testing a leaf for starch.

Safety: Alcohol is flammable (burns easily). Do not collect the alcohol until you have turned off your burner. If your alcohol (or someone else's) does start to burn, put a damp cloth over it immediately.

Method:

1. Cut out a shape from the black paper, which you can attach to a leaf. If you like, you can cut your initials into the paper (Figure B5.13).
2. Attach the paper to a leaf on the potted plant. Do not take the leaf off the plant!
3. Put the plant into a place where it gets plenty of light. Leave it until the next day.
4. Remove the leaf from the plant. Remove the black paper from the leaf.
5. Test the leaf for starch.
6. Record your results by drawing and labelling the leaf after you have added iodine solution to it.

Destarching: Before the experiment, the plant was kept in a dark cupboard for two days. With no light it could not photosynthesise, so it used up its starch stores. **Destarching** means leaving a plant in the dark for long enough for it to use up its starch stores.

Variable or feature	In this investigation
Independent variable	Whether the part of the leaf received light (uncovered) or not (covered by black paper)
Dependent variable	Whether starch is present (colour with iodine solution)
Controlled variables	Same leaf on the same plant, so the same chlorophyll, carbon dioxide, water, temperature and time
Control	The part of the leaf that was not covered

Expected results: The part covered by black paper stays **orange-brown** (no starch). The uncovered part turns **blue-black** (starch). If you cut your initials into the paper, light shone through the holes, so your initials show up blue-black on an orange-brown background.

Conclusion: Only the parts of the leaf that received light made starch, so light is necessary for photosynthesis.

Why only part of the leaf is covered: If you had covered all of the leaf and found that none of it had photosynthesised, you would not be sure that the reason was because it had no light. It could have been because the whole plant was not photosynthesising.

Q: (1) The potted plant was put into a cupboard to make sure it used up all its starch stores and had no light to be able to make more. This is called destarching. Suggest why this was a necessary part of this experiment.

A: Destarching makes sure there is no starch in the leaf at the start. Then any starch found at the end must have been made by photosynthesis during the experiment, in the parts that received light. If the leaf still contained old starch, the covered part might test positive even though it had not photosynthesised, and you could not tell whether light is needed.

Q: (2) Explain why, in this experiment, the leaf had to remain attached to the plant for a day, after you put the black paper onto it.

A: The leaf needed time in the light to photosynthesise and make enough starch to be detected by the iodine test. It had to stay attached to the plant so that it stayed alive and healthy and kept receiving water (through the xylem from the roots), which is a raw material for photosynthesis. A leaf removed from the plant would wilt and would not photosynthesise normally.

Q: (3) Use your results to write a short conclusion to your experiment.

A: The part of the leaf covered by black paper did not contain starch (it stayed orange-brown), but the uncovered part did (it turned blue-black). This shows that light is necessary for photosynthesis.

Controls

A **control** is a standard sample that you use as a comparison, to find the effect of changing a variable. The control is the same as the test sample in every way except for the one factor being investigated.

- In Experimental skills B5.02, you wanted to find out whether chlorophyll is needed for photosynthesis. The white parts of the leaf had no chlorophyll, and you could compare them with the normal, green parts. The **green parts were the control**.
- In Experimental skills B5.03, you wanted to find out whether light is needed. You needed to compare a part of a leaf that did have light with a part that did not. The **part of the leaf that you did not cover was the control**. It allowed you to compare the part that did not have light with this normal part, which did.

Experimental Skills B5.04: Investigating the Necessity for Carbon Dioxide for Photosynthesis

Aim: To compare the ability of two leaves on the same plant — one with carbon dioxide and one without — to make starch. This is not an easy experiment to set up, so you will probably use one set of apparatus for the whole class.

You will need: a healthy plant growing in a pot, which has been destarched (see Experimental skills B5.03); two transparent containers, e.g. conical flasks; two rubber bungs to fit the flasks, each with a slit in one edge in which a leaf stalk can fit; petroleum jelly; a method of supporting the transparent containers; some potassium hydroxide solution; some distilled water; (after one day) the same apparatus as for Experimental skills B5.01 Testing a leaf for starch.

Safety: Alcohol is flammable (burns easily). Do not collect the alcohol until you have turned off your burner. If your alcohol (or someone else's) does start to burn, put a damp cloth over it immediately. Potassium hydroxide solution is also corrosive, so wear eye protection and avoid skin contact.

Method (Figure B5.14):

1. Set up the apparatus as shown in Figure B5.14. Find a way of supporting the flasks so that they do not pull on the leaves — you could use a retort stand and clamps. Put potassium hydroxide solution into one flask and distilled water into the other. Hold one of the leaf stalks carefully, slot it into the slit

in one of the rubber bungs, then gently push the flask over the bung and into its support. Repeat with the other leaf and flask.
2. Rub petroleum jelly over the rubber bungs, to try to prevent any air entering or leaving the flasks (an airtight seal).
3. Place the apparatus in a light place and leave it for at least one day.
4. Test both leaves for starch.
5. Record your results.

Figure B5.14 shows a potted plant with two of its leaves (still attached) each sealed inside a flask: one flask contains potassium hydroxide solution to absorb carbon dioxide, and the other contains distilled water.

Variable or feature	In this investigation
Independent variable	Whether carbon dioxide is present (distilled water flask) or absent (potassium hydroxide flask)
Dependent variable	Whether starch is present in each leaf (colour with iodine solution)
Controlled variables	Same plant, same light, same temperature, same time; both leaves sealed in a flask containing a liquid
Control	The leaf in the flask containing distilled water

Expected results: The leaf in the flask with potassium hydroxide solution stays **orange-brown** (no starch). The leaf in the flask with distilled water turns **blue-black** (starch).

Conclusion: The leaf without carbon dioxide could not make starch, so carbon dioxide is necessary for photosynthesis.

Q: (1) What was the control in this experiment?

A: The leaf in the flask containing distilled water. It had carbon dioxide in the air around it, but was otherwise treated in exactly the same way as the leaf in the potassium hydroxide flask.

Q: (2) Suggest why distilled water was added to one flask, rather than just leaving it empty.

A: So that the two flasks were the same in every way except for the presence of carbon dioxide. Potassium hydroxide solution is mostly water, so it makes the air in its flask humid. Putting the same volume of distilled water (which does not absorb carbon dioxide) in the other flask keeps the conditions, such as humidity, the same. Then any difference between the two leaves can only be caused by the carbon dioxide.

Q: (3) Write a short conclusion for your experiment.

A: The leaf in the flask with potassium hydroxide solution (no carbon dioxide) did not contain starch, while the leaf in the flask with distilled water (carbon dioxide present) did contain starch. This shows that carbon dioxide is necessary for photosynthesis.

Comparing the Three "Necessity" Investigations

Investigation	Factor removed	How it is removed	Control	Result in the test part	Conclusion
B5.02	Chlorophyll	Variegated leaf: the white areas have none	Green areas	No starch in the white areas	Chlorophyll is needed
B5.03	Light	Black paper covers part of	Uncovered part of the leaf	No starch under the paper	Light is needed

		the leaf			
B5.04	Carbon dioxide	Potassium hydroxide solution absorbs it	Leaf in the flask of distilled water	No starch in the leaf with potassium hydroxide	Carbon dioxide is needed

In B5.03 and B5.04 the plant is **destarched** first, so that any starch found must have been made during the experiment.

Experimental Skills B5.05: Investigating the Effect of Varying Light Intensity on the Rate of Photosynthesis

Aim: To find out how light intensity affects the rate of photosynthesis, using an **aquatic plant** (a plant that lives in water). An aquatic plant makes it easy to measure the rate, because you can see **bubbles of oxygen** being released into the water. To measure a rate, you need to measure how much oxygen is given off in **unit time** — a definite length of time, such as one minute.

You will need: a healthy aquatic plant, which has been in the light and has been photosynthesising; the apparatus shown in Figure B5.15; a paperclip; a lamp; a ruler to measure the distance between the lamp and the plant; a stopwatch or timer.

Safety: Keep water away from the lamp.

Before you start: Check that you know how to use your stopwatch or timer, and that it is working properly.

Method:

1. Collect a piece of aquatic plant. Cut off a piece about 7 to 9 cm long. Work out which way up it is. Attach a paperclip to the top of it.
2. Set up the apparatus as shown in Figure B5.15. The paperclip will hold the top of the plant down, so that its cut end floats upwards.
3. Place the lamp as close as possible to the beaker. Measure the distance, and record it in a results table.
4. Start the stopwatch or timer. Count the number of bubbles released from the cut end of the stem in one minute. Record this, and then repeat two more times.
5. Move the lamp a little bit further from the plant. Measure the new distance, and repeat step 4.
6. Repeat step 5 several times, until you have bubble counts for at least five different distances. Do more if you have time.
7. Plot your results as a line graph, with distance on the x-axis and mean number of bubbles on the y-axis. Make sure that the two axes are labelled fully, and that you choose suitable scales. Join the points using ruled, straight lines between the points.

Figure B5.15 shows a lamp on a stand at one end of a metre rule, a transparent heat shield, and a beaker of water containing the cut stem of the aquatic plant, with bubbles containing oxygen produced in photosynthesis rising from the cut end.

The results table in the book has these columns (the last four are all under the heading "Number of bubbles produced in one minute"):

Distance of lamp from beaker / cm	1st try	2nd try	3rd try	Mean

To work out each mean, add the three counts and divide by 3. For example, counts of 24, 26 and 25 bubbles give a mean of (24 + 26 + 25) ÷ 3 = 25 bubbles per minute.

Variable or feature	In this investigation

Independent variable	Light intensity, changed by moving the lamp to different distances from the beaker
Dependent variable	Rate of photosynthesis, measured as the number of bubbles released from the cut stem in one minute
Controlled variables	Temperature (kept steady by the heat shield); carbon dioxide concentration in the water; the same piece of plant; the same lamp; the same counting time; other light in the room kept low
Repeats	Three counts at each distance, then a mean is calculated

Expected results: The closer the lamp, the more bubbles are produced per minute. As the lamp is moved further away, the number of bubbles per minute decreases, so the line on the graph slopes downwards from left to right.

Conclusion: Light intensity decreases as the lamp is moved further away, so the results show that **the higher the light intensity, the faster the rate of photosynthesis**.

Q: (1) Explain why an aquatic plant was used in this experiment.

A: An aquatic plant releases the oxygen made in photosynthesis into the water as visible bubbles, which can be counted (or collected). This makes it easy to measure how much oxygen is given off in unit time, which is the rate of photosynthesis. A land plant releases oxygen into the air, where it cannot be seen or easily measured.

Q: (2) Light intensity decreases as the distance of the lamp from the beaker increases. Write a conclusion to summarise what your results show about the effect of light intensity on the rate of photosynthesis.

A: As the lamp was moved further from the beaker, the mean number of bubbles produced per minute decreased. Because light intensity decreases with distance, this shows that the higher the light intensity, the faster the rate of photosynthesis. Light provides the energy for photosynthesis, so more light energy allows the reaction to happen faster.

Q: (3) Lamps produce heat as well as light. Explain why you would be unable to draw a conclusion about light intensity, if you did not use a heat shield.

A: Without a heat shield, the lamp would warm the water more when it was close to the beaker than when it was far away. Temperature also affects the rate of photosynthesis, because it changes the activity of enzymes. So light intensity and temperature would both be changing at the same time, and you could not tell whether a change in the number of bubbles was caused by the light or by the heat. The transparent heat shield lets light through but stops most of the heat reaching the plant, so light intensity is the only variable that changes.

Q: (4) It is possible to collect the gas given off by the plant and measure its volume. Suggest why this would be an improvement on the method you have used here.

A: Bubbles are not all the same size, so counting them does not measure the amount of oxygen accurately. Bubbles may also come off too fast to count, and very small bubbles may be missed. Collecting the gas (for example in an upturned measuring cylinder, a gas syringe or a capillary tube) and measuring its volume in cm^3 gives a more accurate measurement of how much gas is produced per minute.

Self Assessment: Drawing a Good Line Graph

After drawing your graph, give yourself a mark for each feature below: **2 marks** if you did it really well; **1 mark** if you made a good attempt at it and partly succeeded; **0 marks** if you did not try to do it, or did not succeed.

- The axes are both fully labelled, including units.
- The scales on both axes go up in even steps.

- The scales use at least half of the graph paper.
- The points are plotted as small, neat crosses in exactly the right place.
- The lines between the points are drawn with a ruler, and exactly meet the centres of the crosses.

How many marks do you give yourself out of 10? What will you do next time you draw a line graph, to increase your score?

Experimental Skills B5.06: Investigating the Effect of Temperature and Carbon Dioxide on the Rate of Photosynthesis

Aim: To plan and carry out your own experiment, using a method similar to Experimental skills B5.05, but with **either carbon dioxide concentration or temperature** as the independent variable.

You will need: the apparatus used in Experimental skills B5.05; electric water-baths, or access to ice and hot water, and a thermometer.

Method:

1. Think about how you will vary temperature. You can stand the beaker in a water-bath at different temperatures, or add ice and hot water to the water around the plant. Aim to have at least five values of your independent variable.
2. Think about other variables that you should try to keep constant, and how you will do this.
3. Write out your plan. You could structure it like this: the independent variable and how you will vary it; the values of the independent variable you will use; the variables that you will keep constant, and how you will do this; the variable that you will measure, and how you will do this; possible safety risks, and how you will minimise them; an outline results chart, ready to complete as you work.
4. If possible, exchange your plan with another person in your class who is investigating the other variable. Give them feedback on their plan, and use their feedback to make improvements to your own plan.
5. When you are ready, carry out your experiment and record your results.

An example plan for each version:

Plan heading	Temperature investigation	Carbon dioxide investigation
Independent variable and how to vary it	Temperature: stand the beaker in electric water-baths at different temperatures, or add ice or hot water to the water around the plant	Carbon dioxide concentration: dissolve different amounts of sodium hydrogencarbonate in the water
Values to use (at least five)	For example 10, 20, 30, 40 and 50 °C	For example 0, 0.1, 0.2, 0.3 and 0.4% sodium hydrogencarbonate solution
Variables kept constant, and how	Light intensity (same lamp at a fixed distance, with a heat shield); carbon dioxide (same concentration of sodium hydrogencarbonate); same piece of plant; same counting time	Temperature (water-bath, checked with a thermometer); light intensity (same lamp at a fixed distance); same piece of plant; same counting time
Variable measured, and how	Number of bubbles released in one minute, counted three times to find a mean (or volume of gas collected per minute)	Same as for temperature
Safety risks, and how to minimise them	Hot water can scald, so handle it carefully; keep water away from the lamp and electrical leads; take care with glassware	Keep water away from the lamp; wipe up spills; take care with glassware

Outline results chart	Temperature / °C; number of bubbles in one minute (1st, 2nd and 3rd try); mean	Concentration of sodium hydrogencarbonate / %; number of bubbles in one minute (1st, 2nd and 3rd try); mean

Expected results:

- **Temperature:** the rate increases as the temperature rises, up to an optimum temperature, then falls quickly at higher temperatures because the enzymes that catalyse photosynthesis are denatured.
- **Carbon dioxide:** the rate increases as the carbon dioxide concentration increases, then levels off when another factor (such as light intensity) becomes the limiting factor.

Q: (1) Plot a line graph to display your results.

A: Put the independent variable on the x-axis, labelled "Temperature / °C" or "Concentration of sodium hydrogencarbonate solution / %". Put "Mean number of bubbles per minute" on the y-axis. Use even scales that fill at least half of the grid, plot each mean as a small, neat cross, and join the crosses with ruled straight lines. For temperature, the line should rise to a peak at the optimum and then fall steeply. For carbon dioxide, the line should rise and then level off.

Q: (2) Write a short conclusion.

A: Temperature: "As the temperature increased, the rate of photosynthesis increased up to an optimum temperature (read the exact value from your graph), and then decreased. This is because the enzymes that catalyse photosynthesis work faster as temperature rises, but are denatured at high temperatures." Carbon dioxide: "As the carbon dioxide concentration increased, the rate of photosynthesis increased, because more of this raw material was available. At the highest concentrations the rate levelled off, because another factor, such as light intensity, became limiting."

Q: (3) You probably encountered some difficulties or problems as you did your experiment. What were they, and how did you overcome them?

A: Typical problems and solutions:

- The water temperature drifted during counting — use a thermostatically controlled electric water-bath and check the temperature with a thermometer before and after each count.
- Bubbles came off too fast, unevenly or in different sizes — count for longer, or collect the gas and measure its volume.
- The plant took time to adjust after each change — wait a few minutes at each new temperature or concentration before starting to count.
- The lamp warmed the water — use a heat shield and keep the lamp at a fixed distance.
- Different pieces of plant gave different results — use the same piece of plant throughout.

Q: (4) If you could start again, and had lots of time available, what improvements would you make to your experiment?

A: For example: measure the volume of gas produced instead of counting bubbles (or use an oxygen sensor and data logger); test more values over a wider range, with smaller steps near the optimum; repeat each measurement at least three times, calculate means and repeat any anomalous results; control all the other variables more strictly; and give the plant a few minutes to adjust to each new condition before measuring.

Project B5.01: Designing a Glasshouse

Aim: To design a glasshouse to grow a crop in your country. The glasshouse will provide ideal conditions to help the plants photosynthesise quickly and grow fast, so that they give a **high yield** for the grower.

Method: Work in a group of three or four. You might like to assign different roles to different people (e.g. organiser, researcher, communicator). Research the questions below, then give a presentation to the

rest of the class. You can make your presentation as a video, use PowerPoint, or make models and draw diagrams — or all of these.

Variables to control: temperature, light intensity, carbon dioxide concentration, water supply and mineral ions — each should be kept at a level where it does not limit the rate of photosynthesis, without costing more than the extra crop is worth.

Safety considerations: Burners used to heat the glasshouse and supply carbon dioxide must be well maintained and ventilated, because faulty burners can produce poisonous carbon monoxide. Keep lamps, heaters and electrical cables away from water sprays and irrigation.

Q: What crop will you grow in the glasshouse?

A: Choose a valuable crop that grows well under cover, for example tomatoes, cucumbers, peppers, lettuce, strawberries or flowers.

Q: What conditions does this crop need to grow well?

A: Plenty of light; warm (but not too hot) temperatures — about 20–25 °C for tomatoes during the day; enough carbon dioxide; a steady supply of water; mineral ions such as nitrate ions (for proteins) and magnesium ions (for chlorophyll); and protection from pests and diseases.

Q: How can a glasshouse provide better conditions for the crop, than if it was grown outside?

A: Glass lets light in and traps heat, so it is warmer inside. Temperature, light, carbon dioxide, water and mineral ions can all be controlled, so that none of them limits the rate of photosynthesis. The crop is also protected from wind, frost, heavy rain and many pests, and it can be grown out of season or all year round.

Q: How will you control temperature in the glasshouse? In your country, will you need to heat it, cool it, or do both at different times?

A: To heat it: use electric heaters or paraffin or gas burners (which also release carbon dioxide), and insulate it at night. To cool it: open vents and windows, use fans, shade the glass, or spray a fine water mist. A thermostat can switch these on and off to keep the temperature near the crop's optimum. A cold country mainly needs heating, a hot country mainly needs cooling, and many countries need heating at night or in winter and cooling on hot days.

Q: How will you control light in the glasshouse? Will you need to give extra light, or might you need to shade it sometimes?

A: Lamps (such as LED grow lights) can raise the light intensity on dull days and lengthen the day in winter, so photosynthesis continues for longer. In very sunny, hot places, blinds, shade netting or whitewash on the glass can reduce the light so that the plants do not overheat or lose too much water.

Q: How will you provide carbon dioxide in the glasshouse? Providing it costs money, so what might be a suitable concentration to aim for, that will give a good return for your expenditure?

A: Burn paraffin, propane or natural gas (which supplies heat as well), or release carbon dioxide from cylinders. Normal air contains about 0.04% carbon dioxide. Raising it to about 0.1% (about two and a half times normal) greatly increases the rate of photosynthesis. Above about 0.1%, the rate increases very little because another factor, such as light or temperature, becomes limiting, so the extra cost would not be repaid by a bigger crop.

Expected outcome: A good design keeps every factor close to its best level, so the crop photosynthesises quickly, grows fast and gives a high yield that is worth more than the cost of heating, lighting and carbon dioxide. Remember water (irrigation or drip-feeding) and fertilisers containing nitrate and magnesium ions too.

B5 Key Formulas and Concepts

1. Word equation for photosynthesis: carbon dioxide + water → glucose + oxygen (in the presence of sunlight and chlorophyll)

2. Balanced chemical equation: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
3. The equation balances: 6 carbon, 12 hydrogen and 18 oxygen atoms on each side
4. 6 molecules of CO_2 and 6 of H_2O make 1 molecule of glucose and 6 of O_2, so number of glucose molecules = number of CO_2 molecules ÷ 6 (e.g. 12 ÷ 6 = 2)
5. Chlorophyll absorbs light energy and transfers it into chemical energy stored in glucose
6. Glucose + nitrogen (from nitrate ions) → amino acids → proteins
7. Glucose + nitrogen (from nitrate ions) + magnesium (from magnesium ions) → chlorophyll
8. Glucose is used in respiration (energy) and converted to starch (storage), sucrose (transport in phloem), cellulose (cell walls) and nectar (attracting pollinators)
9. Carbon dioxide route: atmosphere → stomata → air spaces → palisade cells (by diffusion)
10. Water route: soil → roots → xylem vessels in the veins → leaf cells (by osmosis)
11. Starch test: boil in water, then warm alcohol (water bath, no flame), then soften in hot water, then add iodine solution: blue-black = starch, orange-brown = no starch
12. Rate of photosynthesis = quantity of oxygen produced ÷ time (e.g. bubbles per minute, or cm^3 of gas per minute)
13. Mean = (1st try + 2nd try + 3rd try) ÷ 3
14. Light intensity decreases as the lamp is moved further away (useful extension: light intensity is proportional to 1 ÷ distance²)
15. Air contains about 0.04% carbon dioxide; glasshouse growers often raise it to about 0.1%
16. Rate rises with light intensity and carbon dioxide concentration until another factor limits it; rate rises with temperature up to an optimum, then falls as enzymes are denatured

B5 Textbook Questions with Answers

Question B5.01

Explain why animals are dependent on plants for their supply of energy.

Answer: Animals cannot make organic substances (carbohydrates, fats and proteins) from inorganic substances, so they must take them in as food. Plants can make organic substances by photosynthesis: chlorophyll captures light energy from the Sun and transfers it into chemical energy stored in glucose and in the other substances made from it. Animals get this energy by eating plants, or by eating other animals that have eaten plants. So all the energy in an animal's food originally came from sunlight captured by plants — without plants, animals would have no supply of energy.

Question B5.02

Explain how chlorophyll helps photosynthesis to happen.

Answer: Chlorophyll is a green pigment found in chloroplasts. It captures (absorbs) energy from sunlight and immediately transfers this energy to water molecules and carbon dioxide molecules. This energy makes the carbon dioxide and water react together to produce glucose (and oxygen). In this way, chlorophyll transfers energy from light into chemical energy in carbohydrates. Without chlorophyll, light shining on carbon dioxide and water would never produce glucose.

Question B5.03

How many glucose molecules can be made from 12 molecules of carbon dioxide and 12 molecules of water?

Answer: 2 glucose molecules.

Working: the balanced equation is $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$, so 6 molecules of carbon dioxide and 6 molecules of water make 1 molecule of glucose. 12 ÷ 6 = 2, so 12 molecules of carbon dioxide and 12 molecules of water make 2 molecules of glucose (and 12 molecules of oxygen). Check using

carbon atoms: 12 CO_2 molecules contain 12 carbon atoms, and each glucose molecule needs 6, so 12 ÷ 6 = 2.

Question B5.04

Starch can easily be broken down to glucose by the enzyme amylase, which is found in plants. Cellulose is much more difficult to break down to glucose. Suggest how these differences between starch and cellulose relate to their functions in a plant.

Answer: Starch is the plant's energy store. When the plant needs energy (for example at night, or when a seed germinates), it must be able to turn the stored starch back into glucose quickly for respiration, so it is useful that amylase can break starch down easily. Cellulose forms the cell walls, which give each cell strength and support and must last for the whole life of the cell. If cellulose were easy to break down (for example by the plant's own amylase), the cell walls would be weakened or destroyed. Because cellulose is difficult to break down, the walls stay strong and permanent.

Question B5.05

Animals do not make or store starch. What is the substance that animals store, which is made of chains of glucose molecules?

Answer: Glycogen (stored mainly in the liver and muscles).

Question B5.06

Explain why some parts of a plant must have sucrose delivered to them.

Answer: Glucose can only be made by photosynthesis in the parts of a plant that contain chlorophyll, which usually means the leaves. Other parts, such as the roots, flowers, fruits, seeds, storage organs and the inner parts of the stem, have no chlorophyll (or get no light), so they cannot make their own glucose. Their cells still need carbohydrate: for respiration to release energy, to make cellulose for new cell walls as they grow, and to make starch for storage. Plants do not transport glucose — they change it into sucrose, which is carried in the phloem to these parts and changed back into glucose when it arrives.

Question B5.07

Copy and complete this table.

Element	……	magnesium
Mineral salt	nitrate ions	magnesium ions
Why needed	to make ……, and then proteins	……
Deficiency	weak growth, yellow leaves	yellowing of leaves, often especially between the veins

Answer:

Element	nitrogen	magnesium
Mineral salt	nitrate ions	magnesium ions
Why needed	to make amino acids, and then proteins	to make chlorophyll
Deficiency	weak growth, yellow leaves	yellowing of leaves, often especially between the veins

The three missing entries are **nitrogen**, **amino acids** and **to make chlorophyll**.

Question B5.08

Making nectar costs a plant energy because it uses up glucose that the plant has made. Explain why the expense is worthwhile.

Answer: Nectar attracts pollinators — insects, bats or birds — to the flowers. The male gametes of a flower are inside pollen grains and cannot move themselves, so the plant relies on these animals to carry pollen from one flower to another. Animals will only do this if they get a reward, and nectar is that reward. Pollination allows fertilisation to happen, so the plant can reproduce sexually and produce seeds that grow into new plants. Without nectar, far fewer flowers would be pollinated, so the glucose used is a worthwhile price for successful reproduction.

Question B5.09

Figure B5.10 shows a photomicrograph (×400) of a small part of a leaf from a tea plant. Identify structures A–D.

Description of Figure B5.10: label A points to the row of cells forming the top surface of the leaf; label B points to the layer of tall, narrow cells packed side by side just below it; label C points to the region of rounded, loosely arranged cells with spaces between them in the lower part of the leaf; label D points up to a gap in the bottom row of cells, with a large space just above it.

Answer:

A — upper epidermis (covered on the outside by the thin cuticle)

B — palisade mesophyll

C — spongy mesophyll

D — a stoma (the opening between two guard cells in the lower epidermis)

Question B5.10

Explain how a palisade mesophyll cell is adapted for its function.

Answer: The function of a palisade mesophyll cell is photosynthesis. It is adapted in these ways:

- It contains very large numbers of chloroplasts, which contain chlorophyll, so it can absorb as much light energy as possible.
- It is tall and narrow, and the cells are packed tightly side by side, so a large number of cells (and chloroplasts) fit into the top part of the leaf, where the light is brightest.
- It is close to the top of the leaf, just below the transparent cuticle and upper epidermis, so it receives plenty of sunlight.
- It is close to the air spaces of the spongy mesophyll, which supply carbon dioxide by diffusion, and to the veins, which supply water.

Question B5.11

Suggest why the cells in the lower epidermis of a leaf (apart from the guard cells) do not contain chloroplasts.

Answer: Very little light reaches the lower epidermis. Sunlight falls mainly on the upper surface of the leaf, and most of it is absorbed by the chloroplasts in the palisade and spongy mesophyll before it reaches the bottom layer. Chloroplasts in the lower epidermis would therefore carry out very little photosynthesis, so making them would waste the plant's materials and energy. The job of the lower epidermis is to cover and protect the leaf and to contain the stomata for gas exchange, not to photosynthesise.

B5 Practice Exam Questions with Answers

Question B5.P1

Identify the raw materials required for photosynthesis. (1 mark)

A) carbon dioxide and oxygen

B) oxygen and glucose

C) glucose and water

D) water and carbon dioxide

Answer: D — water and carbon dioxide. These are the reactants of photosynthesis. Oxygen and glucose are the products, so options A, B and C are wrong.

Question B5.P2

Identify which ion a plant requires to make amino acids from glucose. (1 mark)

A) calcium

B) iron

C) magnesium

D) nitrate

Answer: D — nitrate. Amino acids contain nitrogen, which glucose does not. Plants absorb nitrogen from the soil as nitrate ions and combine it with glucose to make amino acids. Magnesium ions are needed to make chlorophyll, not amino acids.

Question B5.P3

The diagram shows a three-dimensional drawing of a section through a leaf. (9 marks)

Description of the diagram: a block cut from a leaf, with its layers labelled from top to bottom. A labels the thin layer covering the top surface; B labels the top layer of cells; C labels a layer of tall, narrow cells packed side by side; D labels a deep layer of rounded cells with spaces between them; E labels a bundle of tubes running through the leaf; F labels a space between the rounded cells; G labels a pore in the bottom surface; H labels one of the two cells surrounding that pore; I labels the bottom layer of cells.

a) i) Identify the tissue labelled C. [1]

a) ii) Outline how this tissue is adapted for its function. [2]

b) Give the letters of two cells or tissues that do **not** contain chloroplasts. [2]

c) Describe the function of the cell labelled H. [2]

d) Use the letters on the diagram to describe how water reaches a chloroplast in the leaf. [2]

Answer:

a) i) Palisade mesophyll. [1]

a) ii) Any two of the following [2]:

- The cells contain very large numbers of chloroplasts (containing chlorophyll) to absorb as much light energy as possible.
- The cells are tall, narrow and tightly packed side by side, so many cells fit into the top part of the leaf.
- The layer is close to the upper surface, just below the transparent epidermis, so it receives plenty of sunlight.
- The cells are close to air spaces (supplying carbon dioxide) and veins (supplying water).

b) Any two of: **B** (upper epidermis), **I** (lower epidermis, apart from its guard cells) and **E** (vascular bundle — xylem and phloem). [2] Do not choose A, F or G: they are not cells (the cuticle, an air space and a stoma).

c) H is a guard cell. It changes shape to open and close the stoma, G [1]. This controls the diffusion of carbon dioxide into the leaf and oxygen out of it, and controls how much water vapour is lost from the

leaf [1].

d) Water arrives in the leaf in the xylem vessels inside the vascular bundle, **E** [1]. It then leaves the xylem and moves from cell to cell by osmosis through the spongy mesophyll, **D**, to the palisade mesophyll cells, **C**, where it enters the chloroplasts [1].

Question B5.P4

The bar chart shows the mean number of chloroplasts in four different types of cell in a leaf. (7 marks)

Data read from the bar chart (y-axis: mean number of chloroplasts per cell, scale 0 to 100; x-axis: type of cell):

Type of cell	Mean number of chloroplasts per cell
A	0 (no bar)
B	about 15
C	about 71
Palisade mesophyll	about 100 (the bar reaches the top of the scale)

a) Explain why palisade mesophyll cells have so many chloroplasts. [2]

b) State which cell could be a cell in the upper epidermis. Explain your answer. [2]

c) Guard cells contain chloroplasts, but not as many as spongy mesophyll cells. Suggest which cell could be a guard cell. [1]

d) Chloroplasts contain chlorophyll. Describe the function of chlorophyll in photosynthesis. [2]

Answer:

a) Palisade mesophyll cells are the main site of photosynthesis in the leaf [1]. They are near the upper surface, where they receive the most light, so having many chloroplasts (and so lots of chlorophyll) lets them absorb as much light energy as possible and photosynthesise quickly, making more glucose [1].

b) Cell **A** [1]. Upper epidermis cells contain no chloroplasts (A has a mean of 0); they are transparent so that light passes through them to the palisade cells below [1].

c) Cell **B** — it has some chloroplasts (about 15), but far fewer than cell C (about 71), which must be a spongy mesophyll cell [1].

d) Chlorophyll absorbs (captures) energy from sunlight [1] and transfers it into chemical energy in carbohydrates — the energy makes carbon dioxide and water react to form glucose [1].

Question B5.P5

(8 marks)

a) Describe how plants synthesise carbohydrates. [4]

b) Outline **two** subsequent uses of the carbohydrates made in photosynthesis. [4]

Answer:

a) Any four of the following [4]:

- Plants make carbohydrates by **photosynthesis**, mainly in the palisade mesophyll cells of their leaves.
- Carbon dioxide diffuses into the leaf from the air through the stomata; water is absorbed from the soil by the roots and carried to the leaf in the xylem.
- In the chloroplasts, **chlorophyll** absorbs energy from sunlight.
- Chlorophyll transfers this energy to carbon dioxide and water, making them react; the reactions are catalysed by enzymes.
- The products are **glucose** (a carbohydrate) and **oxygen**, which is released.
- Equation: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

b) Any two uses, each with an explanation [2 + 2]:

- **Respiration** [1] — glucose is broken down to release energy, for example for active transport of mineral ions into root hairs, or for building proteins [1].
- **Converted to starch for storage** [1] — starch is insoluble, so it does not affect osmosis or take part in reactions, and it can be quickly turned back into glucose when needed [1].
- **Converted to sucrose for transport** [1] — sucrose is carried in the phloem to parts that cannot photosynthesise, such as the roots, where it is changed back into glucose [1].
- **Converted to cellulose** [1] — to build cell walls for the new cells made as the plant grows [1].
- **Used to make amino acids and then proteins** [1] — glucose is combined with nitrogen from nitrate ions; the proteins are needed for growth [1].
- **Used to make nectar** [1] — nectar attracts pollinators such as insects, so the flowers are pollinated [1].

Question B5.P6

Two students carried out an experiment to investigate the effect of carbon dioxide concentration on the rate of photosynthesis of an aquatic plant. The photograph shows some of the apparatus that they used. (15 marks)

Description of the photograph: several syringes are clamped upright, side by side, outdoors in bright sunshine, with a long thin transparent tube running down from the nozzle of each syringe.

- The students filled five syringes with five different concentrations of sodium hydrogencarbonate solution. This provides carbon dioxide.
- They placed a piece of aquatic plant in each syringe.
- They attached a long transparent tube to the nozzle of each syringe.
- They pushed the plunger of each syringe in a little way, to force some of the sodium hydrogencarbonate into the tubing.
- They then pulled each plunger out a little way, to introduce some air into the tubing.
- They marked the position of the meniscus on each piece of tubing, as shown in the diagram.

Description of the diagram: a short length of the transparent tubing, with water in the upper part and air in the lower part; the meniscus is the boundary between them, and a mark is drawn on the tubing level with the meniscus.

All five syringes were placed in bright sunshine. As the plants photosynthesised, they released oxygen. The oxygen gas collected above the liquid in the syringe. This pushed the liquid down the tube.

The students recorded the distance moved by the meniscus on each piece of apparatus after 30 minutes.

a) Identify the independent variable and the dependent variable in this investigation. [2]

b) Explain why the five syringes were placed in bright sunshine. [2]

The table shows the results of the experiment.

Percentage concentration of sodium hydrogencarbonate solution	0	0.1	0.2	0.3	0.4
Distance moved by meniscus after 30 minutes / mm	5	8	12	14	17

c) Draw a line graph to display these results. Join the points with ruled straight lines. [5]

d) Explain the results of this investigation. [2]

e) i) Suggest **two** possible sources of error in this investigation. [2]

e) ii) Suggest how the investigation could be modified to reduce these sources of error. [2]

Answer:

a) Independent variable: the concentration of the sodium hydrogencarbonate solution, which determines how much carbon dioxide is available [1]. Dependent variable: the distance moved by the

meniscus in 30 minutes, which shows how much oxygen was produced, and so the rate of photosynthesis [1].

b) Light is needed for photosynthesis, because it provides the energy [1]. Bright sunshine gives a high light intensity, so light is not a limiting factor: the rate is limited by the carbon dioxide concentration, and all five syringes receive the same high light intensity, making it a fair test [1].

c) The correct graph shows [5]:

- The concentration on the x-axis, labelled "Concentration of sodium hydrogencarbonate solution / %", and the distance on the y-axis, labelled "Distance moved by meniscus after 30 minutes / mm" [1].
- Even scales that use at least half of the grid, for example 0 to 0.4 in steps of 0.1 on the x-axis and 0 to 18 mm in steps of 2 mm on the y-axis [1].
- The five points plotted accurately as small neat crosses at (0, 5), (0.1, 8), (0.2, 12), (0.3, 14) and (0.4, 17) [2 for all five correct; 1 for three or four correct].
- The points joined with ruled straight lines from cross to cross, starting at the first point and ending at the last point, not extended to the origin [1].

The finished line rises steadily from left to right and does not level off.

d) As the concentration of sodium hydrogencarbonate increased from 0% to 0.4%, the distance moved by the meniscus increased from 5 mm to 17 mm [1]. Sodium hydrogencarbonate provides carbon dioxide, a raw material for photosynthesis, so a higher concentration allowed a faster rate of photosynthesis (carbon dioxide was the limiting factor). More oxygen was produced in 30 minutes, and this collected in the syringe and pushed the liquid further down the tube [1].

The line does not level off, so carbon dioxide was still limiting even at 0.4%. The meniscus moved 5 mm even at 0%, probably because the water contained some dissolved carbon dioxide (or because the gas warmed and expanded in the sunshine).

e) i) Any two of the following [2]:

- The temperature was not controlled: syringes in bright sunshine may warm up, which makes the gas expand (moving the meniscus) and also changes the rate of photosynthesis.
- The light intensity may not have been the same for every syringe, or constant for the whole 30 minutes (clouds, shadows, syringes shading each other).
- The pieces of aquatic plant may have been different sizes or had different numbers of leaves.
- There was only one syringe for each concentration, so there were no repeats to check the results or calculate a mean.
- Some of the oxygen may have dissolved in the solution or stayed stuck to the plant, so it did not move the meniscus.

e) ii) One matching improvement for each error given in part i) [2], for example:

- Stand the syringes in a thermostatically controlled water-bath, checked with a thermometer, to keep the temperature constant.
- Carry out the experiment indoors, with a lamp at the same fixed distance from every syringe (and a heat shield), instead of in sunshine.
- Use pieces of plant of the same length or mass, cut from the same plant.
- Set up three syringes for each concentration and calculate a mean distance.
- Measure the distance against a ruler marked in millimetres, reading at eye level.

Revision and Exam Preparation

Exam Focus Priorities

B1 Cells and Organisms

1. **Learn MRS GREN with definitions** — movement, respiration, sensitivity, growth, reproduction, excretion and nutrition. Questions often ask which feature is NOT a characteristic of all organisms: photosynthesis, breathing, a blood system and a nervous system are not.
2. **Respiration is not breathing** — respiration is a set of chemical reactions in every living cell that releases energy from glucose; breathing is moving air in and out of the lungs, which plants and bacteria do not do.
3. **Growth must be permanent** — define growth as a permanent increase in size and dry mass; the word permanent earns the mark.
4. **Know which structures are in which cells** — animal and plant cells both have a cell membrane, cytoplasm, a nucleus, ribosomes and mitochondria; only plant cells have a cellulose cell wall, a large permanent vacuole with cell sap and (often) chloroplasts and starch grains.
5. **Bacteria have no nucleus** — their DNA is a single circle in the cytoplasm, often with plasmids; their cell wall is not made of cellulose; they have no mitochondria or chloroplasts.
6. **Partially permeable versus fully permeable** — the cell membrane is partially permeable and controls what enters and leaves the cell; the cell wall is fully permeable, supports the cell and stops it bursting, but does not control what enters.
7. **Mitochondria release energy** — write that mitochondria release energy from glucose by aerobic respiration; never write that they make or produce energy.
8. **Ribosomes make proteins; DNA decides which proteins** — the DNA in the nucleus carries the instructions, and the ribosomes link amino acids together to build the protein.
9. **No chloroplasts does not mean animal cell** — many plant cells, such as root hair cells, have no chloroplasts; the cell wall is the reliable sign of a plant cell.
10. **Cell, tissue, organ, organ system, organism** — a tissue is a group of similar cells; an organ is made of different tissues; be ready to give an example at each level (stomach lining cell, stomach lining tissue, stomach, digestive system, human).
11. **Show the magnification equation and substitution** — write magnification = image size ÷ actual size, convert to the same units, substitute, then give the answer with a × sign and no units; each step can earn a mark.
12. **Divide to find the actual size** — actual size = image size ÷ magnification; multiplying instead of dividing is the most common error in multiple-choice questions.
13. **Convert mm and µm correctly** — multiply by 1000 to go from mm to µm, divide by 1000 to go from µm to mm, and check that your final size is sensible for the structure.
14. **Measure carefully and record the measurement** — measure in mm with a ruler along the length asked for, and write the measurement in your working.
15. **Use the marks as a guide** — a [6] question needs six separate creditworthy points, and a [2] definition usually needs two ideas.

B2 Movement into and out of Cells

1. **Learn the full definition of diffusion** — include "net movement", "from a region of higher concentration to a region of lower concentration", "down a concentration gradient" and "as a result of random movement"; each phrase can earn credit.

2. **Diffusion and osmosis are passive** — their energy comes from the kinetic energy of the particles; never say they use energy from respiration, because only active transport does.
3. **Osmosis is only about water** — it is the diffusion of water molecules through a partially permeable membrane; the solute does not move in osmosis.
4. **Use water potential correctly** — a dilute solution has a high water potential, a concentrated solution has a low water potential, and water moves from high to low water potential.
5. **Explain why plant cells do not burst** — the strong cell wall resists the swelling contents and presses back; use the words turgid and turgor pressure, and say animal cells burst because they have no cell wall.
6. **Do not mix up flaccid and plasmolysed** — a flaccid cell has lost its turgor pressure and is soft; a plasmolysed cell has lost so much water that its cell membrane has torn away from the cell wall.
7. **The cell wall is fully permeable** — it is the cell membrane, not the cell wall, that is partially permeable, so the space between the wall and the membrane of a plasmolysed cell fills with the outside solution.
8. **Active transport answers need three ideas** — movement against a concentration gradient (low to high), carrier proteins in the membrane that change shape, and energy from respiration in mitochondria.
9. **Quote data from graphs and bar charts** — give values with units and compare them (for example "potassium is 8 times more concentrated in the cells"), then use the data to decide which process is happening.
10. **Identify anomalous results properly** — pick the point that is furthest from the line of best fit and does not fit the trend, and say why you chose it.
11. **Plan experiments using variables** — name the independent, dependent and standardised variables, repeat readings and calculate a mean, and put units only in the column headings.
12. **Show working in surface area calculations** — use cm^2 for area and cm^3 for volume; cutting a cube into smaller cubes increases the total surface area but not the total volume.
13. **Common mistake: saying particles want to move** — particles move randomly; the net movement happens because more particles move away from the high concentration than move into it.

B3 Biological Molecules

1. **Know the elements in each molecule** — carbohydrates and fats contain only C, H and O; proteins contain C, H, O and N. Nitrogen is the element found in all proteins but not in carbohydrates or fats.
2. **Link each large molecule to its smaller building blocks** — starch and glycogen are made from glucose; fats and oils from fatty acids and glycerol; proteins from amino acids. A common mistake is to say that fats are made of amino acids, or that starch is made of "sugar" without naming glucose.
3. **Starch is found in plants, glycogen in animals** — glucose is found in both. Starch is the energy store in plant cells and glycogen is the energy store in animal cells.
4. **Only the Benedict's test needs heating** — always write "heat in a water-bath (about 80 °C)". Never heat the biuret test; this is the error in Practice Question 4a(i).
5. **Always give the colour before and after** — iodine orange-brown → blue-black; Benedict's blue → orange-red; biuret blue → violet; ethanol test → milky white emulsion. Write the colour you actually see (e.g. "stayed blue"), never just "no change" or "positive".
6. **Do not mix up the colours** — iodine solution starts orange-brown, not blue. Benedict's and biuret both start blue, but Benedict's ends orange-red and biuret ends violet. Swapping the start and end colours, as in Practice Question 4a(ii), loses the mark.
7. **Describe the ethanol emulsion test in the right order** — shake the food with ethanol first, then pour the ethanol (not the food) into water; a milky white emulsion means fat. Use very clean tubes.
8. **Read results tables carefully** — "contains only fat" means positive for the ethanol emulsion test but negative for every other test (iodine stays brown, Benedict's stays blue, biuret stays blue).

9. **Benedict's tests for reducing sugars, not starch** — the colour reached (green, yellow or orange-red) shows how much reducing sugar is present.
10. **Use the command words** — identify (name, select or recognise), give (produce an answer from a given source or from recall), describe (state the points of a topic or give its main features). For "describe how you could test", give the reagent, what you do and the result.

B4 Enzymes

1. **Learn the definition of a catalyst exactly** — "a substance that increases the rate of a chemical reaction and is not changed by the reaction". You need both parts: it speeds up the reaction and it is unchanged.
2. **Enzymes are proteins** — they are made of amino acids and act as biological catalysts in all metabolic reactions. Without them, reactions would be too slow to sustain life.
3. **Use the word complementary** — the active site has a shape that is complementary to the substrate. Do not say the active site is "the same shape" as the substrate.
4. **Name the enzyme–substrate complex** — in Supplement answers, say that the substrate binds to the active site to form an enzyme–substrate complex, the products are released, and the enzyme is unchanged.
5. **Explain specificity using shape** — only a substrate with a shape complementary to the active site can fit and bind, so each enzyme acts on only one type of substrate.
6. **Enzymes are denatured, not killed** — enzymes are molecules, not living things, so never write that they "die" or are "killed".
7. **Low temperature does not denature enzymes** — at low temperatures enzymes are slow because molecules have little kinetic energy; they work again when warmed. Denaturation happens above the optimum temperature or at a pH far from the optimum.
8. **Use the key words for temperature** — kinetic energy, frequency of effective collisions, enzyme–substrate complex, shape of the active site, complementary, denatured.
9. **Explain pH effects in terms of shape** — pH changes the shape of the enzyme and its active site. Kinetic energy and collisions are the explanation for temperature, not for pH.
10. **Describe graphs with numbers and units** — quote the optimum (for example "about 37 °C") and the values where activity starts and stops. "Describe" means say what happens; "explain" means give the reasons.
11. **Time and rate are opposites** — a shorter time means a faster rate (rate = 1 ÷ time). In the catalase, lipase and amylase experiments, the shortest time is at the optimum.
12. **Know the iodine test colours** — blue-black means starch is present; brown means no starch. Do not say that iodine "turns clear".
13. **Spell enzyme names carefully** — maltose (a sugar) and maltase (an enzyme) look almost the same; lipid is the substrate and lipase is the enzyme.
14. **Make controlled variables specific** — write "temperature", "volume of amylase solution" or "concentration of starch solution", not just "the amount".
15. **Different enzymes have different optimums** — most enzymes about pH 7 but the stomach protease about pH 2; human enzymes about 37 °C but some hot-spring bacteria up to 80 °C.

B5 Plant Nutrition

1. **Learn both photosynthesis equations exactly** — Word equation: carbon dioxide + water → glucose + oxygen. Balanced equation: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. Write light and chlorophyll on the arrow, never as reactants, and check that the atoms balance.
2. **Chlorophyll transfers energy, it does not make it** — Say that chlorophyll absorbs (traps) light energy and transfers it into chemical energy in glucose. Never write that plants "make energy".

3. **Know the starch test and the reason for every step** — Boil in water (breaks cell membranes so iodine can reach the starch), turn off the flame, warm alcohol in a water bath (removes chlorophyll so the colour can be seen), soften in hot water, add iodine solution: blue-black means starch, orange-brown means none.
4. **Explain destarching** — The plant is kept in the dark for about two days so it uses up its starch; any starch found later must have been made during the experiment.
5. **Always identify the control** — Green parts of a variegated leaf (chlorophyll), the uncovered part of the leaf (light), the leaf in the flask of distilled water (carbon dioxide). The control is identical except for the factor being tested.
6. **Name the leaf tissues in order and link each to its function** — Cuticle, upper epidermis, palisade mesophyll, spongy mesophyll with air spaces, vascular bundle (xylem and phloem), lower epidermis with stomata and guard cells.
7. **Know which cells contain chloroplasts** — Palisade mesophyll cells have the most, spongy mesophyll cells have many, guard cells have a few; other epidermis cells and xylem have none. Use this to interpret data such as the bar chart in Practice Question 4.
8. **Get the routes of the raw materials right** — Carbon dioxide diffuses in through the stomata and air spaces; water travels up the xylem and then moves into cells by osmosis. Carbon dioxide does not enter through the roots.
9. **Give a reason with each use of glucose** — Starch (insoluble store, no effect on osmosis), sucrose (transport in phloem), cellulose (cell walls), amino acids and proteins (growth, need nitrate), chlorophyll (needs nitrate and magnesium), nectar (attracts pollinators), respiration (releases energy).
10. **Do not mix up nitrate and magnesium** — Nitrate ions supply nitrogen for amino acids and proteins (deficiency: weak growth, yellow leaves); magnesium ions are needed for chlorophyll (deficiency: yellowing, especially between the veins).
11. **Plan rate experiments with clear variables** — State the independent variable (light intensity via lamp distance, temperature, or carbon dioxide concentration), the dependent variable (bubbles or gas volume per minute), the controlled variables, the heat shield, and repeats with means.
12. **Draw graphs that earn every mark** — Independent variable on the x-axis, both axes labelled with units, even scales filling at least half the grid, small neat crosses, ruled straight lines from point to point, nothing extended beyond the data.
13. **Describe, then explain, data** — Describe the trend with numbers from the table or graph (for example "increased from 5 mm to 17 mm"), then explain it using raw materials, energy from light, enzymes or limiting factors.
14. **Avoid common mistakes** — Photosynthesis does not "change carbon dioxide into oxygen" (the carbon goes into glucose); plants do not get their food from the soil (only water and mineral ions); bubble counts are not a perfect measure of oxygen because bubbles vary in size.

Summary of Key Concepts

B1 Cells and Organisms

Chapter Summary

- **Seven characteristics:** all organisms show seven characteristics: movement, respiration, sensitivity, growth, reproduction, excretion and nutrition.
- **Cells:** all organisms are made of cells, and new cells are always made by the division of existing cells.
- **Animal and plant cells:** both contain cytoplasm, a cell membrane, ribosomes, mitochondria and a nucleus containing DNA in the form of chromosomes. Plant cells also contain a cell wall, a large vacuole and sometimes chloroplasts.

- **Bacterial cells:** they have cytoplasm, a cell membrane, a cell wall and ribosomes, but no nucleus. Their DNA is circular, and they may have extra, small circles of DNA called plasmids.
- **Membrane and wall:** the cell membrane of all cells is partially permeable and controls what enters and leaves the cell. The cell wall is fully permeable and allows all molecules and ions to pass through it.
- **Ribosomes:** the site of protein synthesis in a cell.
- **Mitochondria:** release energy from glucose and other nutrients by aerobic respiration.
- **Chloroplasts:** the site of photosynthesis.
- **Specialised cells:** cells may be specialised for specific functions. Similar cells form tissues, tissues form organs, organs form organ systems, and organ systems make up the organism.
- **Magnification:** magnification = size of image ÷ size of actual object.
- **Units:** convert millimetres to micrometres by multiplying by 1000 (1 mm = 1000 μm).

Exam Technique Tips

- In magnification questions, always write the equation, convert to the same units, substitute your numbers, and give the answer with a × sign and no units.
- When you measure an image, write your measurement down; it often earns the first mark.
- For "explain" questions about cell structures, link each structure to its function and then to the information in the question (for example, a cell that makes a lot of protein needs many ribosomes and many mitochondria).
- Use precise biological words: partially permeable, aerobic respiration, protein synthesis, and release energy (not produce energy).
- Read the question carefully: "not shown by all organisms" and "found in both animal and plant cells" are easy to misread.
- The mark allocation for a question often tells you how many points you need to make to answer it fully.

Command Words Used in This Chapter

- **Identify:** name/select/recognise.
- **Define:** give a precise meaning.
- **Explain:** set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence.
- **Suggest:** apply knowledge and understanding to situations where there is a range of valid responses in order to make proposals/put forward considerations.
- **Calculate:** work out from given facts, figures or information.
- **State:** express in clear terms.

Self-Check

- I can define the seven characteristics of living organisms (B1.01)
- I can describe how all organisms are made of cells, and how new cells are always made by the division of existing cells (B1.02)
- I can describe and compare the structure of a bacterial cell, an animal cell and a plant cell (B1.02)
- I can describe the functions of each of the structures found in an animal cell and a plant cell (B1.02)
- I can identify cell structures in diagrams and micrographs (B1.02)
- I can name the seven specialised cells in Table B1.01 and state their functions (B1.03)
- I can explain the terms cell, tissue, organ, organ system and organism, with examples (B1.03)
- I can calculate magnification, actual size and image size, showing my working (B1.04)
- I can convert between millimetres and micrometres (B1.04)

B2 Movement into and out of Cells

Chapter Summary

- Diffusion is the net movement of molecules or ions from a high concentration to a lower concentration, as a result of their random movement and kinetic energy.
- Oxygen and carbon dioxide enter and leave cells by diffusion through the cell membranes, down a concentration gradient.
- An increase in temperature, concentration gradient or surface area speeds up diffusion into a cell. This can be investigated using pieces of agar jelly.
- Water diffuses through partially permeable membranes, including cell membranes, by osmosis.
- A solution containing a lot of water has a high water potential; in osmosis, water diffuses down a water potential gradient.
- Animal cells burst in pure water and shrink when placed in a concentrated solution. Plant cells do not burst in pure water because their cell wall stops this happening.
- A plant cell in pure water takes up water by osmosis and becomes turgid, as its contents exert outward pressure on the cell wall. A plant cell in a concentrated solution loses water by osmosis. The cell becomes flaccid and may plasmolyse.
- Active transport is the movement of particles across a cell membrane, up the concentration gradient, using energy from respiration.
- Protein carriers in cell membranes change shape to move substances across the cell membrane during active transport.

Self-Evaluation Checklist

Check that you can:

- explain what is meant by a diffusion gradient (B2.01)
- describe how to investigate factors that affect diffusion (B2.01)
- describe osmosis, referring to diffusion of water through a partially permeable membrane (B2.02)
- describe osmosis, using the term water potential (B2.02)
- describe and explain what happens when animal or plant cells are immersed in solutions of different concentrations (B2.02)
- describe active transport (B2.03)
- explain how active transport happens, including the role of protein carriers in membranes (B2.03)

Command Words in This Chapter

- **Identify:** name, select or recognise.
- **Give:** produce an answer from a given source or from recall or memory.
- **Describe:** state the points of a topic, or give its characteristics and main features.
- **Suggest:** apply your knowledge and understanding to a situation where there is a range of valid responses, in order to make proposals or put forward considerations.
- **State:** express in clear terms.
- **Explain:** set out purposes or reasons, make the relationships between things clear, say why and/or how, and support with relevant evidence.
- **Compare:** identify or comment on similarities and/or differences.

Exam Technique Tips

- Match your answer to the command word: "describe" means say what happens (with data if there is a graph); "explain" means say why it happens.
- For "compare" questions, write about both things in each sentence, for example "both cells take in water, but only the animal cell bursts".

- Count the marks: a [3] answer needs three separate, clear points.
- When a graph or chart is given, quote values with units and use comparisons such as "8 times higher".
- Always name the membrane in osmosis answers ("partially permeable cell membrane") and the energy source in active transport answers ("energy from respiration").
- In experiment questions, use the words independent variable, dependent variable and standardised variables, and describe how you would keep each standardised variable the same.

B3 Biological Molecules

From the Chapter Summary

- Carbohydrates are made of the elements C, H and O. Sugars, such as glucose, have relatively small molecules. Starch and glycogen are made of many glucose molecules linked in a long chain.
- Starch can be detected using the iodine test. Benedict's solution is used to test for reducing sugars.
- Fats and oils are also made of the elements C, H and O. Another name for fats and oils is lipids. A lipid molecule is made of three fatty acid molecules combined with one glycerol molecule.
- The ethanol emulsion test is used to detect the presence of fats and oils.
- Proteins contain the elements C, H, O and N. A protein molecule is made of many amino acid molecules combined in a long chain. There are 20 different amino acids, and their specific sequence in the molecule determines the specific protein that is formed.
- Biuret reagent is used to test for proteins.

Self-Evaluation Checklist

All of these skills belong to topic B3.01. Make sure you can:

- List the chemical elements that make up carbohydrates, fats and proteins
- State that starch and glycogen are made from glucose
- State that fats and oils are made from fatty acids and glycerol
- State that proteins are made from amino acids
- Describe how to use the Benedict's test for reducing sugars
- Describe how to use the iodine test for starch
- Describe how to use the biuret test for proteins
- Describe how to use the ethanol emulsion test for fats and oils

Exam Technique Tips

- For any "describe how you could test" question, include four things: the reagent, what you do (heat in a water-bath only for the Benedict's test), the colour for a positive result and the colour for a negative result
- Match the number of points to the marks: a 3-mark "describe" question needs three separate, clear points
- For an "identify" question, a single word or letter is enough; for a "give" question, a short answer from memory is enough
- With a results table, decide whether each test is positive or negative before you choose an answer
- Use precise colour words: blue-black, orange-red (brick red), violet (purple) and milky white; avoid vague phrases such as "changes colour" or "goes darker"
- Do not confuse glycogen (a carbohydrate energy store made from glucose) with glycerol (part of a fat molecule)

B4 Enzymes

Chapter Summary

- **Enzymes are proteins** that function as biological catalysts and increase the rate of metabolic reactions.
- **Without enzymes**, these reactions would not take place fast enough to sustain life.
- **Each enzyme has an active site** that has a complementary shape to its substrate.
- **When the substrate binds with the active site** of the enzyme, the substrate is converted to product. The enzyme is unchanged.
- **The binding of substrate to the active site** of the enzyme forms an enzyme–substrate complex.
- **Enzymes have an optimum temperature and pH** at which they work most quickly. Higher temperatures or a very different pH cause the enzyme to change shape and denature.
- **As temperature increases**, the kinetic energy of enzyme and substrate molecules increases, and they collide more frequently. This increases the rate of activity of the enzyme. However, above the optimum temperature, the enzyme begins to lose its shape, until it can no longer combine with the substrate. The shape of the active site is no longer complementary to the shape of the substrate.

Command Words Used in This Chapter

- **Identify** — name, select or recognise.
- **Describe** — state the points of a topic; give characteristics and main features.
- **Explain** — set out purposes or reasons; make the relationships between things clear; say why and/or how, and support with relevant evidence.
- **State** — express in clear terms.

Exam Technique Tips

- **Graph questions:** "describe" means say what the graph shows, with numbers and units (for example, "the activity of enzyme A increases from zero at 10 °C to a maximum at 40 °C, then falls to zero at about 58 °C"). "Explain" means give the biological reasons (kinetic energy, collisions, shape of the active site, denaturation).
- **Compare questions:** write about both enzymes in every sentence, using words like "whereas" and "both", and include at least one similarity and one difference.
- **Time versus rate:** if a table gives times, remember that the shortest time shows the highest activity.
- **Experiment questions:** name specific controlled variables, give a reason for any anomalous result, and suggest realistic improvements (repeats, smaller intervals, a water-bath, fresh solutions).
- **Diagrams of enzyme action:** label the substrate, active site, enzyme–substrate complex and products, and say that the enzyme is unchanged at the end.

Self-Evaluation Checklist

Check that you can:

- explain what a catalyst is (B4.01)
- explain why enzymes are said to be biological catalysts, and why they are important to sustain life (B4.01)
- describe how enzymes work, using the terms active site, substrate, complementary shape and product (B4.01)
- use the term enzyme–substrate complex (B4.01)
- explain why enzymes are specific (B4.01)
- describe how to investigate the effect of temperature and pH on enzyme activity, and describe these effects, using the words optimum and denaturation (B4.02)
- explain the effect of temperature on enzyme activity, using the words kinetic energy, frequency of effective collision, complementary shape and denaturation (B4.02)
- explain the effect of pH on enzyme activity, using the words complementary shape and denaturation (B4.02)

B5 Plant Nutrition

Chapter Summary

- **Photosynthesis** is a reaction in which plants make carbohydrates from carbon dioxide and water, using light as a source of energy.
- The word equation for photosynthesis is: carbon dioxide + water → glucose + oxygen
- The balanced equation is: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
- **Chlorophyll** is a green pigment present in the chloroplasts of some plant cells. Chlorophyll transfers energy from light into energy in carbohydrates.
- The carbohydrate **glucose** is made in photosynthesis. It can be used to release energy in **respiration**. Glucose can be used to make **starch** for storage of energy and **sucrose** for transport.
- Glucose can be used to make **amino acids and proteins**, using nitrogen obtained from **nitrate ions**. It can also be used to make **chlorophyll**, using nitrogen and also **magnesium ions**.
- **Cellulose** is used to build cell walls, **sucrose** is used for transport in the phloem, and **nectar** attracts insects for pollination.
- Leaves have a **large surface area**, to capture energy from sunlight and to increase absorption of carbon dioxide from the air. **Stomata** on the lower surface allow carbon dioxide to diffuse into the **air spaces**, and then to the cells in the **palisade mesophyll** layer, where most photosynthesis takes place.
- To **test a leaf for starch**, it is first boiled to break down cell membranes, then heated in alcohol to remove the chlorophyll. Finally, iodine solution is added.
- The starch test shows that leaves, or parts of a leaf, that do not have light, chlorophyll or carbon dioxide are not able to photosynthesise and make starch. **Controls** are needed so that we can compare the results with them.
- We can use an **aquatic plant** to measure the rate of photosynthesis, by counting the number of bubbles given off in unit time, or by measuring the volume of gas given off in unit time.
- This method can be used to investigate the effect of altering **light intensity, temperature or carbon dioxide concentration** on the rate of photosynthesis.

Command Words Used in This Chapter

- **Identify** — name, select or recognise.
- **Give** — produce an answer from a given source or from recall or memory.
- **Outline** — set out the main points.
- **Describe** — state the points of a topic, or give its characteristics and main features.
- **Explain** — set out purposes or reasons; make the relationships between things clear; say why and/or how, and support with relevant evidence.
- **Suggest** — apply your knowledge and understanding to situations where there is a range of valid responses, in order to make proposals or put forward considerations.

Exam Technique Tips

- For "identify" multiple-choice questions, rule out options that contain a product (glucose or oxygen) when the question asks for raw materials.
- For "outline" and "describe" questions, give one clear point per mark — a 2-mark adaptation question needs two separate adaptations.
- For "explain" questions, always link the feature to the function using "so that" or "because", e.g. "many chloroplasts, so that more light energy is absorbed".
- When a question says "use the letters on the diagram", put the letters into your answer (for example "xylem in E, then by osmosis through D to C").
- When describing results, quote numbers with units from the data, then explain the pattern.

- In planning questions, always state the independent variable, the dependent variable, at least two controlled variables, repeats and a safety point.
- On a graph, read values carefully against the gridlines and say "about" if a value lies between lines.

Self-Evaluation Checklist

I can:

- state the word equation for photosynthesis (B5.01)
- write the balanced equation for photosynthesis (B5.01)
- describe the role of chlorophyll in transferring energy from light to carbohydrates, in photosynthesis (B5.01)
- outline the uses of starch, cellulose, sucrose and nectar, made from the carbohydrates made in photosynthesis (B5.01)
- state the functions of nitrate ions and magnesium ions in a plant (B5.01)
- name the different cell types and layers of a leaf (B5.02)
- explain how the features of a leaf are adaptations for photosynthesis (B5.02)
- describe how to investigate the necessity for chlorophyll, light and carbon dioxide for photosynthesis (B5.03)
- explain what a control is, in an experiment (B5.03)
- describe how to investigate the effect of varying light intensity and temperature on the rate of photosynthesis of an aquatic plant (B5.03)